



DEPARTMENT OF LABOR AND EMPLOYMENT

Division of Workers' Compensation

WORKERS' COMPENSATION RULES OF PROCEDURE WITH TREATMENT GUIDELINES

RULE 17, EXHIBIT 1 LOW BACK PAIN MEDICAL TREATMENT GUIDELINES

7 CCR 1101-3 Rule 17 - Exhibit 1

[Editor's Notes follow the text of the rules at the end of this CCR Document.]

Section 1. Guidelines Introduction

Section 1.a. Context and Use

This document has been prepared by the Colorado Department of Labor and Employment, Division of Workers' Compensation (Division) and should be interpreted within the context of guidelines for physicians/providers treating individuals who qualify as injured workers with low back pain under the Colorado Workers' Compensation Act.

Although the primary purposes of this document for practitioners are advisory and educational, these guidelines are enforceable under the Workers' Compensation Rules of Procedure, 7 CCR 1101-3. The Division recognizes that acceptable medical practice may include deviations from these guidelines, as individual cases dictate. Therefore, these guidelines are not relevant as evidence of a provider's legal standard of professional care.

To properly utilize this document, the reader should not skip or overlook any sections.

Section 1.b. Application of the Guidelines

The Division provides procedures to implement medical treatment guidelines (MTGs) and to foster communication to resolve disputes among the provider, payer, and patient through the Workers' Compensation Rules of Procedure. In lieu of more costly litigation, parties may wish to seek administrative dispute resolution services through the Division or the Office of Administrative Courts.

Section 1.c. Guidelines Recommendations and Inclusion of Medical Evidence

All recommendations are based on available evidence and/or consensus judgment. A Division staff methodologist (MD, MSPH) researched and adopted literature critique criteria for certain study designs (e.g., randomized controlled trial [RCT]). Literature critiques were performed in a manner congruent with national standards and were completed independent of the multidisciplinary task force group that drafted initial recommendations. The methodology is described in detail on the Division's MTGs page at cdle.colorado.gov/medical-treatment-guidelines. Please also refer to the Division's website for evidence tables and study critiques that provide details on the studies used to develop the evidence statements.

When possible, guidelines recommendations link to evidence tables that include the level of evidence supporting the treatment recommendation. An evidence summary is included in tables where multiple evidence statements address the same topic. It is generally recognized that early reports of a positive treatment effect are frequently weakened or overturned by subsequent research. When interpreting medical evidence statements in the guidelines, the following apply:

- “Some evidence” means the recommendation considered at least 1 adequate scientific study, which reported that a treatment was effective. The Division recognizes that further research is likely to have an impact on the intervention’s effect.
- “Good evidence” means the recommendation considered the availability of multiple adequate scientific studies or at least 1 relevant high-quality scientific study, which reported that a treatment was effective. The Division recognizes that further research may have an impact on the intervention’s effect.
- “Strong evidence” means the recommendation considered the availability of multiple relevant and high-quality scientific studies, which arrived at similar conclusions about the effectiveness of a treatment. The Division recognizes that further research is unlikely to have an important impact on the intervention’s effect.

Recommendations not linked to evidence tables are based on consensus; consensus means the judgment of experienced professionals based on general medical principles.

All recommendations in the guidelines are considered to represent reasonable care in appropriately selected cases, irrespective of the level of evidence or consensus statement attached to them. Those procedures considered inappropriate, unreasonable, or unnecessary are designated in the guidelines as “not recommended.”

Section 1.d. Recommended Citation for This Document

Colorado Division of Workers' Compensation. (2021). Low Back Pain Medical Treatment Guidelines. Colorado Department of Labor and Employment. <https://cdle.colorado.gov/medical-providers/medical-treatment-guidelines>.

Section 1.e. Glossary of Abbreviations

- ADLs - Activities of daily living
- ALIF - Anterior lumbar interbody fusion
- BBHI 2 - Brief Battery for Health Improvement, 2nd Edition
- BCIA - Biofeedback Certification International Alliance
- BDI-II - Beck Depression Inventory-II
- BHI-MV - Behavioral Health Index-Multimedia Version
- BMI - Body mass index
- BMP - Bone morphogenetic protein
- CBT - Cognitive behavioral therapy
- CES-D - Center for Epidemiological Studies Depression Scale
- COX-2 - Cyclooxygenase-2
- CT - Computed tomography
- DRAM - Distress and Risk Assessment Method
- DSM - Diagnostic and Statistical Manual of Mental Disorders
- ESI - Epidural steroid injection
- FCE - Functional capacity evaluation
- G6PD - Glucose-6-phosphate-dehydrogenase
- GAD-7 - Generalized Anxiety Disorder Scale
- GI - Gastrointestinal
- H2-blocker - Histamine H2-receptor antagonists
- MME - Morphine milligram equivalents
- MMI - Maximum medical improvement
- MRI - Magnetic resonance imaging
- MTGs - Medical treatment guidelines

- NMES - Neuromuscular electrical stimulation
- NRS - Numeric rating scale
- NSAIDs - Non-steroidal anti-inflammatory drugs
- ODI - Oswestry Disability Index, or Oswestry Low Back Pain Disability Questionnaire
- ORT - Opioid Risk Tool
- PDMP - Prescription Drug Monitoring Program
- PENS - Percutaneous electrical nerve stimulation
- PHQ - Patient Health Questionnaire
- PLIF - Posterior lumbar interbody fusion
- PMMA - Polymethylmethacrylate
- PRIME-MD - Primary Care Evaluation for Mental Disorders
- QPOP - Quality Performance and Outcomes Payments
- RCT - Randomized controlled trial
- RF - Radiofrequency
- rhBMP-2 - Recombinant human bone morphogenetic protein-2
- ROM - Range of motion
- SF-12 - 12-item Short Form Health Survey
- SF-36 - 36-item Short Form Health Survey
- SI - Sacroiliac
- SNRI - Serotonin norepinephrine reuptake inhibitor
- SSRI - Selective serotonin reuptake inhibitors
- TDR - Total disc replacement
- TENS - Transcutaneous electrical nerve stimulation
- TLIF - Transforaminal lumbar interbody fusion
- VAS - Visual analogue scale

Section 2. General Guidelines Principles

This document should be interpreted within the parameters of the following guidelines principles that may lead to more optimal medical and functional outcomes for injured workers.

Section 2.a. Education

Education of the individual and family and/or support system, as well as the employer, insurer, policy makers, and the community, should be the primary emphasis in the treatment of low back pain and disability. Currently, practitioners often think of education last, after medications, manual therapy, and surgery. Practitioners must implement strategies to educate individuals with low back pain, employers, insurance systems, policy makers, and the community as a whole. An education-based paradigm should always start with inexpensive communication that provides recovery, function-focused, patient-centered, and evidence-based information to the individual with low back pain. More in-depth education is currently a component of treatment regimens that employ functional, restorative, preventive, and rehabilitative programs. No treatment plan is complete without addressing issues of individual and/or group patient education as a means of facilitating self-management of symptoms and prevention. Facilitation through language interpretation, when necessary, is a priority and part of the medical care treatment protocol.

Section 2.b. Shared Decision Making

Providers should implement shared decision making as a crucial element of a successful treatment plan. Patients, with the assistance of their health care practitioner and support system, should identify their personal and professional functional goals of treatment at the first visit. Progress towards the individual's identified functional goals should be addressed by all members of the health care team at subsequent visits and throughout the established treatment plan. Nurse case managers, psychologists, physical therapists, and other members of the health care team play an integral role in shared decision making and achievement of functional goals. Patient education and shared decision making should facilitate self-management of symptoms and prevention of further injury.

Section 2.c. Return to Work

Return to work is therapeutic, assuming the work is not likely to aggravate the basic problem. The practitioner must provide specific written physical limitations, and the patient should never be released to work with non-specific and vague descriptions such as "sedentary" or "light duty." The following physical limitations should be considered and modified as recommended: lifting, pushing, pulling, crouching, carrying, walking, using stairs, bending at the waist, awkward and/or sustained postures, tolerance for sitting or standing, hot and cold environments, repetitive motion tasks, sustained grip, tool usage, and vibration factors. Even if there is residual chronic pain, return to work is not usually contraindicated.

The practitioner should understand all of the physical demands of the patient's job position before returning the patient to full duty and should request clarification of the patient's job duties. Clarification should be obtained from the employer or, if necessary, including, but not limited to, an occupational health nurse, occupational therapist, vocational rehabilitation specialist, or an industrial hygienist.

Section 2.d. Treatment Parameter Duration

Time frames for specific interventions commence once treatments have been initiated, not on the date of injury. Duration will be impacted by the individual's adherence, as well as availability of services. Clinical judgment may substantiate the need to accelerate or decelerate the time frames discussed in this document.

Section 2.e. Active Interventions

Active interventions emphasizing patient responsibility, such as therapeutic exercise and/or functional treatment, are generally utilized over passive interventions, especially as treatment progresses. Generally, passive interventions are viewed as a means to facilitate progress in an active rehabilitation program with concomitant attainment of objective functional gains.

Section 2.f. Active Therapeutic Exercise Program

Exercise program goals should incorporate patient strength, endurance, flexibility, coordination, and education. This includes functional application in vocational or community settings.

Section 2.g. Positive Patient Response

Positive results are defined primarily as functional gains that can be objectively measured. Objective functional gains include, but are not limited to: positional tolerances, range of motion (ROM), strength, endurance, activities of daily living (ADLs), ability to function at work, cognition and communication, psychological behavior, and efficiency/velocity measures that can be quantified. Subjective reports of pain and function should be considered and given relative weight when the pain has anatomic and physiologic correlation. Anatomic correlation must be based on objective findings. Patient completed functional questionnaires can provide useful additional confirmation, such as those recommended by the Colorado Division of Workers' Compensation (Division) Quality Performance and Outcomes Payments program (QPOP, see Rule 18), the Patient-Specific Functional Scale, or other validated function scales.

Section 2.h. Re-evaluation of Treatment Effectiveness

Re-evaluation should occur every 3 to 4 weeks or within the time to produce effect for a given treatment. Treatment should be modified or discontinued if there is no evidence of positive results. Before discontinuing the treatment, the provider should have a detailed discussion with the patient to determine the reason for failure to produce positive results. Reconsideration of diagnosis should also occur in the event of a poor response to a seemingly rational intervention.

Section 2.i. Surgical Interventions

Surgery should be contemplated within the context of expected functional outcome and not purely for the purpose of pain relief. The concept of "cure" with respect to surgical treatment by itself is generally a misnomer. All operative interventions must be based upon positive correlation of clinical findings, clinical course, and diagnostic tests. A comprehensive assimilation of these factors must lead to a specific diagnosis with positive identification of pathologic conditions.

Section 2.j. 6-month Time Frame

The prognosis drops precipitously for returning an injured worker to work once they have been temporarily totally disabled for more than 6 months. The emphasis within these guidelines is to move patients along a continuum of care and return to work within a 6-month time frame, whenever possible. It is important to note that time frames may be less pertinent for injuries that do not involve work-time loss or are not occupationally related.

Section 2.k. Delayed Recovery

For patients who are failing to make expected progress 6 to 12 weeks after initiation of treatment of an injury, strongly consider a psychological evaluation, if not previously provided, as well as initiating interdisciplinary rehabilitation treatment and vocational goal setting. The Division recognizes that 3 to 10% of all industrially injured patients will not recover within the timelines outlined in this document, despite optimal care. Such individuals may require treatments beyond the timelines discussed within this document, but such treatment requires clear documentation by the authorized treating practitioner focusing on objective functional gains afforded by further treatment and impact upon prognosis.

Section 2.l. Post Maximum Medical Improvement (MMI) Care

Maximum medical improvement (MMI) should be declared when a patient's condition has plateaued to the point where the authorized treating physician no longer believes further medical intervention is likely to result in improved function. However, some patients may require treatment after MMI has been declared in order to maintain their functional state. The recommendations in these guidelines are for pre-MMI care and are not intended to limit post-MMI treatment.

Section 3. Overview of Care

Introduction. Low back pain is a common condition with a lifetime prevalence of 84% and a high recurrence rate. However, only about 15% of the population has severe pain with functional/disability limitations. Most low back pain responds to therapy and self-management and does not require invasive measures. These guidelines outline a biopsychosocial approach to low back pain care and integrate biological, psychological, and social elements.

Significant trauma resulting in fractures and/or spinal cord dysfunction are not covered in these guidelines. Early imaging and surgical evaluation should be reserved for more severe issues outlined in [recommendation 1](#). Most patients will respond to conservative care as outlined in recommendations 2 through 5 and will not require invasive treatment. Additional interventions, as indicated for less common cases, are covered by the remaining recommendations.

Recommendations.

Core Requirements.

Recommendation 1. Early imaging and surgical evaluation is required for patients with evidence of any of the following:

- trauma with concern for acute fracture or dislocation,
- epidural abscess,
- myelopathy,
- cauda equina syndrome, or
- progressive neurologic deficits (e.g., motor weakness and abnormal reflexes) that specifically correlate with spinal cord or nerve root impingement.

Recommendation 2. Initial conservative management without imaging is strongly recommended for all low back pain patients who do not meet the criteria in [recommendation 1](#), focusing on the following:

- education that prolonged periods of immobility are detrimental,
- education on the positive impact of movement on recovery,
- appropriate use of over the counter medications, and
- application of ice or heat.

Recommendation 3. Bed rest is not recommended ([table 1](#)).

Recommendation 4. Close follow-up and subsequent neurologic examinations are required for patients with radicular low back pain who do not meet the criteria in [recommendation 1](#) or [recommendation 8](#).

Recommendation 5. Participation in self-directed exercise or an active therapy program that includes core stabilization, strengthening, and endurance is recommended as a principal low back pain treatment. See the [Active Therapies section](#).

Recommendation 6. Individuals with barriers to functional recovery may benefit from an interdisciplinary approach to care. See the [Diagnosis section](#).

Recommendation 7. Referral for epidural steroid injection (ESI) evaluation within 6 weeks of pain onset is permitted when all of the following criteria are met:

- severe function-limiting pain in legs greater than the back that interferes with return to work, activities of daily living (ADLs), and/or participation in active therapy; and
- positive correlation among clinical findings, the clinical course, and diagnostic tests.

See the [Epidural Steroid Injections section](#).

Recommendation 8. Referral for discectomy evaluation within 6 weeks of pain onset is permitted when all of the following criteria are met:

- function-limiting pain in legs greater than the back that interferes with return to work, ADLs, and/or participation in active therapy;
- physical exam findings of abnormal reflexes, motor weakness, or radicular sensory deficits;
- findings on magnetic resonance imaging (MRI) indicate impingement of nerves or the spinal cord that specifically correlate with reproducible physical exam findings; and
- the observed pathology is amenable to surgical intervention.

See the [Discectomy section](#).

Recommendation 9. Referral for spinal injections is indicated when all of the following criteria are met:

- positive correlation among clinical findings, the clinical course, and diagnostic tests;
- positive functional response to diagnostic injection, if required;
- persistent functional impairment despite engagement in 6 weeks of active therapy; and
- confounding psychosocial risk factors have been screened for and clinically addressed ([see the Behavioral and Psychological Interventions section](#)).

See the [Injections section](#).

Recommendation 10. Referral for surgical evaluation at 6 weeks or greater is indicated when the expected functional outcome of surgery is better than non-operative management and all of the following criteria are met:

- symptomatic and functional improvement has plateaued with unacceptable functional disability;
- greater pain in the legs relative to back that interferes with function, return to work, and/or active therapy;
- physical exam findings of abnormal reflexes, motor weakness, or radicular sensation deficits;
- findings on MRI that indicate impingement of nerves or the spinal cord that specifically correlate with reproducible physical examination findings; and
- diagnostic elimination of confounding psychological or physical conditions that may respond to non-surgical techniques but may be refractory to surgical intervention ([see the Behavioral and Psychological Interventions section](#)). Also [see the Surgical Interventions section](#).

Evidence Table.

Table 1.

Evidence Table: Return to Daily Activities and Avoidance of Bed Rest in Low Back Pain

Summary:

Bed rest is not helpful for treating uncomplicated low back pain. Mobilization in subacute low back pain reduces disability.

Strong evidence	Evidence statement	Design
	There is strong evidence against the use of bed rest in acute low back pain cases without neurologic symptoms.	RCT
Good evidence	Evidence statement	Design
	Education and mobilization of subacute low back pain reduces disability.	RCT
Some evidence	Evidence statement	Design
	Among a cohort of 33,908 healthy study volunteers, new cases of depression were potentially preventable by participation in at least 1 hour of weekly exercise.	Cohort study

See related sections for additional evidence tables.

Section 4. **Diagnosis**

Introduction. Initial and follow-up evaluation of low back pain may include a clinical history, physical examination, laboratory studies, imaging studies, and electrodiagnostic studies. Please see specific sections for diagnostic injections.

Recommendations.

Initial Diagnostic Studies Requirements.

Recommendation 11. A detailed history of pain onset, past medical history, and a detailed neurologic examination are required at the initial low back pain evaluation (tables 2, 3).

Recommendation 12. Medical causation must establish that low back pain is a result of a specific injury, an aggravation of an underlying condition, or a previously asymptomatic condition made symptomatic by a work-related exposure (tables 4, 5).

Recommendation 13. Initial functional assessment is strongly recommended. See the Appendix Functional Screens.

Recommendation 14. A psychological screen is encouraged as a routine part of clinical care and is required as soon as any of the following barriers to functional recovery are identified (Appendix Psychological Screens):

- limited patient engagement in recovery,
- activity avoidance or catastrophization due to pain,
- avoidance of essential recovery activities,
- inappropriate expression of a sense of injustice,
- exaggeration of symptoms/situation,
- low expectations of recovery,
- ineffective coping skills, or
- loss of vocational connection.

Individuals with barriers to functional recovery may benefit from an interdisciplinary approach to care (table 6, 7).

Recommendation 15. Imaging studies before 6 weeks are not recommended for patients with uncomplicated low back pain (table 8).

Recommendation 16. Imaging studies are recommended if the patient is 55 years or older, if there is a history of significant trauma, if there is persistent or unexplained pain after 6 weeks, or if there is concern for any of the following:

- fracture,
- instability,
- occult cancer,
- weakness,
- pain worse at rest,
- cauda equina syndrome,
- epidural abscess, or
- myelopathy.

Documentation shall include the specific findings under investigation and how the imaging test will influence treatment.

Recommendation 17. When clinically indicated, early laboratory studies may be ordered to evaluate for systemic illness, infection, neoplasia, underlying rheumatologic, or connective tissue disorder.

Follow-Up Diagnostic Studies Requirements.

Recommendation 18. A formal psychological or psychosocial evaluation is required for patients not making expected progress within 6 weeks of injury and whose subjective symptoms do not correlate with objective signs and tests. See the Behavioral and Psychological Interventions section.

Recommendation 19. Magnetic resonance imaging (MRI) is indicated when there is concern for myelopathy, masses, infection, metastatic disease, or cord contusion. MRI is recommended for patients who have responded poorly to initial care and there is clinical suspicion for disc herniation, nerve root compression, or annular tear. Contrast MRI can be used when there is a history of lumbar surgery or if there is possible infection, malignancy, or tumor (table 8).

Recommendation 20. Computed tomography (CT) scan is permitted to better visualize bone and to further evaluate masses and suspected fractures not clearly identified on radiographic evaluation.

Recommendation 21. Myelography is a diagnostic option when:

- CT or MRI are contraindicated or not available, and
- when other tests have proven non-diagnostic in the surgical candidate.

Recommendation 22. Radioisotope bone scanning is permitted when there is clinical suspicion for metastatic or primary bone tumors, occult or stress fractures, osteomyelitis, infection, or other inflammatory lesions.

Recommendation 23. Discography is not generally recommended, but it is permitted when confirmatory information is necessary prior to proceeding with lumbar fusion or total disc replacement (TDR). See the Spinal Fusion or Total Disc Replacement sections.

Recommendation 24. Electromyography and nerve conduction studies are recommended for patients with suspected neural involvement and persistent symptoms that are unresponsive to initial conservative treatment.

Recommendation 25. Somatosensory evoked potentials may be used to evaluate myelopathy, neurogenic bladder, and sexual dysfunction, but it is not recommended to identify radiculopathy.

Recommendation 26. Surface electromyography and current perception threshold evaluation are not recommended.

Evidence Tables.

Table 2.

Evidence Table: Self-reported Body Pain Diagrams in Describing Low Back Pain

Summary:

Pain diagrams, either manual or computerized, reliably identify pain location.

Some evidence	Evidence statement	Design
	Using a computer program (electronic body surface estimation method) to measure pain distribution was found to have both high intra- and inter-examiner reliability in patients with chronic nonmalignant pain.	Systematic review
	Using either segmental classification or Quebec Task-Force classification to measure pain location was found to have both adequate intra- and inter-examiner reliability in patients with low back pain or lumbar radiculopathy.	Systematic review
	A body diagram, in which the patients use a pencil to shade in painful areas of the body, has acceptable reliability for designating the regions in which their pain is felt.	Systematic review

Table 3.

Evidence Table: Physical Examination Findings and Low Back Pain

Summary:

A single neurologic test in isolation has limited diagnostic value.

Good evidence	Evidence statement	Design
	In the setting of suspected herniated discs, a single neurologic test in isolation (such as sensory testing of specific dermatomes, weakness of specific muscles, or tendon reflexes) is of limited diagnostic value in identifying lumbar spinal disc herniation with radiculopathy. A single level disc herniation may not cause major sensory or motor deficits unless there is a large conduction block of the nerve root axons.	Systematic review
Some evidence	Evidence statement	Design
	In the initial clinical examination findings for adults with low back pain (who presented with radicular findings), the following prognostic factors were found to be consistent for evidence; symptom response classification (i.e., centralization) and non-organic signs. Specifically, centralization was the only factor with an association of less short-term pain. In addition, non-organic signs were the only factors associated with poor long-term outcome of return to work.	Systematic review

Table 4.

Evidence Table: Non-Occupational Risk Factors for Low Back Pain

Summary:

Smoking is a risk factor for lumbar radicular pain. Severe obesity is associated with low back pain.

Strong evidence	Evidence statement	Design
	Current and ever smoking are both moderate risk factors for patients experiencing lumbar radicular pain.	Systematic review
Good evidence	Evidence statement	Design
	Out of many modifiable lifestyle factors, severe obesity (defined as a body mass index [BMI] ≥ 35 kg/m ²) was found to be most associated with lifetime chronic back pain, independent of genetic and familial factors.	Cohort study

Table 5.

Evidence Table: Occupational Risk Factors for Low Back Pain

Summary:

Standing and walking in the workplace do not cause low back pain. Occupational sitting is unlikely to cause low back pain. Lifting or trunk rotation in combination with bending the trunk are risk factors for low back pain, with or without radiculopathy. There is good evidence that the combination of trunk flexion, rotation, and lifting in the workplace cumulatively is associated with low back pain. A combination of trunk flexion while repetitively lifting 50 pounds or more is a cumulative risk factor for low back pain (e.g., lifting a 50 pound box 3 feet more than 50 times per work week). Cumulative lifting (e.g., ≥ 7 hours per week for > 9.5 years), particularly when combined with other exposures, may result in low back pain.

Good evidence	Evidence statement	Design
	Standing and walking do not cause low back pain.	Systematic review
	Trunk flexion, rotation, and lifting in the workplace cumulatively is associated with low back pain.	Cohort study
	Work-related factors, such as lifting and bending of the trunk or bending and twisting of the trunk, increase a worker's risk of developing lumbosacral radiculopathy.	Systematic review
Some evidence	Evidence statement	Design
	Whole body vibration has no important association with disc pathology in the lumbar spine in professional drivers.	Case control study
	Cumulative exposure to lifting in the workplace is associated with the development of low back pain. Exposures of 7 hours per week or greater, over more than 9.5 years, is associated with low back pain in an apparent dose-response relationship. The effects of lifting may only become apparent when considered in combination with other work exposures.	Case control study
	Sitting does not cause low back pain.	Systematic review

Table 6.

Evidence Table: Interdisciplinary Back Pain Care

Good evidence	Evidence statement	Design
	Interdisciplinary programs that include screening for psychological issues, identification of fear-avoidance beliefs and treatment barriers, and establishment of individual functional and work goals will improve function and decrease disability.	RCT

Table 6 continued.

Evidence Table: Interdisciplinary Back Pain Care

Good evidence cont.	Evidence statement	Design
	Interventions addressing fear-avoidance beliefs in populations with high baseline scores on Fear-Avoidance Behavior Questionnaire or Tampa Scale of Kinesiophobia are associated with reduction in pain (or disability) and improvements in return to work in patients with low back pain of less than 6 months duration (i.e., an acute or subacute population).	Systematic review

Table 7.

Evidence Table: Multidisciplinary Biopsychosocial Rehabilitation

Summary:

Multidisciplinary interventions lead to greater improvements in function, pain, and return to work time compared to usual care or singular interventions for patients with subacute and chronic low back pain. Patients with psychosocial risk factors are more likely to benefit from a multidisciplinary approach.

Good evidence	Evidence statement	Design
	Among people with subacute (6-12 weeks) low back pain, multidisciplinary biopsychosocial rehabilitation was significantly more effective than usual care for short (≤ 3 months) and long-term (≥ 12 months) pain and disability improvement, but it was no more effective than other types of treatment. For this systematic review, multidisciplinary biopsychosocial rehabilitation included a physical component (e.g., pharmacological, physical therapy, massage, manual therapy, etc.) in combination with either a psychological, social or occupational component (e.g., cognitive behavioral interventions, fear-avoidance treatment, motivational interviewing).	Systematic review
	Among people with subacute low back pain, multidisciplinary biopsychosocial rehabilitation was significantly more effective than usual care for return to work at the long term (≥ 12 months), but it was no more effective than other types of treatment in reducing sick leave periods at the long term. For this systematic review, multidisciplinary biopsychosocial rehabilitation included a physical component (e.g., pharmacological, physical therapy, massage, manual therapy, etc.) in combination with either a psychological, social or occupational component (e.g., cognitive behavioral interventions, fear-avoidance treatment, motivational interviewing).	Systematic review

	Multidisciplinary biopsychosocial rehabilitation shows small effects in reducing pain and improving disability compared to usual care, and that multidisciplinary biopsychosocial rehabilitation was more effective than physical treatment for return to work after 12 months of treatment in patients with chronic low back pain. Patients with a significant psychosocial impact are most likely to benefit.	Meta analysis
Table 7 continued. <i>Evidence Table: Multidisciplinary Biopsychosocial Rehabilitation</i>		
Good evidence cont.	Patients with chronic low back pain who received multidisciplinary rehabilitation had lower short-term pain intensity and greater improvement in function than those who received usual care, no rehab, and non-multidisciplinary physical therapy.	Systematic review
Some evidence	Evidence statement	Design
	In patients with symptomatic degenerative disc disease at L4-5 or at L5-S1 lasting more than 1 year, persisting after 6 months of physical therapy or chiropractic treatment, an initial treatment regimen consisting of rehabilitation with a multidisciplinary team of specialists may effectively improve function, defined as a 15 point improvement in the Oswestry Disability Index (ODI), in up to half of patients, making it a reasonable initial treatment option. However, a TDR is overall more effective than multidisciplinary rehabilitation for pain and function, and may improve function by the same amount in about 70% of patients.	RCT
	Patients with persistent low back pain and subthreshold depression are at risk for developing a major depressive episode over the course of the next 12 months. A program of online self-help, based on cognitive-behavioral principles in which each participant completes weekly homework assignments and receives weekly guidance and feedback from a trained psychologist, may prevent the progression from subthreshold depression to a major depressive episode.	RCT

Table 8. <i>Evidence Table: Imaging Findings and Low Back Pain</i>		
Summary: Imaging findings occur frequently in asymptomatic patients and therefore are a poor indicator for clinically significant pathology. In addition, imaging may fail to detect problems identified by clinical predictors, such as physical exam findings, pain drawing, and depression.		
Good evidence	Evidence statement	Design
	In the asymptomatic population, disc bulges, disc protrusions, annular tears, high intensity zone areas, and disc height loss are prevalent 40–60% of the time, depending on the condition, study, and age of the patient.	Cohort study

Table 8 continued.

Evidence Table: Imaging Findings and Low Back Pain

Some evidence	Evidence statement	Design
	Extruded discs are uncommon in asymptomatic persons and are associated with low back pain.	Cohort study
	In patients with long-standing nerve root symptoms, MRI significantly underestimates the presence of nerve involvement otherwise detected by a combination of physical examination and simplified pain drawing.	Diagnostic study
	Asymptomatic subjects of all ages frequently show common degenerative changes in the cervical and lumbar spine as seen on MRI. These changes are more common with increasing age. The cervical and lumbar spine are equally affected, suggesting that disc degeneration occurs in tandem in the lumbar and cervical spine.	Cohort study
	The majority of workers over 60 show evidence of disc degeneration, and posterior disc protrusions are present in the majority of asymptomatic workers over 40 years of age.	Cohort study
	Type 1 Modic changes are at an increased risk for no improvement in pain or function and for unsuccessful return to work at 1 year.	Cohort study
	Depression is a more accurate predictor of the development of low back pain than many common MRI findings, such as disc bulges, disc protrusions, Modic endplate changes, disc height loss, annular tears, and facet degeneration, which are common in asymptomatic persons and are not associated with the development of low back pain.	Cohort study
	Early radiographic imaging without clear indications is associated with prolonged care, although it does not change functional outcomes.	RCT

Section 5. Return to Activity and Work Considerations

Introduction. Return to activity and/or work-related activities is a primary therapeutic component of low back pain management and rehabilitation. There are special tests performed as a part of a skilled assessment of the patient's capacity to return to work, their strength capacities, and/or physical work demand classifications and tolerance. Terminology relevant to the activity and return to work discussion includes the following:

- computer enhanced evaluations,
- immobility,
- functional capacity evaluation (FCE),
- jobsite evaluation or analysis, and
- work tolerance screenings.

Communication among the patient, treating provider, employer, and insurer is essential for successful return to work efforts. Refer to the Active Therapies section for work conditioning and work simulation.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Return to Work.

- Absolute and relative contraindications include performance of job activities that pose a medical risk to the recovering worker.

Side Effects and Adverse Events Related to Return to Work.

- Side effects and adverse events include a temporary increase in discomfort or muscle soreness when correcting deconditioning after time away from typical tasks.

Recommendations.

Core Requirements.

Recommendation 27. A job history interview should be completed at the time of the initial evaluation. A thorough job history generally includes:

- duties and demands of job held at time of injury;
- duties and demands of current job, if different;
- stressors; and
- cognitive and social issues, with treatment incorporated into the plan of care.

Recommendation 28. A formal job description for the injured worker is required to identify physical demands at work and to assist in the creation of modified duty.

Recommendation 29. Ergonomic or adaptive equipment, therapeutic breaks, and workplace interventions are recommended to maintain employment (table 9).

Recommendation 30. Nurse case management is recommended in medically complex cases to facilitate communication between the primary provider, referral providers, insurer, employer, and employee. The patient may decline nurse case management services.

Workplace Tests Requirements.

Recommendation 31. A jobsite evaluation is permitted to fully understand the physical demands of a working day; this information can be used when determining medical causation, the need for ergonomic changes, and/or ability to return to work.

Recommendation 32. A work tolerance screening is recommended to determine a patient's tolerance for performing a specific job activity or task from a cardiovascular, postural tolerance, and physical fitness perspective.

Recommendation 33. An FCE may be used to determine an individual's capacity to work. A formal job description and jobsite evaluation, if performed, should be made available to the FCE evaluator prior to having the FCE performed.

Recommendation 34. Follow-up evaluation with the treating therapist and/or the treating physician is required between 1 to 3 days after the FCE to assess the patient's status.

Recommendation 35. FCEs are not recommended as the sole tool for the development of temporary or permanent work restrictions (table 10).

Recommendation 36. Performance during FCEs cannot be used as the sole criteria in diagnosing malingering.

Recommendation 37. Computer-enhanced strength evaluations or performance on FCEs cannot be used alone to determine return to work restrictions.

Return to Work Requirements.

Recommendation 38. It is strongly recommended that the patient return to work in a modified capacity as soon as it is medically appropriate.

Recommendation 39. Interdisciplinary services are permitted to assist the injured worker in return to work efforts (e.g., behavioral and/or psychological support, active therapy).

Recommendation 40. A graduated return to work strategy should be incorporated into a successful medical treatment plan and rehabilitation after injury with a goal of return to full duty, if medically feasible (e.g., nature and frequency of activities, and hours worked).

Recommendation 41. Permanent work restrictions should be developed based on objective information available, including:

- history;
- findings on physical examination and diagnostic testing; and
- functional response to active therapy, work conditioning, and/or modified duty.

Timing of Tests.

Timing of Tests and Maximum Allowed	
FCE	1 time to determine baseline status, and 1 time to determine permanent work restrictions at case closure if it is clear that the patient cannot return to the position held at the time of injury.
Jobsite evaluations	1 time for initial evaluation, 1 for mid-treatment assessment, and 1 at final evaluation.
Computer enhanced evaluations	1 time for initial evaluation, 1 for mid-treatment assessment, and 1 at final evaluation.
Work tolerance screening	1 time for the initial screen. May monitor improvements in strength every 3 to 4 weeks up to a total of 6 visits.

Evidence Tables.

Table 9.

Evidence Table: Return to Work and Spine Pain

Summary:

For workers with musculoskeletal conditions, including low back pain, workplace interventions reduce the recurrence of further injury and reduce time away from work. Such interventions involve the worker, employer, and an occupational physician, and they include ergonomic adjustments and work modification, as needed.

	Evidence statement	Design
Strong evidence	For workers who have been absent from work due to musculoskeletal conditions, the cumulative time away from work in the first 12 months is likely to be reduced if there is a workplace intervention in which several stakeholders, including the workers, the employer, and an occupational physician, formulate a plan to reduce the recurrence of further injury; this generally will involve an ergonomic evaluation of the worksite and plans for work modification as needed.	Systematic review
Good evidence	Workplace interventions in which several stakeholders, including the workers, the employer, and an occupational physician formulate a plan to reduce the recurrence of further injury are also likely to shorten the time needed for a first return to work, and are likely to be beneficial in reducing the risk of a recurrence of sickness absence due to the underlying musculoskeletal condition.	Systematic review
Some evidence	An integrated care program, consisting of workplace interventions and graded activity teaching that pain need not limit activity, is effective in returning patients with chronic low back pain to work, even with minimal reported reduction of pain.	RCT

Table 10.

Evidence Table: FCE

Summary:

FCEs are poor predictors of return to work, and there is no evidence that they predict re-injury. A short form FCE, designed for completion in 1 day, may have similar predictive value to standard FCEs.

Good evidence	Evidence statement	Design
	Performance-based measures, such as those assessed in an FCE, are weakly predictive of return to work variables. Most of the variation in time for return to work is explained by factors other than the physical ability to perform specific tasks.	Systematic review
Some evidence	Evidence statement	Design
	Time off work and gender are important predictors for return to work, and floor-to-waist lifting may also help predict return to work; however, the strength of that relationship has not been determined.	Cohort study
	In chronic low back pain patients, (1) FCE task performance is weakly related to time on disability and time for claim closure and (2) even claimants who fail on numerous physical performance FCE tasks may be able to return to work.	Cohort study
Table 10 continued.		
<i>Evidence Table: FCE</i>		
Some evidence cont.	An FCE fails to predict which injured workers with chronic low back pain will have sustained return to work.	Cohort study
	A short form FCE reduced to a few tests produces a similar predictive quality compared to the longer 2-day version of the FCE regarding length of disability and recurrence of a claim after return to work.	RCT
Lack of evidence	Lack of evidence statement	Design
	There is a lack of evidence supporting the validity of FCE for prediction of re-injury following return to work.	Systematic review

Section 6. Essential First Line Treatment

Introduction. Education about low back pain and associated disability may involve the patient, patient's family, employer, insurer, policymakers, and community. Informed consent and shared decision making are 2 important components of patient education. Informed consent is an individual's autonomous authorization of a medical intervention that has resulted from a conversation between a patient and provider. It is also a formal process that healthcare institutions require before permitting procedures. Shared decision making involves an exchange of ideas between the patient and provider and collaboration in the decision itself.

Recommendations.

Core Requirements.

Recommendation 42. Patient education is required as a primary component of low back pain treatment, beginning with the expected natural history of low back pain, with and without radicular symptoms.

Recommendation 43. In the setting of benign clinical examination, it is strongly recommended that the provider educates and reassures the patient that there is a high likelihood their condition will improve and that it is essential to self-manage their symptoms.

Recommendation 44. Patient education should include a thorough discussion of how behavioral health evaluation and treatment is an essential component to support recovery from physical injury.

Recommendation 45. Shared decision making is required when 2 or more medically reasonable treatment choices exist, regardless of whether the degree of risk is high or low. Elements of shared decision making for spine interventions must include the following:

- the patient's experience with treatment;
- creation of individualized functional goals of treatment and anticipated barriers to success;
- documentation of expected results of diagnostic testing and possible plan of action in response to test results;
- a discussion of the continuum of treatment from the least invasive to the most invasive, with the intent of identifying a treatment along this continuum that most completely addresses the condition;
- expectation regarding the functional impact of the proposed treatment, including a discussion regarding return to work and expected timeframe for treatment;
- specific measurable and clinically meaningful criteria for determining treatment success or failure;
- confirmation of the patient's commitment to perform active therapy to optimize treatment outcomes; and
- documentation and consideration of the patient's unique risks and benefits based on comorbid medical conditions.

Shared decision making discussions should be tailored to the patient's health literacy.

Recommendation 46. Informed consent is required when a high risk treatment is under consideration. It includes the following elements:

- discussion of the proposed treatment's purpose;
- benefits and risks of the proposed treatment, alternative treatments, and nontreatment; and
- explicit patient agreement or refusal.

Evidence Tables. None. The above recommendations were based on consensus.

Section 7. Second Line Treatment

Section 7.a. Core Second Line Treatment

Section 7.a.i. Active Therapies

Introduction. Active therapies are therapeutic exercises and/or activities that intend to restore flexibility, strength, endurance, function, and range of motion (ROM) and to alleviate discomfort. They require intrinsic motivation by the patient to complete a specific exercise or task. The following active therapies are common treatments:

- activities of daily living (ADLs) therapy,
- aquatic therapy,
- back schools,
- functional activities therapy,
- functional electrical stimulation or neuromuscular electrical stimulation (NMES),
- neuromuscular re-education,
- Pilates,
- tai chi,
- therapeutic exercise,
- yoga,
- work conditioning, and
- work simulation.

Active therapies also include developing functional goal-setting, maintaining or returning to usual activities and exercise in a graded fashion, and providing patient education and key recovery messages. Active therapies may coincide with pain neuroscience education. Pain neuroscience education involves educating patients about the biological and physiological processes involved in their pain experience and, importantly, deemphasizing the issues associated with anatomical structures, pain generators, and tissue damage.

Individual patients may benefit from additional education approaches. Education may include, but is not limited to: a favorable prognosis for recovery, the importance of continuing daily activities, promotion of self-efficacy, problem-solving, engagement of support systems, and relaxation techniques.

Contraindications / Side Effects and Adverse Events.

Relative Contraindications to Active Therapies.

- Motor, sensory, or reflex abnormalities are relative contraindications to the use of the McKenzie Method of Mechanical Diagnosis and Therapy.
- Pulmonary or cardiac conditions limiting participation are relative contraindications to active therapy.

Side Effects and Adverse Events Related to Active Therapies.

- Exercise therapy may result in temporary thoracic or lumbar muscle soreness. This side effect, within reasonable limits, should not deter continued active therapy.

Recommendations.

Core Requirements.

Recommendation 47. Active interventions, including therapeutic exercise and/or functional treatment, are recommended in the treatment of low back pain ([table 13](#)).

Recommendation 48. Medical clearance is required prior to participation in active therapies if a patient has any of the following:

- angina/dyspnea on exertion or at rest,
- paroxysmal nocturnal dyspnea and/or orthopnea,
- syncope or presyncope,
- arrhythmia or palpitations, or
- cardiac murmur ([see Contraindications to Active Therapies](#)).

Recommendation 49. Patients in active therapy must:

- demonstrate functional progress that is documented through validated sequential functional assessment measures,
- return to work with decreased restrictions, and/or
- have improvement in clinical measures (e.g., strength, ROM, and ADLs).

If there is no documented evidence of functional progress after 6 treatments, the therapy will be discontinued and the patient must be referred back to their treating provider for further evaluation. Each patient is limited to a maximum of 4 discrete active therapy trials without documented functional progress ([Appendix Functional Screens](#)).

Recommendation 50. A patient is allowed up to 6 active therapy visits to advance their active home exercise program. These visits are contingent on documented demonstration of previously instructed exercises, performance of their home program at the recommended frequency, and progress in their exercise program.

Recommendation 51. Adjunct passive therapy can occur concurrently with active therapy and the frequency of passive therapies will decrease over time. [See the Passive Therapies section](#).

- **Recommendation 52.** Durations of care beyond those listed as “time to produce effect” and “maximum” are warranted in certain circumstances when treatment to date has resulted in measurable and clinically meaningful functional improvement. These circumstances include:
 - re-injury;
 - interrupted continuity of care;
 - after surgery, particularly after multiple surgeries;
 - injuries (e.g., fracture); or
 - comorbidities.

Recommendation 53. Functional electrical stimulation or NMES home units require prior authorization, documenting medical justification for home use.

Time Frames.

Time Frames for Active Therapies			
Time to produce effect	Frequency	Optimum duration	Maximum duration
6 treatments	up to 4 times per week	6 weeks	8 weeks

Evidence Tables.

Table 11.
Evidence Table: Therapeutic Response Considerations in Low Back Pain

Some evidence	Evidence statement	Design
	Stratification of patients with low back pain using the STarT Back questionnaire can identify patients at low risk of poor outcomes and prevent unnecessary referrals for physical therapy which adds little incremental benefit over minimal care. Patients at medium and high risk may also benefit from receiving appropriate referrals for physical therapy.	RCT

Table 12.
Evidence Table: Timing of Physical Therapy Initiation and Low Back Pain

Some evidence	Evidence statement	Design
	In patients with subacute low back pain and symptoms distal to the knee, early physical therapy produces improvements in function and pain similar to those seen with a single education session and follow-up with a primary care physician. A modified version of the ODI was used in this study.	RCT
	In patients with subacute low back pain and symptoms distal to the knee, early physical therapy does not reduce the utilization of advanced imaging, spinal injection, emergency department visits, or spine surgery in the year following initial consultation. A modified version of the ODI was used in this study.	RCT

Table 13.
Evidence Summary: Active Therapies

Active therapies may provide modest functional benefit and symptomatic relief in patients with subacute and chronic low back pain. No evidence demonstrates that 1 form of active therapy is superior to any other, and the selection of an active therapy may be guided by local availability and the patient's preference and capability. While the evidence does not support the use of active therapies in acute low back pain, they are generally accepted interventions for patients at risk for immobility or deconditioning. Patients with barriers to functional recovery may benefit from the incorporation of pain education into therapeutic exercise.

Table 14.
Evidence Table: Exercise and Low Back Pain

Summary:

For patients with subacute and chronic low back pain, exercise-based interventions may reduce pain and disability.

Good evidence	Evidence statement	Design
	For acute or subacute low back pain, no difference was found between exercise therapy and no exercise on pain.	Systematic review
	Exercise alone or as part of a multi-disciplinary program results in decreased disability for workers with non-acute low back pain.	Systematic review
	A web-based 11 minute postural and exercise program that can be used daily by sedentary workers with nonspecific low back pain who work at a computer is likely to lead to clinically important improvements in back pain disability and in the endurance of abdominal and lumbar muscles, while reducing the risk of new episodes of back pain.	RCT

Table 14 continued.
Evidence Table: Exercise and Low Back Pain

Good evidence cont.	Massage therapy in combination with exercise reduces pain and improves function short-term for patients with subacute low back pain.	Systematic review
Some evidence	Evidence statement	Design
	In patients with acute low back pain lasting less than 15 days and with a modified Oswestry Disability Index (ODI) score of 20 or higher and no symptoms distal to the knee, early intervention with physical therapy consisting of manipulation and exercise produces improvements similar to those seen with providing an instructional booklet and follow-up with a primary care physician.	RCT
	In patients with acute low back pain lasting less than 15 days and with a modified ODI score of 20 or higher and no symptoms distal to the knee, early physical therapy does not appreciably reduce the utilization of advanced imaging, spinal injection, emergency department visits, and spine surgery in the year following the onset of acute low back pain.	RCT
	An unsupervised, 12-week, periodized musculoskeletal rehabilitation program of weight training conducted 2, 3, or 4 days a week is effective at improving musculoskeletal strength and quality of life and at reducing pain and disability in untrained persons with chronic low back pain. The 4 days a week training volume is most effective. The volume (total number of reps) of periodized musculoskeletal rehabilitation exercise prescribed is important.	RCT
	Trunk balance exercises combined with flexibility exercises are more effective than a combination of strength and flexibility exercises in reducing disability and improving physical function in patients with chronic low back pain.	RCT
	There is no significant difference in the effectiveness of an 8-week supervised walking program, an evidence-based group exercise class, and usual physiotherapy for improvement in functional disability after 6 months for people	RCT

	with chronic low back pain; all 3 interventions resulted in small, significant improvements in physical function, reduction of pain, quality of life, and fear avoidance over time.	
	In those who have chronic low back pain, 12 weeks of supervised high-dose exercise, spinal manipulative therapy, or low-dose home exercise with advice are all equally effective for reducing pain in the short- and long-term (1 year).	RCT
	There is a modest benefit from adding a back school to other treatments such as non-steroidal anti-inflammatory drugs (NSAIDs), massage, transcutaneous electrical nerve stimulation (TENS), and other physical therapy modalities.	Systematic review
	Disability improved in patients with lumbar spinal stenosis who underwent 6 or 12 sessions (3 and 6 weeks respectively) of cycling as measured by the Roland Morris Questionnaire.	RCT

Table 14 continued.

Evidence Table: Exercise and Low Back Pain

Some evidence cont.	Aquatic exercises can be encouraged as part of an exercise program for patients with low back pain. Aquatic exercise could increase physical function in patients with low back pain. However, there is no evidence that aquatic exercise is superior to other forms of active physical therapy. The participation of the patient is likely to influence the outcome of the exercise program, and it is reasonable to offer aquatic exercise to patients who have a distinct preference for it over other forms of exercise.	Systematic review
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Table 15.

Evidence Table: Motor Control Exercise and Low Back Pain

Summary:

Motor control exercises are more effective than no exercise in reducing pain and improving disability in patients with chronic low back pain.

Strong evidence	Evidence statement	Design
	In the short, intermediate, and long-term, motor control exercises that emphasize the transversus abdominis and multifidus are at least as effective as other forms of exercise and manual therapy. They are possibly more effective than other minimal interventions in reducing pain and improving disability in patients for the treatment of chronic non-specific low back pain.	Systematic review
Good evidence	Evidence statement	Design
	For chronic low back pain, exercise and motor control exercise were associated with greater pain relief than no exercise.	Systematic review

Some evidence	Evidence statement	Design
	Among patients with recurrent non-specific low back pain and associated motor control impairment of at least 6 weeks duration, an intervention consisting of 5 sessions of specific movement control exercise and manual therapy may be as effective as a general exercise regimen and manual therapy in reducing disability at 12 months.	RCT

Table 16.
Evidence Table: The McKenzie Method and Low Back Pain

Summary:
For patients with subacute back pain, the McKenzie method results in similar disability reduction as compared to chiropractic manipulation.

Table 16 continued.
Evidence Table: The McKenzie Method and Low Back Pain

Good evidence	Evidence statement	Design
	Centralization is a favorable prognostic factor in low back pain with and without sciatica.	Systematic review
	A 12-week course of McKenzie therapy is at least as effective as, and may have modestly superior results to, chiropractic manipulation in reducing disability from nonspecific low back pain lasting 6 weeks or more.	RCT
Some evidence	Evidence statement	Design
	Referral of patients for chiropractic care or McKenzie physical therapy in the first weeks of uncomplicated low back pain adds little to the otherwise favorable prognosis for acute low back pain and does incur additional short-term costs of care.	RCT
	The McKenzie approach provides similar outcomes in improving pain, disability, and ability to carry out work activities in comparison with cognitive behavioral therapy (CBT).	RCT
	The McKenzie method is as effective as intensive dynamic strengthening training in reducing short-term back and leg pain intensity in nonspecific low back pain.	RCT

Table 17.
Evidence Table: Yoga and Low Back Pain

Summary:
Yoga results in improved pain and disability for patients with low back pain as compared to education alone or usual care.

Strong evidence	Evidence statement	Design
	Yoga has small to moderate advantages over providing only a booklet in reducing low back pain and back-specific disability, but there is no evidence	Systematic review

	that yoga is superior to stretching and strengthening classes led by a licensed physical therapist.	
Good evidence	Evidence statement	Design
	In the setting of chronic low back pain, an 8-week mindfulness-based stress reduction meditation program with yoga or 8 weeks of CBT resulted in small, significant improvements in physical function and reduction in pain compared to usual care at 26 weeks with no significant differences in outcomes between the 2 treatments.	RCT
Table 17 continued. <i>Evidence Table: Yoga and Low Back Pain</i>		
Good evidence cont.	Compared with education, yoga was associated with lower pain intensity and better function at short-term for chronic low back pain.	Systematic review
	Yoga is comparable to or of small additional benefit when added to usual care for chronic low back pain.	Systematic review
Some evidence	Evidence statement	Design
	Iyengar yoga, which avoids back bending, results in improved function and decreased chronic mechanical low back pain for up to 6 months. Instruction occurred 2 times per week for 24 weeks and was coupled with home exercise. One quarter of the participants dropped out.	RCT
	Yoga emphasizing mobility, strength, and posture to relieve pain may be more effective than usual care for chronic and recurrent low back pain.	RCT

Table 18. <i>Evidence Table: Pilates and Tai Chi in Low Back Pain</i>		
Summary: Pilates is as effective as other forms of exercise in improving disability for patients with chronic low back pain. Tai chi may reduce pain and improve function for patients with chronic low back pain.		
Good evidence	Evidence statement	Design
	Pilates exercise shows statistically and clinically significant reductions in pain and statistically significant improvements in function in the short-term (maximum follow-up 15 weeks) compared with usual care, no exercise, education, or physical activity for the treatment of patients with chronic low back pain.	Systematic review
	Pilates is more effective in reducing pain and improving disability compared with a minimal intervention at intermediate term follow-up, but pilates is equally as effective as other forms of exercise in improving disability at short- or intermediate-term follow-up for the treatment of patients with chronic non-specific low back pain.	Systematic review

Some evidence	Evidence statement	Design
	Tai chi had a small effect on reducing pain and improving function in chronic low back pain as compared to wait-list or no tai chi.	Systematic review

Table 19.
Evidence Table: Active Therapies with Education for Back Pain

Summary:

Active therapies with education lead to greater improvements in function, pain, and return to activity, compared to active therapy alone. The professional delivering such interventions should have training to do so.

Good evidence	Evidence statement	Design
	Patients with at least 6 weeks of non-specific low back pain can achieve sustained reductions in pain and disability over a 12-month period after participating in physiotherapist-led cognitive behavioral interventions. Physiotherapist-led cognitive behavioral interventions included in this review focused on a selection of cognitive behavioral techniques designed to reduce the fear of movement and pain-related disability such as pacing, goal setting, problem-solving, relaxation, and challenging unhelpful thoughts relevant to low back pain. Furthermore, the physiotherapist in the study received 16 to 72 hours of additional training. However, the details of the training program are not well-known, potentially limiting the reproducibility of study findings.	Systematic review
Some evidence	Evidence statement	Design
	Among patients with non-radicular, nonspecific and uncomplicated low back pain (<8 weeks duration), disability observed at 4 weeks and 6 months post-treatment are dependent on an interaction between the type of treatment received and the baseline level of fear-avoidance beliefs. That is, the effectiveness of fear-avoidance-based physical therapy depends on the level of the individual patient's fear-avoidance beliefs regarding back pain. Patients who avoid physical activity because they fear exacerbating their pain are more likely to benefit from physical therapy if that physical therapy includes elements of the accentuated physical therapy described in this study: a graded exercise program that places special emphasis on seeing back pain as a common condition with a favorable prognosis and encourages the patient to play an active role in recovery from the current back pain episode. Patients who do not limit their activities because of these beliefs are probably better served by a standard physical therapy program.	RCT
	A 2-day course focusing on the biopsychosocial model with an emphasis on the goals of returning to usual activities and fitness is as effective in reducing disability as 6 sessions of manual therapy sessions provided by physiotherapists and more limited patient education.	RCT
	A cognitive intervention consisting of 2 consultations lasting 1 hour each with a physical medicine specialist and a physical therapist covering coping strategies and patient education on motion produces short-term reductions in sub-acute back disability.	RCT

Table 19 continued. <i>Evidence Table: Active Therapies with Education for Back Pain</i>		
Some evidence cont.	12 sessions of either graded activity or physiotherapy exercises over 6 weeks were similarly effective in reducing short-term pain and disability, improving quality of life, global perceived effect, return to work, physical activity, physical capacity, and kinesiophobia in patients with chronic nonspecific low back pain.	RCT
	Intensive exercise for approximately 25 hours per week for 4 weeks, combined with cognitive interventions emphasizing the benefits of maintaining usual activity, produces functional results similar to those of posterolateral fusion in patients with chronic non-radicular back pain and no stenosis or instability after 1 year.	RCT
	Among patients with chronic nonspecific low back pain, a 12-week physiotherapist-led cognitive-based program emphasizing specific movement exercises, targeted functional integration activities into daily life, and physical activities suited to the patient's preference was found to be more effective than a combination of manual therapy and exercise in improving disability and pain for up to 12 months. Physiotherapists underwent an extensive training program prior to delivering the intervention, averaging 106 hours that consisted of workshops, patient examinations, pilot study, and clinical manuals.	RCT
	An integrated care program, consisting of workplace interventions and graded activity teaching that pain need not limit activity, is effective in returning patients with chronic low back pain to work, even with minimal reported reduction of pain.	RCT
	Incorporating telephone-delivered cognitive-behavioral strategies into postoperative physical therapy for patients at-risk for poor outcomes after lumbar laminectomy with or without fusion for degenerative conditions produced significant and clinically meaningful improvements in pain, disability, general health, psychosocial risk factors, and physical performance outcomes compared with telephone-delivered education and physical therapy at 3-months post intervention.	RCT

Table 20. <i>Evidence Table: Pain Neuroscience Education</i>		
Summary: For patients with chronic musculoskeletal pain, including low back pain, active therapy with pain neuroscience education reduces pain and increases function as compared to active therapy alone.		
Good evidence	Evidence statement	Design
	Pain neuroscience education combined with a physical intervention is more effective in reducing pain, improving disability, and reducing healthcare utilization compared with either usual care, exercise, other education, or another control group for the treatment of patients with chronic musculoskeletal pain.	Systematic review

Table 20 continued.

Evidence Table: Pain Neuroscience Education

Some evidence	Evidence statement	Design
	Patients with nonspecific, nonradicular, uncomplicated chronic low back pain may or may not benefit from a 6-week program of biweekly aquatic supervised exercise. They are more likely to experience pain relief and functional benefit from an aquatic program if they first receive two 90-minute sessions of pain neurophysiology education dealing with the role of the brain in pain perceptions, how pain originates in the nervous system, how pain goes from being acute to being chronic, central sensitization, cognitive and behavioral responses related to pain, psychosocial factors affecting pain, and how to manage flare-ups of pain.	RCT
	The difference between the pain-education and control group for patients with chronic neck or low back pain reaches clinical relevance for both physical and mental-health domains in 36-item Short Form Health Survey (SF-36) scale and persists at 12 months.	RCT
	Among motivated patients presenting with chronic non-specific low back pain, 2 group-based sessions of pain-neurophysiology education (which focused on the explanation and discussion of neurophysiology of pain) in combination with a multimodal exercise program was found to reduce pain and disability up to 3 months post treatment as compared to those participating in a multimodal exercise program alone.	RCT

Section 7.a.ii. Behavioral and Psychological Interventions

Introduction. Psychological therapeutic and diagnostic interventions have selected use in acute pain problems and more widespread use in sub-acute and chronic pain populations. Psychosocial interventions include psychotherapeutic treatments for mental health conditions, as well behavioral medicine treatments. Therapeutic psychological interventions include, but are not limited to, individual counseling and group therapy. Treatment can occur within an individualized model, a multi-disciplinary model, or a structured pain management program.

These interventions may similarly benefit patients without psychiatric conditions but who may need to make major life changes in order to cope with pain or adjust to disability. Health behavior assessment and intervention services are used to identify and address the psychological, behavioral, emotional, cognitive, and interpersonal factors important to the assessment, treatment, or management of physical health problems.

The following commonly used terminology appears in the context of addressing behavioral and psychological interventions:

- biofeedback training, including:
 - electromyogram,
 - skin temperature,
 - respiration feedback,
 - respiratory sinus arrhythmia,
 - heart rate variability,
 - electrodermal response, and
 - electroencephalograph;
- cognitive behavioral therapy (CBT);
- mindfulness-based stress reduction;

- progressive relaxation;
- relaxation therapy; and
- sleep hygiene training.

CBT is a psychological therapy that integrates theories of cognition and learning with treatment techniques derived from cognitive therapy and behavior therapy. It assumes that cognitive, emotional, and behavioral variables are functionally interrelated. Treatment is aimed at identifying and modifying the patient's unproductive thought processes and behaviors through cognitive restructuring and behavioral techniques to achieve change. Variations of CBT methods can be used to treat a variety of conditions, including chronic pain, depression, anxiety, phobias, and post-traumatic stress disorder. For patients with multiple diagnoses, more than 1 type of CBT might be needed. A "manualized" approach to CBT follows a specific protocol in a manual. See the Chronic Pain Disorder Medical Treatment Guidelines (MTGs) for additional information on psychological evaluation.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Behavioral and Psychological Interventions.

- Contraindications include active suicidality, homicidality, active psychosis, or major untreated psychological comorbidity.
- Relative contraindication includes a lack of patient engagement despite interventions targeting initial resistance.
- Relative contraindications to CBT include literacy below the 6th grade level, lack of English language proficiency, and cognitive impairment.

Side Effects and Adverse Events Related to Behavioral and Psychological Interventions.

- Side effects and adverse events include emotional discomfort, cognitive dissonance, irritability, interpersonal difficulties, or temporary increase in stress.

Recommendations.

Core Requirements.

Recommendation 54. Formal psychological or psychosocial evaluation is recommended as a component of the biopsychosocial approach to patient recovery from physical injury and should attempt to identify both primary psychological risk factors (e.g., psychosis and active suicidality) and secondary psychological risk factors (e.g., moderate depression and job dissatisfaction) (table 21). The evaluation should distinguish between pre-existing, aggravated, and/or purely causative psychological conditions.

Recommendation 55. Comprehensive psychological evaluation must be performed by a psychologist with PhD, PsyD, or EdD credentials or a physician with psychiatric MD/DO credentials.

Recommendation 56. Psychometric testing, distinct from screening psychological questionnaires, must be administered by a psychologist with a PhD, PsyD, or EdD or a health professional working under the supervision of a doctorate level psychologist. A physician with appropriate training may also administer such testing, but interpretation of the tests should be done by a properly credentialed mental health professional.

Recommendation 57. Evaluation for psychiatric medication is permitted if there is an established diagnosis consistent with the standards of the American Psychiatric Association's Diagnostic and Statistical Manual of Mental Disorders (DSM). See the Medications section.

Recommendation 58. An initial assessment, ongoing assessment, and treatment plan with measurable and clinically meaningful behavioral goals, time frames, and specific planned interventions are required before initiating psychological/psychiatric interventions. Ongoing requirements include:

- a biweekly status report during initial, more frequent treatment and monthly thereafter, documenting progress toward functional recovery;
- a discussion of the psychosocial issues affecting the patient's ability to participate in treatment; and
- addressing pertinent issues such as pre-existing, aggravated, and/or causative issues, as well as realistic functional prognosis.

Recommendation 59. The following is recommended when obtaining a psychological evaluation:

- a full release from the patient prior to evaluation acknowledging that information that may go to the employer, and
- exclusion of sensitive health information not directly related to the work-related conditions in reports sent to the insurer is essential.

CBT Requirements.

Recommendation 60. CBT is recommended for low back pain patients who catastrophize, cope ineffectively with pain, or avoid activity out of fear of re-injury (tables 22, 23).

Recommendation 61. A full psychological evaluation is required before CBT can be initiated.

Recommendation 62. A CBT provider must be a:

- psychologist with PhD, PsyD, or EdD credentials;
- psychiatric MD/DO; or
- licensed mental health or licensed health care provider with training in CBT or certified as a CBT therapist with experience in treating chronic pain disorders who works in consultation with a psychologist with a PhD, PsyD, EdD, or psychiatric MD/DO.

Recommendation 63. A manualized approach to CBT is strongly recommended if it is being performed by a non-mental health professional who is supervised by a psychologist with a PhD, PsyD, EdD, or psychiatric MD/DO provider.

Recommendation 64. Candidates for CBT will have all of the following characteristics:

- adequate literacy level to complete homework that is used to teach inductive rational thinking,
- adequate cognitive and education abilities to meet the requirements of a CBT protocol, and
- otherwise stable social circumstances.

The selection of CBT methods should be based on the individual's literacy level, English proficiency, and cognitive capabilities, as assessed by the behavioral health provider. Individuals who are not candidates for CBT may benefit from other behavioral and psychological approaches.

Biofeedback Requirements.

Recommendation 65. Biofeedback as an adjunct to psychological therapy is recommended for patients who meet 1 of the following indications:

- musculoskeletal injury, in which muscle dysfunction or other physiological indicators of excessive prolonged stress response affects and/or delays recovery;

- a need for training to improve self-management of pain, anxiety, panic, anger or emotional distress, opioid withdrawal, insomnia, sleep disturbance, and/or other central and autonomic nervous system imbalances; and/or

motivation to learn and practice biofeedback and self-regulation techniques ([table 24](#)).

Recommendation 66. If biofeedback treatment is indicated, it must be done in conjunction with the patient's other psychosocial or medical interventions.

Recommendation 67. Psychologists or psychiatrists who provide psycho-physiological therapy, which integrates biofeedback with psychotherapy, must be either Biofeedback Certification International Alliance (BCIA) certified or practicing within the scope of their training.

- Non-licensed health care providers of biofeedback for chronic pain patients must be BCIA certified and shall have their biofeedback treatment plan approved by the authorized treating psychologist or psychiatrist.
- Biofeedback can also be provided by health care providers who follow a set treatment and educational protocol. Such treatment may utilize standardized material, relaxation recordings, or app-based interventions. App-based interventions must meet the criteria in Rule 18.

Substance Use Disorder Requirements.

Recommendation 68. Patients with substance use disorder, high-dose opioid use, or use of other drugs of abuse may require inpatient and/or outpatient chemical dependency treatment programs before or in conjunction with other interdisciplinary rehabilitation. [See the Chronic Pain Disorder Medical Treatment Guidelines \(MTGs\)](#).

Time Frames.

Time Frames for Behavioral and Psychological Interventions			
	Time to produce effect (sessions)	Frequency (sessions/week)	Maximum duration
Group CBT	up to 8 (2-hours)	up to 2	16 sessions
Individual CBT	up to 8 (1-hour)	up to 2	16 sessions
Biofeedback	up to 4 sessions	up to 2	12 sessions [▮]
Relaxation	up to 4 sessions	up to 2	12 sessions [▮]
Other psychological interventions	up to 8	up to 2 [*]	6 months [‡]
[▮] Treatment beyond 12 sessions must be documented with respect to need, expectation, and ability to facilitate positive symptomatic and functional gains. [*] For the first 2 weeks of treatment, excluding hospitalization, if required; decreasing to weekly in the 2nd month of treatment and then 2-4 times per month (with the exception of exacerbations, which may require increased frequency of visits). [‡] Longer supervised psychological/psychiatric treatment may be required, especially if there are ongoing medical procedures or complications. If counseling beyond 6 months is indicated, the management of psychosocial risks or functional progress must be documented.			

Evidence Tables.

Table 21. <i>Evidence Table: Psychological Assessments and Outcomes</i>		
Summary: Psychological testing identifies patients most likely to benefit from surgery.		
Good evidence	Evidence statement	Design
	Psychometric testing can predict medical treatment outcomes.	Cohort study
Some evidence	Evidence statement	Design
	Psychological and medical risk factor assessment prior to surgery can identify patients unlikely to benefit from surgery.	Cohort study

Table 22. <i>Evidence Table: CBT</i>		
Summary: CBT results in reduced pain and disability for patients with subacute and chronic low back pain.		
Good evidence	Evidence statement	Design
	CBT may reduce pain and disability in patients with chronic pain, but the magnitude of the benefit is uncertain.	Meta analysis
	CBT, but not behavioral therapy, shows weak to small effects in reducing pain, and small effects on improving disability, mood, and catastrophizing in the treatment of patients with chronic pain.	Meta analysis
	Psychological interventions, especially CBT, are superior to no psychological intervention for chronic low back pain.	Meta analysis
	6 group therapy sessions lasting 90 minutes each focused on CBT skills improved function and alleviated pain in uncomplicated sub-acute and chronic low back pain patients.	RCT
	A stepped care program is more effective than usual care in veterans with chronic musculoskeletal pain. The stepped care program consisted of 12 weeks during which nurse case managers took a medication use history and adjusted medication dosage and scheduling through telephone contacts with patients every other week, followed by a 12-week step in which cognitive behavioral treatment was administered by 45 minute individual sessions by telephone every other week. Disability and pain interference with daily activity with stepped care were both superior to usual care in which patients were given printed handouts and were followed for all care by their primary treating physicians.	Meta analysis

Some evidence	Evidence statement	Design
	A 6-week program of cognitive-behavioral group intervention with or without physical therapy can reduce sick leave, health care utilization, and the risk for developing long-term sick leave disability (> 15 days) in workers with nonspecific low back or neck pain compared with simple verbal instruction by a physician.	RCT
	CBT provided in seven 2-hour small group sessions can reduce the severity of insomnia in chronic pain patients.	RCT
Table 22 continued. <i>Evidence Table: CBT</i>		
Some evidence cont.	Among patients with low back pain who participated in a brief cognitive intervention based on a non-injury model addressing pain and fear avoidance provided over 4 sessions, the addition of either a manualized CBT program or the use of nutritional supplements did not improve on the demonstrated return to work benefit at 12 months post-intervention resulting from the brief cognitive intervention alone.	RCT
	10 sessions of group-based multidisciplinary rehabilitation focusing on task-oriented exercises plus 5 sessions of CBT aimed at managing fear-avoidance beliefs over 5 weeks was significantly more effective than traditional exercises in reducing disability, kinesiophobia, pain, catastrophizing, and enhancing the quality of life in patients with chronic low back pain, and the effects lasted for at least 2 years after the end of the intervention.	RCT
	The McKenzie approach provides similar outcomes in improving pain, disability, and ability to carry out work activities in comparison with CBT.	RCT
	The benefits of CBT continue to be present 3 or more years after the initial intervention, although the differences between CBT and the control group are likely to become narrower with extended follow-up.	RCT
	Behavioral modification, such as patient education and group or individual counseling with CBT, can be effective in reversing the effects of insomnia.	RCT
	Intensive exercise coupled with CBT is as effective as posterolateral fusion for chronic un-operated low back pain.	RCT

Table 23.
Evidence Table: CBT in the Pre- and Post-Operative Setting

Summary:

A pre-operative program of multidisciplinary CBT focused on information about the fusion operation, pain perceptions, coping strategies, and return to activity and work may improve early mobilization and result in more rapid reduction in disability up to 3 months after surgery.

Some evidence	Evidence statement	Design
	In patients scheduled for lumbar spinal fusion surgery, a 4-week preoperative program of multidisciplinary CBT, emphasizing the interactions between cognition and pain perception, coping strategies for dealing with pain, pacing principles, return to work, and details about the fusion operation, does not appear to reduce pain scores in the immediate postoperative period. However, such a CBT program as preparation for surgery does appear to be advantageous in the first 3 days after surgery by leading to earlier mobilization and independent ambulation, even though the intensity of pain scores is not significantly reduced.	RCT

Table 23 continued.
Evidence Table: CBT in the Pre- and Post-Operative Setting

Some evidence cont.		
	CBT also appears to give spinal fusion patients an advantage later in the postoperative surgical period, with a faster decrease in disability scores for at least 3 months following the operation. At 1 year, the difference in disability between CBT and no CBT is no longer significant, but the functional gains from the early postoperative period are maintained.	RCT

Table 24.
Evidence Table: Biofeedback and Relaxation Training

Summary:

Both biofeedback and relaxation therapy reduce pain and disability for patients with chronic low back pain as compared to CBT.

Good evidence	Evidence statement	Design
	Biofeedback or relaxation therapy is equal in effect to CBT for chronic low back pain.	Meta analysis
	Self-regulatory interventions, such as biofeedback and relaxation training, may be equally effective.	Meta analysis
Some evidence	Evidence statement	Design
	Progressive relaxation was the only psychological therapy associated with improvement in function as well as pain intensity compared to wait-list control in chronic low back pain.	Systematic review

Table 25.

Evidence Table: Mindfulness-based Stress Reduction

Summary:

For patients with chronic low back pain, mindfulness-based stress reduction improves function and reduces pain as compared to usual care. A mindfulness-based stress reduction meditation program with yoga results in similar functional improvement and pain reduction as CBT, when each intervention is compared to usual care.

Good evidence	Evidence statement	Design
	As compared to usual care for chronic low back pain, mindfulness-based stress reduction reduced pain by 30% and improved function as well.	Systematic review
Table 25 continued. <i>Evidence Table: Mindfulness-based Stress Reduction</i>		
Good evidence cont.	In the setting of chronic low back pain, an 8-week mindfulness-based stress reduction meditation program with yoga or 8 weeks of CBT resulted in small, significant improvements in physical function and reduction in pain compared to usual care at 26 weeks with no significant differences in outcomes between the 2 treatments.	RCT
Some evidence	Evidence statement	Design
	In the setting of chronic low back pain, 8 weeks of CBT resulted in small, significant improvements in physical function compared to usual care at 2 years post treatment with no significant differences between CBT and a mindfulness-based stress reduction meditation program with yoga. No significant differences in pain bothersomeness were shown between CBT, a mindfulness-based stress reduction meditation program with yoga, and usual care at 2 years post treatment.	RCT

Section 7.b. Adjunct Second Line Treatments, as Indicated

Section 7.b.i. Passive Therapies

Introduction. Passive therapies include treatments that do not require a patient's energy expenditure. They are most helpful early in treatment, and they are directed at controlling symptoms and improving the rate of healing soft tissue injuries. They can be used adjunctively with active therapies to help control swelling, pain, and inflammation. The following passive therapies are common treatments:

- acupuncture,
- acupuncture with electrical stimulation,
- iontophoresis,
- joint mobilization,
- low level laser,
- manipulation,
- manual treatment,
- manual traction,
- massage,
- mechanical or motorized traction,
- phonophoresis,
- soft tissue mobilization,

- short-wave diathermy,
- superficial heat and cold therapy,
- trigger point dry needling,
- ultrasound, and
- unattended electrical stimulation (e.g., transcutaneous electrical nerve stimulation [TENS]).

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Passive Therapies.

- High velocity / low amplitude manipulation is contraindicated in those with joint instability, fractures, severe osteoporosis, infection, metastatic cancer, active inflammatory arthritis, aortic aneurysm, and signs of progressive neurologic deficits.
- Relative contraindications to manipulation include spinal stenosis, spondylosis, and disc herniation.
- Manual traction is contraindicated in those with tumor, infection, fracture, or fracture dislocation.
- Non-oscillating inversion traction methods are contraindicated in patients with glaucoma or hypertension.

Complications of Passive Therapies.

- Rare complications of dry needling or acupuncture include pneumothorax, infection, or syncope.

Side Effects and Adverse Events Related to Passive Therapies.

- Acupuncture may result in treatment-related pain or bruising associated with needle insertion.

Recommendations.

Core Requirements.

Recommendation 69. Patients in passive therapy must demonstrate functional progress through validated functional assessment measures. If there is no evidence of functional progress within the time to produce effect, the therapy shall be discontinued and the patient must be referred back to their treating provider for evaluation. Each patient is limited to a maximum of 4 discrete passive therapy trials.

Recommendation 70. Passive therapies must occur concurrently with self-directed exercise or formal active therapy programs (table 26).

Recommendation 71. The frequency of passive therapy must decrease over time.

Recommendation 72. Durations of care beyond those listed as “time to produce effect” and “maximum” are warranted in certain circumstances when treatment to date has resulted in measurable and clinically meaningful functional improvement. These circumstances include:

- re-injury;
- interrupted continuity of care;
- after surgery, particularly multiple surgeries;
- injuries (e.g., fracture); or
- comorbidities.

Specific goals with objective measures of functional improvement must be cited to justify extended durations of care.

Recommendation 73. Manipulation under general anesthesia and manipulation under joint anesthesia are not recommended.

Recommendation 74. Mechanical or motorized traction is not recommended for low back pain patients with or without radicular symptoms ([table 31](#)).

Recommendation 75. Due to the absence of quality evidence supporting their use, ultrasound, phonophoresis, low level laser, kinesiotaping, and iontophoresis are not recommended for low back pain.

Time Frames.

Time Frames for Passive Therapies			
	Time to produce effect (sessions)	Frequency* (sessions/week)	Maximum duration
Mobilization and Manipulation	up to 6	up to 3 times/week	8 weeks
Massage	1	up to 2 times/week	8 weeks
Acupuncture	up to 6	up to 3 times/week	15 treatments
Heat/cold, short-wave diathermy, unattended electrical stimulation (e.g., TENS**)	up to 4	up to 3 times/week	8 weeks
Trigger point / dry needling	up to 4	up to 2 times/week	8 weeks
Traction (manual)***	up to 3	up to 3 times/week	4 weeks
*See recommendation 71 regarding the expected decreasing frequency over time. **If TENS treatment results in documented functional benefit and is anticipated to extend beyond 4 treatments, consider purchase of a home TENS unit. ***If response is negative after 3 thirty minute treatments, discontinue.			

Evidence Tables.

Table 26.
<i>Evidence Summary: Passive Therapies</i>
Passive therapies may provide functional benefit and symptomatic relief in patients with low back pain, especially when paired with an active intervention. The decision to refer a patient for 1 passive therapy over another should be based on patient preference and relative safety, not on the expectation of a greater treatment effect among passive therapy options.

Table 27.
Evidence Table: Manipulation and Mobilization

Summary:

Manipulation and/or mobilization, particularly when combined with exercise, may reduce pain and improve function for patients with low back pain.

Good evidence	Evidence statement	Design
	2 sessions of thrust manipulation of the thoracolumbar spine followed by an exercise regimen leads to better low back function at 6 months than oscillatory non-thrust manipulation in patients with subacute low back pain. The study found patients with the following characteristics were likely to benefit from the program: segmental hypomobility, no symptoms distal to the knee, low fear avoidance scores, and preservation of at least 35 degrees of internal rotation in at least 1 hip.	RCT
	For acute low back pain, spinal manipulation had greater effects on function than sham manipulation. For acute low back pain, spinal manipulation did not result in greater pain relief than treatments considered inactive (e.g., an educational booklet, bed-rest) at short term (1-week). For radicular low back pain, spinal manipulation in combination with home exercise resulted in greater improvement in leg and back pain at 12 weeks as compared to home exercise and advice alone.	Systematic review
	Spinal manipulative therapy is comparable to exercise, standard medical care, and physiotherapy in reducing chronic low back pain, and spinal manipulative therapy does not provide clinically important, superior pain relief over these interventions.	Meta analysis
	Manipulation does not have a clinically greater therapeutic effect on acute, 6 weeks or less, nonspecific low back pain than other interventions including physical therapy.	Meta analysis
	Among people with acute (≤ 6 weeks) low back pain, spinal manipulative therapy was significantly more effective than usual care, physiotherapy, and exercise for short-term pain and functional improvements. However, effects were small, with only transient minor musculoskeletal harms.	Systematic review

Table 27 continued.
Evidence Table: Manipulation and Mobilization

Some evidence	Evidence statement	Design
	A combination of spinal manipulation and exercise is more effective than exercise alone in reducing pain and improving function of low back pain for 1 year.	RCT
	Referral of patients for chiropractic care or McKenzie physical therapy in the first weeks of uncomplicated low back pain adds little to the otherwise favorable prognosis for acute low back pain and does incur additional short-term costs of care.	RCT

	6 sessions of osteopathic manual treatment, delivered over a period of 8 weeks, is likely to be an effective intervention for nonspecific, nonradicular, non-traumatic low back pain which has lasted more than 3 months.	RCT
	In those who have chronic low back pain, 12 weeks of supervised high-dose exercise, spinal manipulative therapy, or low-dose home exercise with advice are all equally effective for reducing pain in the short- and long-term (1 year).	RCT
	Spinal manipulation/mobilization, followed by active exercises, may be effective for the reduction of disability from nonspecific low back pain lasting more than 12 weeks.	RCT
	12 sessions of spinal manipulation in 6 weeks from a chiropractor yielded the most favorable pain reduction and functional disability improvement compared to a hands-on control in the short-term (12 weeks) for chronic nonspecific low back pain. However, differences between dosage groups did not meet the predetermined minimally clinical significance standards. There was no significant difference between 12 and 18 sessions of spinal manipulation. Also, at 1 year, there was no significant difference between groups.	RCT

Table 28.
Evidence Table: Massage

Summary:

Massage therapy may reduce pain and improve function in patients with subacute and chronic low back pain, particularly when combined with exercise.

Good evidence	Evidence statement	Design
	Massage therapy in combination with exercise reduces pain and improves function short-term for patients with subacute low back pain.	Systematic review
Some evidence	Evidence statement	Design
	In the setting of acute uncomplicated low back pain, treatment with manual pressure at myofascial trigger points up to six 15-minute sessions over 2 weeks reduces back pain more effectively than manual pressure 30 mm away from myofascial trigger points and more effectively than conventional low back massage (effleurage).	RCT
	In patients with chronic low back pain, craniosacral therapy or classic massage are equally effective in reducing disability after 10 weekly sessions of treatment and at 1 additional month of follow-up. However, since both groups received an intervention, the trial does not address the question of the effect of these treatments in comparison to usual care.	RCT
	Among those with chronic low back pain, a 10-week course of relaxation massage is equally effective when compared to structural massage in improving functional disability and reducing pain. Both techniques are more effective than usual care and benefits persist for at least 6 months.	RCT
	There were no clear differences in pain or function between foot reflexology and usual care (or sham foot massage) in chronic low back pain.	Systematic review

Table 29.
Evidence Table: Acupuncture

Summary:

True and sham acupuncture improves function in patients with chronic low back pain as compared to usual care. Acupuncture reduces pain in patients with acute low back pain, as compared to sham acupuncture. It also provides short-term relief in patients with chronic low back pain, as compared to sham acupuncture. Individuals with positive expectations of acupuncture likely experience enhanced treatment benefit.

Good evidence	Evidence statement	Design
	For acute low back pain, acupuncture decreased pain intensity more than sham acupuncture with nonpenetrating needles. Acupuncture was associated with a greater likelihood of improvement at the end of treatment as compared to NSAIDs. For chronic low back pain, acupuncture was associated with lower pain intensity and better function after the intervention as compared to no acupuncture. For chronic low back pain, acupuncture also decreased pain intensity more than sham acupuncture in the immediate time-frame.	Systematic review
	There is a likely, small clinical benefit of acupuncture for acute low back pain, and it may be considered an alternative for some patients.	Systematic review
	Acupuncture is effective in the treatment of low back pain in patients with positive expectations of acupuncture.	RCT
Table 29 continued.		
<i>Evidence Table: Acupuncture</i>		
Good evidence cont.	Acupuncture, true or sham, is superior to usual care for the reduction of disability and pain in patients with chronic nonspecific low back pain, but true and sham acupuncture are likely to be equally effective.	RCT
	Acupuncturists must be properly trained in aseptic technique.	Systematic review
Some evidence	Evidence statement	Design
	Acupuncture is better than no acupuncture for axial chronic low back pain.	RCT
	Among those with nonspecific low back pain, the use of integrated therapy consisting of acupuncture combined with conventional medical care appears to be more effective than conventional medical care alone.	Systematic review

Table 30.

Evidence Table: Laser Acupuncture

Good evidence	Evidence statement	Design
	In the setting of chronic nonspecific low back pain, 8 weekly sessions of high dose laser acupuncture at a dose of 0.8 Joules per acupuncture point are not more effective than a low dose laser at a dose of 0.2 Joules per acupuncture point or a sham laser treatment at a dose of 0 Joules.	RCT

Table 31.

Evidence Table: Traction

Summary:

Traction offers no clinical benefit for patients with radicular or non-radicular low back pain.

Good evidence	Evidence statement	Design
	Mechanical traction is not useful for low back pain patients with sciatica nor those with low back pain without radicular symptoms.	Systematic review
Some evidence	Evidence statement	Design
	Intervertebral lumbar traction does not add significantly to a graded activity program for resolution of non-radicular chronic low back pain.	RCT

Table 32.

Evidence Table: Electrical Therapies

Good evidence	Evidence statement	Design
	Percutaneous electrical nerve stimulation (PENS) produces improvement of pain and function compared to placebo; however, there is no evidence that the effect is prolonged after the initial 3 week treatment episode.	RCT
Some evidence	Evidence statement	Design
	Among patients with chronic nonspecific low back pain, 5 days of anodal transcranial direct current stimulation compared with sham stimulation did not result in pain or disability reduction, alone or when combined with a 4-week cognitive behavioral management program.	RCT

Section 7.b.ii. Durable Medical Equipment

Introduction. Durable medical equipment includes devices that can be used repeatedly to serve a medical purpose. These include braces or splints for supporting, immobilizing, or treating muscles, joint, or skeletal parts that are weak, ineffective, deformed, or injured.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications of Durable Medical Equipment.

- Deconditioning of the lumbar musculature and associated fatigue or pain.

- Skin irritation resulting from contact with the orthotic.

Recommendations.

Foot Orthotics Requirements.

Recommendation 76. Foot orthotics and shoe inserts are appropriate for patients with spinal disorders attributed to aggravated mechanical abnormalities, such as leg length discrepancy, scoliosis, or lower extremity misalignment.

Lumbar Support Devices Requirements.

Recommendation 77. If lumbosacral support devices (e.g., lumbar corsets, braces, sacroiliac [SI] belts, and back belts) are used, the provider must educate the patient about potential for disuse and deconditioning of the lumbar musculature, skin irritation, and general discomfort. Documentation should include directions for duration and frequency of use.

Recommendation 78. Thoracolumbar orthotics are acceptable in the treatment of burst fractures ([table 33](#)).

Evidence Table.

Table 33.

Evidence Table: Orthotics in the Treatment of Burst Fractures

Summary:

For thoracolumbar burst fractures or compression fractures with intact posterior elements and no neurological compromise, nonoperative management may be superior to instrumented fusion. Treatment with or without a brace may lead to similar improvements. Treatment should include appropriate restrictions in the first 4-8 weeks, progressing to active therapies and returning to normal activity.

Table 33 continued.

Evidence Table: Orthotics in the Treatment of Burst Fractures

Some evidence	Evidence statement	Design
	In the setting of stable thoracolumbar burst fractures when there is no neurological compromise of lower extremities, bowel, or bladder, there is a lack of evidence that surgical intervention with multilevel posterior or anterior instrumented stabilization offer any advantages over conservative management with postural reduction and early bracing and mobilization in the first 2 to 4 years of follow-up. However, surgical treatment has a higher rate of subsequent surgery than conservative treatment.	Systematic review
	In the long term of 16 to 20 years, pain and disability are distinctly superior with nonoperative management with a brace or cast than with instrumented fusion.	Systematic review
	For neurologically intact patients who have sustained an isolated thoracolumbar burst fracture grade AO-A3, which results in retropulsion of bone into the spinal canal but spares the posterior longitudinal ligament and other posterior elements of the spine, treatment with and without a thoracolumbosacral orthosis offer equivalent benefits. Treatment without an orthosis involves immediate mobilization as tolerated, supervised by a physical therapist, using restrictions to limit bending and rotating through the trunk, with encouragement to return to normal activities	RCT

	within 8 weeks. Lifting and carrying restrictions of 5 kg and restrictions of 90° hip flexion should be done for the first 8 weeks. Isometric spine stabilization exercises should begin at 4 weeks, progressing to isotonic exercises at 8 weeks, and occupation-specific rehabilitation at 9 weeks.	
	In the setting of acute vertebral compression fractures with intact posterior elements and the absence of neurological deficits, treatment with no brace is not inferior to treatment with a soft brace or treatment with a rigid brace; all 3 treatments lead to similar improvements in disability and pain during the 12 weeks after the onset of the fracture.	RCT

Section 8. Third Line Treatment, as Indicated

Section 8.a. Injections - Diagnostic and Therapeutic

Introduction. Spinal injections are considered when function remains limited despite appropriate non-operative treatment options. Diagnostic spinal injections are used to localize a source of pain when the results would impact the course of treatment. Therapeutic injections are used to temporarily reduce functional impairment. Most individuals with low back pain will not need a spinal injection. See the Overview of Care section.

Contraindications / Complications / Side Effects and Adverse Events.

- See the specific injection section.

Core Requirements.

Recommendation 79. Spinal injections are indicated when all of the following criteria are met:

- positive correlation among clinical findings, the clinical course, and diagnostic tests;
- positive functional response to a diagnostic injection, if required;
- persistent functional impairment despite engagement in 6 weeks of active therapy; and

confounding psychosocial risk factors have been screened for and clinically addressed. See the Behavioral and Psychological Interventions section.

Recommendation 80. All spinal injections should be preceded by magnetic resonance imaging (MRI) or computed tomography (CT) scan.

Recommendation 81. Informed consent is required before all invasive procedures. See the Essential First Line Treatment section.

Recommendation 82. Routine use of sedation is not recommended for diagnostic or therapeutic spinal injections. Minimal to moderate sedation and pain relief may be needed for some patients but are only permitted when there is documentation of extreme circumstances.

Recommendation 83. All injections must be accompanied by self-directed exercise or formal physical therapy.

Recommendation 84. Multi-planar fluoroscopic imaging is required to create permanent images that confirm proper needle placement and contrast flow pattern. CT needle guidance may be required in rare circumstances.

Recommendation 85. Total steroid injections at all sites, including the extremities, should be limited to 4 per year to avoid side effects from steroids. Prior authorization is required for additional injections with appropriate documentation of medical reasoning and functional improvement.

Recommendation 86. Due to the absence of quality evidence supporting their use, the injection of substances other than steroids, anesthetic, or contrast solution is not recommended (e.g., orthobiologics such as platelet rich plasma, stem cells, or amniotic fluid; ozone; etc.).

Evidence Tables. None. The above recommendations were based on consensus.

Section 8.a.i. Epidural Steroid Injection (ESI) and Transforaminal Nerve Root Block

Introduction. Epidural steroid injections (ESIs) are corticosteroid injections into the lumbar epidural space by a caudal, transforaminal, or interlaminar approach. A transforaminal nerve root block involves injecting steroid medication near where the nerve root exits the spinal column. The therapeutic goals of injections are to restore range of motion (ROM) and facilitate progress in active therapy programs by temporarily reducing pain and inflammation in the acute or subacute phases of injury. ESIs may result in small, short-lived reduction in leg pain and disability for individuals with lumbar radiculopathy. ESIs do not result in clinically meaningful long-term improvements in leg pain, back pain, or disability in patients with lumbar radicular pain. ESIs do not have a role in the treatment of non-radicular low back pain.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to ESI.

- Poorly controlled diabetes mellitus, systemic or localized bacterial infection, known anatomic abnormalities, and bleeding diathesis are absolute contraindications to ESI.
- Patients with existing osteoporosis or other risk factors for osteoporosis should rarely receive ESIs.
- Patients on medications that affect coagulation require special consideration prior to ESI.
- Allergy to contrast material, somatization disorder, poorly controlled congestive heart failure, and uncontrolled hypertension are relative contraindications to ESI.

Complications of ESI.

- Long-term morbidity can occur as a result of ESI, although complications that lead to long-term morbidity are rare.
- Permanent paresis, anaphylaxis, and arachnoiditis have been rarely reported with the use of epidural steroids.
- Direct needle trauma is a reported complication (recommendation 89).
- Although major neurological complications are rare with fluoroscopically-guided lumbar ESIs, there is a much greater risk with particulate steroid injections (recommendation 90; table 35).
- Osteoporotic fractures, Cushing's syndrome, hypopituitarism, spinal epidural lipomatosis, and growth hormone deficiency have been associated with ESI.

Side Effects and Adverse Events Related to Injectate.

- Increased blood glucose levels, decreased plasma cortisol levels, and impaired bowel or bladder function are possible side effects of ESI.
- There is a dose-dependent suppression of the hypothalamic-pituitary-adrenal axis, leading to a transient secondary adrenal insufficiency in the first few weeks after ESI (table 34).

Recommendations.

Core Requirements.

Recommendation 87. For a patient to be a candidate for an ESI, the provider must document all of the following:

- subjective report of severe radicular pain that correlates with objective findings;
- positive straight leg raise test, femoral stretch test, and/or reflex, motor, or sensory changes on examination that specifically correlate with imaging findings; and
- imaging findings that demonstrate impingement of nerve(s) or spinal cord.

Recommendation 88. ESI is not recommended for the treatment of axial low back pain in the absence of radicular symptoms ([table 39](#)).

Recommendation 89. Multi-planar fluoroscopic imaging is required to create permanent images confirming proper needle placement and contrast flow pattern.

Recommendation 90. Particulate corticosteroid preparations are not recommended for epidural injection ([table 35](#)).

Recommendation 91. A patient who meets the ESI indications can trial an epidural injection of local anesthetic alone as an alternative to an injection containing steroid ([table 36](#)).

Recommendation 92. All patients with diabetes must be instructed to monitor their glucose levels carefully over the 7 days after steroid injection. Clinicians can pursue diabetic screening tests for those who may be at risk for Type 2 diabetes ([table 37](#)).

Recommendation 93. A diagnostic response to a selective nerve root block must be documented to show its value, including:

- improvement in at least 3 objective functional measures (e.g., spinal ROM; straight leg raise; tolerance and time limits for sitting, standing, walking, and lifting); and
- at least 80% improvement in an accepted pain scale (e.g., visual analog scale [VAS] or numeric rating scale [NRS]) that is consistent with:
 - the expected duration of the injected local anesthetic phase, and
 - a pain diary recording an hourly response for at least 8 hours, but preferably for 1 week, post-injection. Documentation of pain response may be indicated for up to 3 weeks, depending on the nature of the injection.

A successful response to a diagnostic injection requires documentation of positive functional changes by trained personnel and may include nurses, physician assistants, medical assistants, therapists, or non-injectionist physicians. Functional progress supersedes pain improvement.

Acute Phase Requirements (Within 6 Weeks of Initial Care).

Recommendation 94. Early referral within 6 weeks for ESI is permitted when the provider documents all criteria in [recommendation 87](#) and at least 1 of the following:

- radicular functional impairment prevents participation in active therapy and activities of daily living (ADLs), or
- an opioid treatment is being considered ([table 40](#)).

Recommendation 95. A patient who meets the ESI indications in [recommendation 87](#), but wants to avoid ESI, may trial oral gabapentin ([table 38](#)).

Subacute and Chronic Phase Requirements (After 6 Weeks of Care).

Recommendation 96. ESI or referral for surgical consultation are permitted for patients who meet the criteria in [recommendation 87](#) and both of the following:

- functional impairment despite 6 weeks of optimal medical management and participation in active therapy, and
- confounding psychosocial risk factors have been screened for and are being clinically addressed. [See the Behavioral and Psychological Interventions section \(table 41\).](#)

Repeat ESI Requirements.

Recommendation 97. Repeat ESI can occur as soon as 2 weeks after the first injection when the initial injection resulted in a sustained therapeutic response for 2 weeks or more and is characterized by the following:

- improvement in at least 3 objective functional measures (e.g., spinal ROM; tolerance and time limits for sitting, standing, walking, and lifting), and
- 80% radicular pain improvement as measured using an accepted pain scale (e.g., VAS or NRS).

Functional progress supersedes pain improvement. The third injection can then be repeated after a hiatus of 6 months (Appendix Functional Screens). See the Chronic Pain Disorder Medical Treatment Guidelines (MTGs) for information on subsequent injections.

Recommendation 98. Ordering a “series” of ESIs is prohibited. If the first injection fails to result in functional gain and pain relief as outlined in recommendation 97, the injection cannot be repeated at that level with the same approach.

Recommendation 99. If the decision is made to proceed with a different type of injection, the specific criteria for that injection must be satisfied.

Recommendation 100. Testing morning cortisol levels is recommended prior to the third or fourth steroid injection.

Evidence Tables.

Table 34. <i>Evidence Table: ESI and the Hypothalamic-Pituitary-Adrenal Axis</i>		
Some evidence	Evidence statement	Design
	In the setting of ESI given for low back pain with radiculopathy, there is a dose-dependent suppression of the hypothalamic-pituitary-adrenal axis leading to a transient secondary adrenal insufficiency in the first few weeks after the procedure, with an 80 mg dose of methylprednisolone acetate leading to secondary adrenal insufficiency in about 80% of patients the first week after the injection, and in about 50% of patients receiving a 40 mg dose. The cortisol response to an injection of adrenocorticotrophic hormone (ACTH) gradually returns to normal in the weeks after the injection, but secondary adrenal insufficiency remains detectable in some patients 4 weeks later.	RCT

Table 35.

Evidence Table: Risk of Particulate ESI

Summary:

ESIs with particulate steroids, such as methylprednisolone and triamcinolone, have systemic absorption effects and a much higher risk of major neurological complications than ESIs of water soluble steroids, such as betamethasone and dexamethasone.

Strong evidence	Evidence statement	Design
	Although major neurological complications are rare with fluoroscopically-guided cervical and lumbar ESIs, the risk is much greater with particulate steroid injections.	Systematic review
Some evidence	Evidence statement	Design
	Some steroid preparations have a systemic absorption effect, which is reflected in suppression of morning cortisol measurements 3 and 6 weeks after the injection. This appears to be the case with the particulate steroids, methylprednisolone and triamcinolone, but not with the water soluble steroids, betamethasone and dexamethasone.	RCT

Table 36.

Evidence table: Use of Steroid in Epidural Injections

Summary:

For patients with lumbar radicular pain, the addition of a corticosteroid does not provide added improvement in function or pain over epidural injections with local anesthetic alone. While the evidence does not support a benefit of adding steroids to local anesthetic spinal injections, steroids are routinely used in therapeutic spinal injections due to a presumed physiologic effect.

Strong evidence	Evidence statement	Design
	In the setting of low back radicular pain, there are no significant clinical differences between injections of local anesthetic alone and injections of local anesthetic plus steroid.	Systematic review
Good evidence	Evidence statement	Design
	Among patients who achieve clinically important benefit from an epidural injection at 6 weeks, the benefits continue for 12 months after the injection.	RCT
	The addition of steroids to a transforaminal bupivacaine injection has a small, clinically unimportant, and statistically insignificant effect on outcomes of pain and disability in patients with lumbar radicular pain resulting from disc herniation or spinal stenosis.	RCT
	Particulate steroids offer no advantage over nonparticulate steroids in the setting of transforaminal or interlaminar ESIs for patients with lumbar or cervical radiculopathy, either with respect to pain relief or improvement of function.	Systematic review

Some evidence	Evidence statement	Design
	Epidural injections of solutions not containing steroids (such as normal saline or local anesthetic in saline) are likely to have therapeutic effects in the setting of radicular pain, possibly arising from lavage effects and other effects not depending on the steroid component of the injected solution. The interpretation of clinical trials in which epidural steroid and non-steroid injections appear to have similar effects should be influenced by this principle.	Systematic review
	In the setting of lumbar radiculopathy lasting 12 weeks or longer, there is no clinically important difference between a sham subcutaneous injection of 2 ml of normal saline and a 30 ml caudal injection of saline with or without triamcinolone.	RCT
	Bupivacaine alone is equal to bupivacaine plus steroid, with no effect of steroid on Oswestry Disability Index (ODI), leg pain, or back pain.	RCT

Table 37.
Evidence Table: Corticosteroid-Containing ESI in Diabetic Patients

Good evidence	Evidence statement	Design
	In the setting of ESI for patients with adequately controlled diabetes mellitus, significant increases in fasting and postprandial blood glucose can be expected on the day of the procedure and for the first 3 days thereafter; however, the increase in blood glucose are less with a 20 mg than with a 40 mg dose of triamcinolone. There is no difference in pain relief between the 2 steroid doses, making the 20 mg dose preferable to a 40 mg dose.	RCT

Table 38.
Evidence Table: Early Oral Gabapentin for Radicular Pain

Good evidence	Evidence statement	Design
	In the setting of lumbosacral radiculopathy from herniated disc or spinal stenosis, oral gabapentin without physical therapy and an ESI without physical therapy are approximately equally effective in the first 3 months, but no conclusions can be supported that either intervention is superior to placebo.	RCT

Table 39.
Evidence Table: ESI for Axial Low Back Pain

Summary:
ESIs are not beneficial in the treatment of axial low back pain.

Strong evidence	Evidence statement	Design
	ESIs have no short-term or long-term benefit for low back pain.	Systematic review

Lack of evidence	Lack of evidence statement	Design
	A high quality meta-analysis did not provide evidence for any injection therapy (epidural, facet, or local) for non-radicular low back pain.	Meta analysis

Table 40.
Evidence Table: ESI for Radicular Pain

Summary:

ESIs result in a small reduction in leg pain and disability for patients with lumbar radiculopathy, but the benefits are short-lived. ESIs do not result in clinically meaningful long-term improvements in leg pain, back pain, or disability in patients with lumbar radicular pain or radiculopathy. ESI for radicular low back pain may reduce the frequency of surgery in the first year after use.

Strong evidence	Evidence statement	Design
	ESIs have a small average short-term benefit for leg pain and disability for those with sciatica.	Systematic review
	ESIs do not, on average, provide clinically meaningful long-term improvements in leg pain, back pain, or disability in patients with undefined sciatica (lumbar radicular pain or radiculopathy).	Systematic review
Some evidence	Evidence statement	Design
	After 6 weeks of conservative therapy for large herniated discs, an epidural injection may be attempted, as it does not compromise the results of a discectomy at a later date. One half of the patients in this study who were randomized to ESIs did not have surgery and this benefit persisted. Because this study did not have a control group that received neither treatment nor a group that received injections without steroids, one cannot make definite conclusions regarding the efficacy of ESI injections in this setting.	RCT
	The addition of steroids to a transforaminal bupivacaine injection may reduce the frequency of surgery in the first year after treatment in patients with neurologic compression and corresponding imaging findings who also are strong candidates for surgery and have completed 6 weeks of therapy without adequate benefit. The benefits for the non-surgical group persisted for at least 5 years in most patients, regardless of the type of block given.	RCT

Table 40 continued.
Evidence Table: ESI for Radicular Pain

Some evidence cont.	Evidence statement	Design
	The addition of steroids to a transforaminal bupivacaine injection may reduce the frequency of surgery in the first year after treatment.	RCT
	Evidence statement	Design
	In the setting of lumbar transforaminal ESIs, the volume of injectate plays a role in the analgesic response; an injection of 8 ml of solution with 4 mg of dexamethasone provides greater pain relief than a 3 ml injection with 4 mg of dexamethasone. However, there was no group which received an injection without a steroid component, and no inference can be made about a "pure" lavage effect of a transforaminal injection.	RCT

Table 41.
Evidence Table: Referral for Surgical Consultation

Summary:

In patients with disc herniation who are persistently symptomatic at 6 weeks, ESI may prevent the need for future surgery. The delay in surgery does not appear to reduce later surgical success.

Good evidence	Evidence statement	Design
	Among patients with herniated discs who have remained symptomatic after 6 weeks of noninvasive care and who are likely candidates for surgery, surgical discectomy successfully leads to symptom resolution in a large majority of patients. However, patients from the same population who receive ESI after 6 weeks of unsuccessful noninvasive care frequently experience symptom resolution sufficiently well to be able to avoid surgery. The magnitude of the surgery-sparing effect of ESI has not been established, but as many as half of patients may avoid surgery for up to 3 years. Patients who have an unsuccessful ESI after 6 weeks of noninvasive care may undergo discectomy at a later time, with no evidence that the delay in surgery reduces the success of the later operation.	Systematic review

Table 42.
Evidence Table: ESI for Spinal Stenosis

Summary:

For patients with symptomatic lumbar spinal stenosis, the addition of a corticosteroid does not provide added benefit over epidural injections with local anesthetic alone. For patients with lumbar spinal stenosis and neurogenic claudication accompanied by confirmed hypertrophy of the ligamentum flavum, a minimally invasive percutaneous partial resection of the ligamentum flavum leads to greater functional improvement and reduced pain compared with ESI.

Good evidence	Evidence statement	Design
	In patients with neurogenic claudication arising from lumbar central spinal stenosis, there is little difference in pain and functional benefit at 6 weeks and 12 months between an injection of a local anesthetic and an injection of a local anesthetic plus a corticosteroid.	RCT
	There are no significant differences between epidural injections with corticosteroid plus local anesthetic versus local anesthetic alone in patients with symptomatic spinal stenosis.	RCT
	Compared to a fluoroscopically guided interlaminar ESI, a minimally invasive percutaneous partial resection of the ligamentum flavum leads to greater functional improvement and reduced pain at 6 and 12 months in patients with lumbar spinal stenosis and neurogenic claudication accompanied by confirmed hypertrophy of the ligamentum flavum. There is good evidence that these improvements remain stable 24 months following the procedure. However, there is no evidence that the surgical procedure reduces medication use compared to the ESI.	RCT

Some evidence	Evidence statement	Design
	Translaminar steroid injections do not increase walking tolerance in those with spinal stenosis compared to local anesthetic.	RCT
Lack of evidence	Lack of evidence statement	Design
	There is a lack of evidence of a surgery-sparing effect in symptomatic lumbar spinal stenosis.	Systematic review

Table 43.
Evidence Table: Repeat ESI in Central Spinal Stenosis

Some evidence	Evidence statement	Design
	Among patients who repeat an ESI after having received 1 injection, there is very little additional benefit to be achieved from the second or subsequent injections.	RCT

Section 8.a.ii. Zygapophyseal (Facet) Injection

Introduction. Facet joint injection consists of the intra-articular or pericapsular injection of local anesthetic and corticosteroid. It has very limited uses.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Facet Joint Injection.

- Systemic or localized bacterial infection and bleeding diathesis are absolute contraindications to steroid injections. Poorly controlled diabetes is a contraindication if steroids are used.
- Patients with existing osteoporosis or other risk factors for osteoporosis should rarely receive steroid injections. It is unknown if facet joint injections contribute to increased risk of vertebral fractures, but appropriate precautions should be taken for at-risk patients.
- Patients on medications that affect coagulation require special consideration prior to these injections.
- Somatization disorder, poorly controlled congestive heart failure, and uncontrolled hypertension are relative contraindications to these injections. Allergy to contrast material may be a contraindication if alternative contrast material is unavailable.

Complications of Facet Joint Injection.

- Complications include nerve injury, bleeding, infection, contrast allergy, localized pain, or irritation.
- Extravasation of local anesthetic to the lumbar nerves may result in difficulty in ambulation.
- Extravasation to the sacral nerves may result in bowel and bladder impairment.

Side Effects and Adverse Events Related to Facet Joint Injection.

- Diabetic patients may notice a steroid-induced effect on blood glucose levels in the post-injection period.

Recommendations.

Core Requirements.

Recommendation 101. Facet joint injections are of very limited use. One of the following sets of requirements must be met prior to proceeding with a facet joint injection:

either

- at least 3 months of pain, unresponsive to 6 weeks of conservative therapies, including manual therapy; and
- confounding psychosocial risk factors have been screened for and clinically addressed; and
- physical examination findings are consistent with facet origin pain (e.g., pain on extension with lateral bending and referral patterns consistent with the expected pathologic level) that is affecting activity;

or

- the patient has refused a rhizotomy despite facet origin pain on clinical exam;

or

- the patient has facet findings with a thoracic component (tables [44](#), [45](#)).

Recommendation 102. Repeat facet joint injections are only recommended when the first injection results in a sustained therapeutic response for 3 months or more and is characterized by the following:

- improvement in at least 3 physical examination findings consistent with facet origin pain (e.g., pain on extension with lateral bending and referral patterns consistent with the expected pathologic level), and
- at least 80% improvement in an accepted pain scale (e.g., VAS or NRS).

Functional progress supersedes pain improvement.

Recommendation 103. There is no indication for combined facet injection and medial branch block. Intraarticular facet joint steroid injections are not of diagnostic utility in determining appropriateness for radiofrequency (RF) neurotomy.

Recommendation 104. Facet injections cannot be performed at more than 2 levels, unilaterally or bilaterally.

Time Frames.

Time Frames for Facet Joint Injections	
Time to produce effect	Maximum duration
Approximately 30 minutes for local anesthetic; 48 to 72 hours for corticosteroid.	2 injections per level per year when at least 3 months of functional benefit is documented. Prior authorization must be obtained for injections beyond 2 levels.

Evidence Tables.

Table 44. <i>Evidence Table: Diagnosis of Facet-Mediated Pain</i>		
Some evidence	Evidence statement	Design
	Revel's criteria are unsuitable as a clinical screening test to select chronic low back pain patients for initial zygapophyseal joint blocks.	RCT

Table 45. <i>Evidence Table: Effectiveness of Facet Joint Injection</i>		
Summary: There is a lack of evidence for therapeutic benefit of intraarticular facet joint injections in treating low back pain.		
Some evidence	Evidence statement	Design
	In the setting of axial back pain accompanied by paraspinal tenderness, in which the facet joint is suspected to be involved in the etiology of the pain, the therapeutic effects at 1 month of an intra-articular injection of local anesthetic plus a depot steroid are similar to the effects of a medial branch block using a local anesthetic plus a depot steroid, and both interventions are similar to the injection of a saline solution at the anatomic site used for a medial branch block. In summary, there is some evidence that both interventions lack therapeutic benefit at 1 month.	RCT
Lack of evidence	Lack of evidence statement	Design
	A high quality meta-analysis did not provide evidence for any injection therapy (epidural, facet, or local) for non-radicular low back pain.	Meta analysis

Section 8.a.iii. Sacroiliac (SI) Joint Injection

Introduction. Sacroiliac (SI) joint injection is an injection of local anesthetic and/or corticosteroids in an intra-articular fashion into the SI joint under fluoroscopic guidance.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to SI Joint Injection.

- Poorly controlled diabetes mellitus, systemic or localized bacterial infection, and bleeding diathesis are absolute contraindications to steroid injections.
- Patients with existing osteoporosis or other risk factors for osteoporosis should rarely receive steroid injections. It is unknown if SI joint injections contribute to increased risk of fractures, but appropriate precautions should be taken for at-risk patients.
- Patients on medications that affect coagulation require special consideration prior to these injections.
- Somatization disorder, poorly controlled congestive heart failure, and uncontrolled hypertension are relative contraindications to these injections. Allergy to contrast material may be a contraindication if alternative contrast material is unavailable.

Complications of SI Joint Injection.

- Complications include nerve injury, bleeding, infection, localized pain, or irritation.
- Extravasation of local anesthetic to the lumbar nerves may result in difficulty in ambulation.
- Extravasation to the sacral nerves may result in bowel and bladder impairment.

Side Effects and Adverse Events Related to SI Joint Injection.

- Diabetic patients may notice a steroid-induced effect on blood glucose levels in the post-injection period.

Recommendations.

Core Requirements.

Recommendation 105. The following requirements must be met prior to proceeding with a SI joint injection:

- at least 3 months of pain, unresponsive to 6 weeks of conservative therapies, including manual therapy;
- confounding psychosocial risk factors have been screened for and clinically addressed; and
- 3 positive physical examination findings are consistent with SI joint origin pain (i.e., compression test, thigh thrust test, sacral thrust test, Patrick's, Gaenslen test, distraction test, Gillett test) for SI joint pain that is affecting activity.

Recommendation 106. Repeat SI joint injections are only recommended when the first injection resulted in a sustained therapeutic response for 3 months or more and is characterized by the following:

- improvement in at least 3 physical examination findings consistent with SI joint origin pain (e.g., Patrick's sign, Gaenslen, distraction or gapping, or compression test), and
- at least 80% improvement in an accepted pain scale (e.g., VAS or NRS).

Functional progress supersedes pain improvement.

Time Frames.

Time Frames for SI Joint Injections		
Time to produce effect	Frequency	Maximum duration
Approximately 30 minutes for local anesthetic; 48 to 72 hours for corticosteroid.	2 to 3 injections per year.	3 injections per year.

Evidence Table.

Table 46.		
<i>Evidence Table: Imaging Guidance for SI Joint Injection</i>		
Summary: For SI joint injections, there may be minimal difference between the use of fluoroscopic guidance and the use of ultrasound guidance in the effects on function or pain.		
Some evidence	Evidence statement	Design
	SI joint injection with ultrasound is likely to be less accurate than fluoroscopic guidance in terms of intra-articular placement, since some of the injected solution may reach the periarticular structures. However, the significance of this difference in terms of pain relief and functional improvement is not clear.	RCT
	In the setting of SI joint pain in nonobese patients with pain duration of approximately 6 months, intra-articular injection of lidocaine and dexamethasone is similarly effective when guidance of the needle is done with either ultrasound or fluoroscopy, and radiation exposure is absent with ultrasound.	RCT
	In the setting of steroid and local anesthetic injection into the SI joint, the difference in pain relief at 1 month is likely to be no greater than 30% between guidance using ultrasound and guidance using fluoroscopy; however, there is a lack of evidence that either method of injection improves disability associated with SI joint pain. There is a lack of evidence that pain relief persists beyond 1 month.	RCT

Section 8.a.iv. Intradiscal Injection

Introduction. Intradiscal steroid injection is a direct injection of a steroid-containing solution into an intervertebral disc.

Contraindications / Complications / Side Effects and Adverse Events.

Complications of Intradiscal Steroid Injection.

- Complications include infection, pain, bleeding, nerve injury, and allergic reaction to contrast dye.

Recommendations.

Core Requirements.

Recommendation 107. Intradiscal injections are not recommended (table 47).

Evidence Table.

Table 47. <i>Evidence Table: Intradiscal Steroid Injection</i>		
Good evidence	Evidence statement	Design
	Intradiscal steroid injection is unlikely to relieve pain or provide functional benefit in patients with non-radicular back pain and therefore, they are not recommended.	RCT

Section 8.a.v. Medial Branch Block and Radiofrequency (RF) Denervation

Introduction. Medial branch blocks are diagnostic injections used to determine whether a patient is a candidate for radiofrequency (RF) medial branch neurotomy. Medial branch neurotomy, also known as facet rhizotomy, is a procedure designed to denervate the facet joint by ablating the corresponding sensory medial branches. Continuous thermal percutaneous RF is the method frequently used. See the [Chronic Pain Disorder Medical Treatment Guidelines \(MTGs\)](#) for information on the use of repeat RF neurotomy in chronic pain maintenance management.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Medial Branch Block or RF Neurotomy.

- Systemic or localized bacterial infection and bleeding diathesis are absolute contraindications to medial branch blocks and RF neurotomy. Poorly controlled diabetes is a relative contraindication if steroids are used post-neurotomy.
- Allergy to contrast material may be a contraindication if alternative contrast material is unavailable.

Complications of Medial Branch Block or RF Neurotomy.

- Complications of medial branch block or RF neurotomy include bleeding, infection, neural injury, localized neuritis, or rarely, a deafferentation centralized pain syndrome.
- Complications of repeat neurotomy include atrophy of spinal musculature.

Side Effects and Adverse Events Related to Medial Branch Block or RF Neurotomy.

- Side effects and adverse events related to medial branch block or RF neurotomy include localized pain at the injection and/or RF site and post-injection dysesthesia.

Recommendations.

Diagnostics Medial Branch Block Requirements.

Recommendation 108. Diagnostic medial branch blocks are recommended when all of the following are present:

- physical examination findings consistent with facet origin pain (e.g., pain with extension greater than flexion, pain on extension with lateral bending, and referral patterns consistent with the expected pathologic level);
- at least 3 months of pain, unresponsive to 6 to 8 weeks of conservative therapies, including manual therapy; and

- confounding psychosocial risk factors have been screened for and clinically addressed. See the Behavioral and Psychological Interventions section.

Recommendation 109. Diagnostic medial branch blocks are limited to 2 anatomic facet joint levels or 3 medial branch levels. Bilateral controlled blocks are permitted, if performed in a way that preserves diagnostic accuracy.

Recommendation 110. A diagnostic response to medial branch blocks must be documented to show its value, including:

- improvement in at least 3 objective functional measures (e.g., spinal ROM; tolerance and time limits for sitting, standing, walking, and lifting); and
- at least 80% improvement in an accepted pain scale (e.g., VAS or NRS) reported with post-injection provocative testing that is consistent with:
 - the expected duration of the injected local anesthetic phase, and
 - a post-injection pain diary with at least 8 hourly response recordings or until the block has clearly worn off.

A successful response to a diagnostic injection requires documentation of positive functional changes by trained personnel and may include nurses, physician assistants, medical assistants, therapists, or non-injectionist physicians. Functional progress supersedes pain improvement (table 48).

Recommendation 111. If the diagnostic response to the initial block is considered positive, then a separate confirmatory block using a local anesthetic of different duration on a different date must be performed to confirm the level of involvement prior to rhizotomy.

RF Neurotomy Requirements.

Recommendation 112. RF neurotomy is not recommended for patients with non-facetogenic pain generators or involvement of more than 3 levels of medial branch nerves.

Recommendation 113. RF neurotomy is only indicated for those with proven facetogenic pain who have met the criteria for a positive response to 2 controlled medial branch blocks, as defined in recommendation 110. Intra-articular facet joint steroid injections are not considered diagnostic blocks for the purposes of this criteria.

Recommendation 114. Cooled RF denervation, pulsed RF, dorsal nerve root ganglion RF ablation, and transdiscal biacuplasty are not recommended for the lumbar spine.

Repeat RF Neurotomy Requirements.

Recommendation 115. Repeat RF neurotomy may be indicated if the patient experiences sustained (6 months or more), measurable, and clinically meaningful improvement in at least 3 objective functional measures (e.g., spinal ROM; tolerance and time limits for sitting, standing, walking, and lifting) and improvement in an accepted pain scale (e.g., VAS or NRS) after initial RF neurotomy (Appendix Functional Screen). Functional progress supersedes pain improvement. The potential for atrophy of the spinal musculature should be factored into clinical decision making.

Recommendation 116. If the patient's pain presents differently than the initial rhizotomy, a confirmatory medial branch block, as outlined in recommendation 110, will be necessary.

Evidence Table.

Table 48. <i>Evidence Table: Medial Branch Blocks and RF Neurotomy at the Lumbar Spine</i>		
Summary: Patients with substantial (80%) pain relief after controlled and blinded medial branch blocks who proceed to RF neurotomy experience decreased impairment and improved pain relief over 6 months as compared to sham procedures. Patients who experience higher levels of pain relief with medial branch blocks are more likely to benefit from RF neurotomy.		
Good evidence	Evidence statement	Design
	In the lumbar spine, carefully selected patients who had 80% relief with medial branch controlled blinded blocks and then had RF neurotomy had improved pain relief over 6 months and decreased impairment compared to those who had sham procedures. Pain relief was defined as 1 hour of 80% relief from the lidocaine injection and 2 hours of 80% relief with bupivacaine.	RCT
	RF denervation of the lumbar facet joint with a 22 gauge probe in patients who are selected on the basis of a single medial branch block does not decrease low back pain 3 months after the procedure.	RCT
Some evidence	Evidence statement	Design
	RF is more effective than placebo for suspected facet joint pain in the short term, but there is a lack of evidence for its effects on pain and function at other time points.	Meta analysis
	In the setting of axial back pain accompanied by paraspinal tenderness, in which the facet joint is suspected to be involved in the etiology of the pain, the therapeutic effects at 1 month of an intra-articular injection of local anesthetic plus a depot steroid are similar to the effects of a medial branch block using a local anesthetic plus a depot steroid, and both interventions are similar to the injection of a saline solution at the anatomic site used for a medial branch block. There is some evidence that both interventions lack therapeutic benefit at 1 month.	RCT

Section 8.a.vi. Lateral Branch Block and Radiofrequency (RF) Denervation

Introduction. Lateral branch blocks are diagnostic injections used to determine whether a patient is a candidate for radiofrequency (RF) lateral branch neurotomy. Lateral branch neurotomy is a procedure designed to denervate the SI joint by ablating the corresponding sensory lateral branches. See the Chronic Pain Disorder MTGs for information on the use of repeat RF neurotomy in chronic pain maintenance management.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Lateral Branch Blocks or RF Neurotomy.

- Systemic or localized bacterial infection and bleeding diathesis are absolute contraindications to lateral branch blocks and RF neurotomy. Poorly controlled diabetes is a relative contraindication if steroids are used post-neurotomy.

- Allergy to contrast material may be a contraindication if alternative contrast material is unavailable.

Complications of Lateral Branch Blocks or RF Neurotomy.

- Complications of lateral branch block or RF neurotomy include bleeding, infection, neural injury, localized neuritis, or rarely, a deafferentation centralized pain syndrome.
- Complications of repeat neurotomy include atrophy of spinal musculature.

Side Effects and Adverse Events Related to Lateral Branch Blocks and RF Neurotomy.

- Side effects and adverse events related to lateral branch block or RF neurotomy include localized pain at the injection and/or RF site and post-injection dysesthesia.

Recommendations.

Diagnostics Lateral Branch Blocks Requirements.

Recommendation 117. Diagnostic lateral branch blocks are recommended when all of the following are present:

- physical examination findings consistent with SI joint origin pain, including 3 positive physical exam maneuvers (e.g., compression test, thigh thrust test, sacral thrust test, Patrick's, Gaenslen test, distraction test, and Gillett test);
- at least 3 months of pain, unresponsive to 6 to 8 weeks of conservative therapies, including manual therapy; and
- confounding psychosocial risk factors are screened for and clinically addressed. See the Behavioral and Psychological Interventions section.

Recommendation 118. Diagnostic lateral branch blocks should be performed in a manner consistent with the planned RF procedure without anesthetizing the nerve root or allowing intraarticular SI joint flow of medication. Bilateral controlled blocks are permitted, if performed in a way that preserves diagnostic accuracy.

Recommendation 119. A diagnostic response to lateral branch blocks must be documented to show its value, including:

- improvement in at least 3 objective functional measures (e.g., tolerance and time limits for sitting, standing, walking, and lifting); and
- at least 80% improvement in an accepted pain scale (e.g., VAS or NRS) reported with post-injection provocative testing that is consistent with:
 - the expected duration of the injected local anesthetic phase, and
 - a post-injection pain diary with at least 8 hourly response recordings or until the block has clearly worn off.

A successful response to a diagnostic injection requires documentation of positive functional changes by trained personnel and may include nurses, physician assistants, medical assistants, therapists, or non-injectionist physicians. Functional progress supersedes pain improvement (table 49).

Recommendation 120. If the initial block is considered positive, then a separate confirmatory block using a local anesthetic of different duration on a different date must be performed to confirm the level of involvement prior to rhizotomy.

RF Neurotomy Requirements.

Recommendation 121. RF neurotomy is not recommended for patients without sacral nerve branch involvement.

Recommendation 122. RF neurotomy is only indicated for those with proven sacral nerve branch pain who have met the criteria for a positive response to 2 controlled lateral branch blocks, as defined in [recommendation 119](#). Intra-articular SI joint steroid injections are not considered as a diagnostic block for the purposes of this criteria ([table 49](#)).

Recommendation 123. Pulsed RF, dorsal nerve root ganglion RF ablation, and transdiscal biacuplasty are not recommended for the lumbar spine.

Repeat RF Neurotomy Requirements.

Recommendation 124. Repeat RF neurotomy may be indicated if the patient experiences sustained (6 months or more), measurable, and clinically meaningful functional improvement in at least 3 objective functional measures (e.g., tolerance and time limits for sitting, standing, walking, and lifting) and improvement in an accepted pain scale (e.g., VAS or NRS) after initial RF neurotomy ([Appendix Functional Screens](#)).

Recommendation 125. If the patient's pain presents differently than the initial rhizotomy, a confirmatory lateral branch block, as outlined in [recommendation 119](#), will be necessary.

Evidence Table.

Table 49. <i>Evidence Table: RF Neurotomy at the SI Joint</i>		
Summary: Patients with chronic SI joint pain who experience substantial pain relief after local anesthetic lateral branch blocks and proceed to cooled RF neurotomy may have clinical benefit at 3 months post-procedure.		
Some evidence	Evidence statement	Design
	RF neurotomy with a cooled probe may benefit patients with chronic SI joint pain who demonstrate at least a 75% pain response to repeated nerve blocks done under fluoroscopic guidance 3 months after the procedure.	RCT
	RF neurotomy with a cooled probe of the SI joint provides more pain relief than placebo RF at 3 months.	Meta analysis

Section 8.a.vii. Prolotherapy

Introduction. Prolotherapy, also known as sclerotherapy, consists of a series of injections of hypertonic dextrose, with or without glycerine and phenol, into the ligaments of the low back.

Contraindications / Complications / Side Effects and Adverse Events.

Side Effects and Adverse Events Related to SI Joint Prolotherapy.

- Side effects and adverse events include a temporary increase in low back pain and stiffness post-injection.

Recommendations.

Core Requirements.

Recommendation 126. Prolotherapy is not recommended for nonspecific low back pain ([table 50](#)).

Recommendation 127. Prolotherapy for SI joint pain is generally not recommended. For a patient with SI joint pain to be a candidate for prolotherapy, the provider must document all of the following:

- at least 6 months of persistent functional impairment, unresponsive to intensive conservative therapies;
- localization of reported pain at the posterior superior iliac spine;
- 3 positive physical examination findings consistent with SI joint origin pain (e.g., compression test, thigh thrust test, sacral thrust test, Patrick's, Gaenslen test, distraction test, or Gillett test);
- imaging and/or laboratory studies fail to reveal other sources that might better explain the symptoms;
- psychological evaluation and treatment, as appropriate; and
- positive response to initial and confirmatory fluoroscopically-guided SI joint local anesthetic blocks, consisting of:
 - documented improvement in previously impaired SI joint function and provocative physical examination maneuvers within expected time frame of local anesthetic (e.g., spinal ROM; tolerance and time limits for sitting, standing, walking, and lifting; Patrick's sign; Gaenslen; distraction; gapping and compression tests); and
 - 80% improvement in accepted pain scales (e.g., VAS or NRS), consistent with:
 - the expected duration of the injected local anesthetic phase, and
 - a post-injection pain diary with at least 8 hourly response recordings or until the block has clearly worn off.

Functional progress supersedes pain improvement.

Evidence Table.

Table 50.		
Evidence Table: Prolotherapy		
Summary: Prolotherapy is not effective for treating non-specific low back pain, but it may provide a longer duration of pain relief than intra-articular steroid injections for SI joint pain.		
Good evidence	Evidence statement	Design
	Prolotherapy alone is not an effective treatment for chronic low back pain.	Systematic review
Some evidence	Evidence statement	Design
	Prolotherapy of the SI joint is longer lasting, up to 15 months, than intra-articular steroid injections. The study was relatively small and long term blinding was unclear; however, all injections were done under fluoroscopic guidance.	RCT

Section 8.a.viii. Trigger Point Injection

Introduction. Trigger point injections consist of injection of local anesthetic, with or without corticosteroid, into highly localized, extremely sensitive bands of skeletal muscle fibers that produce local and referred pain when activated.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Trigger Point Injection.

- Contraindications include underlying structural issues that more reasonably account for myofascial pain or infection.

Complications of Trigger Point Injection.

- Complications include infection, pneumothorax, anaphylaxis, penetration of the viscera, neurapraxia, neuropathy, syncope, and/or local myopathy (when corticosteroid injected).

Side Effects and Adverse Events Related to Trigger Point Injection.

- Side effects and adverse events include localized pain.
- If a patient experiences severe pain, there is a possibility of intraneural injection and the needle should be immediately repositioned.

Recommendations.

Core Requirements.

Recommendation 128. Trigger point injections should be reserved for patients who meet all of the following criteria:

- consistent, well circumscribed trigger points with a local twitch response;
- characteristic radiation of pain pattern and local autonomic reaction (e.g., persistent hyperemia following palpation); and
- trigger points are not responding to specific, noninvasive, myofascial interventions within a 6-week time frame ([table 51](#)).

Recommendation 129. Concurrent participation in a therapeutic exercise program is required while undergoing myofascial interventions. [See the Active Therapies section.](#)

Recommendation 130. Conscious sedation is not recommended for patients receiving trigger point injections.

Time Frames.

Time Frames for Trigger Point Injection			
Time to produce effect	Frequency	Optimum duration	Maximum duration
Local anesthetic - up to 30 mins No local anesthetic - up to 48 hrs	Weekly*	4 weeks	8 weeks†
*No more than 4 injections per session per week to avoid significant post-injection soreness. †Occasional patients may require 2 to 4 repetitions of trigger point injections series over a 1 to 2 year period.			

Evidence Table.

Table 51. <i>Evidence Table: Trigger Point Injection</i>		
Lack of evidence	Lack of evidence statement	Design
	A high quality meta-analysis did not provide evidence for any injection therapy (epidural, facet, or local) for non-radicular low back pain.	Meta analysis

Section 8.a.ix. Botulinum Toxin Injection

Introduction. Botulinum toxin injections are used to temporarily weaken or paralyze muscles. These injections may reduce muscle pain in conditions associated with spasticity or dystonia.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute Contraindications to Botulinum Toxin Injection.

- Absolute contraindications include hypersensitivity to any botulinum toxin preparation or any of the components in the formulation.
- Infection at the proposed injection site.
- Motor neuropathy or neuromuscular junction disorders.

Complications of Botulinum Toxin Injection.

- Complications of infection at the injection site.

Side Effects and Adverse Events Related to Botulinum Toxin Injection.

- Side effects and adverse events include flu-like symptoms, dry mouth, dysphagia, weakening of the paralumbar musculature, and localized muscular tenderness at the site of injection.
- Neutralizing antibodies develop in at least 4% of patients treated with botulinum toxin type A, rendering it ineffective. Several antigenic types of botulinum toxin have been described. Botulinum type B appears to be effective in patients who have become resistant to the type A toxin. The immune responses to botulinum toxins type A and B are not cross-reactive, allowing type B toxin to be used when type A action is blocked by antibodies.

Recommendations.

Core Requirements.

Recommendation 131. Botulinum injections are not recommended for nonspecific low back pain.

Evidence Table. None. The above recommendation was based on consensus.

Section 8.b. Surgical Interventions

Introduction. Operative treatments are considered when the natural history for surgically treated lesions is better than that of non-operatively treated lesions. Most individuals with low back pain will not need a surgical intervention. See the Overview of Care section.

Contraindications / Complications / Side Effects and Adverse Events.

- See specific surgical procedure section.

Recommendations.

Core Requirements.

Recommendation 132. Early surgical evaluation is required for patients with evidence of any of the following:

- acute fracture or dislocation,
- epidural abscess,
- myelopathy,
- cauda equina syndrome, or
- progressive neurologic deficits (e.g., motor weakness and reflexes) that specifically correlate with spinal cord or nerve root impingement.

Recommendation 133. Referral for surgical evaluation at 6 weeks or greater is indicated when the expected functional outcome of surgery is better than non-operative management and all of the following criteria are met:

- symptomatic and functional improvement has plateaued with unacceptable functional disability;
- greater pain in the legs relative to back that interferes with function, return to work, and/or active therapy;
- physical exam findings of abnormal reflexes, motor weakness, or radicular sensation deficits;
- findings on MRI that indicate impingement of nerves or the spinal cord that specifically correlate with reproducible physical examination findings; and
- diagnostic elimination of confounding psychological or physical conditions that may respond to non-surgical techniques or may be refractory to surgical intervention (see the Behavioral and Psychological Interventions section).

Recommendation 134. Continuation of exercise and/or active therapy while awaiting a surgical evaluation or surgical procedure is recommended to maintain and optimize physical conditioning.

Recommendation 135. Informed consent is required before all invasive procedures. See the Essential First Line Treatment section.

Recommendation 136. Participation in a postoperative active therapy program that includes core stabilization, strengthening, and endurance is strongly recommended. See the Active Therapies section.

Recommendation 137. Diabetes control and related routine lab work, including hemoglobin A1c, are recommended prior to surgery.

Recommendation 138. A multi-modal approach to postoperative pain management is recommended to avoid overuse and misuse of opioid medications. This approach may include any of the following:

- cognitive behavioral therapy (CBT) (see the Behavioral and Psychological Interventions section);

- pain neuroscience education ([see the Active Therapies section](#));
- use of non-opioid medication strategies as adjunct to anesthesia and pre- or postoperative pain management ([see the Medications section](#));
- use of peripheral regional analgesia; and/or
- use of passive therapy for postoperative pain ([see the Passive Therapies section](#)).

Recommendation 139. Intraoperative neuromonitoring is permitted to evaluate spinal cord integrity and screw placement during the operative procedure.

Recommendation 140. Percutaneous RF disc decompression and intradiscal electrothermal therapy are not recommended.

Evidence Table.

Table 52.		
<i>Evidence Table: Factors Influencing Surgical Outcomes</i>		
Strong evidence	Evidence statement	Design
	Surgical interventions for low back pain with radiculopathy produce large and clinically important reductions in pain and disability 3 months after surgery. However, pain and disability are not entirely eliminated in the long term, and mild to moderate pain and disability, approximately on the order of 15 to 20 points on a 100 point scale, can be expected to be present 5 years after surgery for most patients who have a successful surgical procedure for low back pain with radiculopathy.	Systematic review
Good evidence	Evidence statement	Design
	Functional improvement and relief of back pain from most back surgery is similar between patients with a body mass index (BMI) under 25 and overweight or mildly obese patients with a BMI between 25 and 35. Mild obesity does not appear to have an adverse effect on the responsiveness to surgery for these clinical outcomes.	Systematic review

Section 8.b.i. Discectomy

Introduction. Discectomy is a surgical treatment for ruptured or herniated discs of the lumbar spine. There are several procedures in use to decompress extruded disc material around nerve roots, including several minimally invasive techniques. These include:

- lumbar microscopic discectomy,
- percutaneous discectomy,
- microdiscectomy, and
- laser discectomy.

Contraindications / Complications / Side Effects and Adverse Events.

Complications of Lumbar Discectomy.

- Complications of open discectomy include nerve damage, spinal fluid leakage, disc reherniation requiring additional surgery, spinal instability, persistent disc related back pain, infection, and hemorrhage.

- Complications of percutaneous discectomy include nerve or vessel injury, infection, hematoma, incomplete nerve root dissection, disc reherniation requiring additional surgery, spinal instability, and persistent disc related back pain.

Recommendations.

Core Requirements.

Recommendation 141. A psychological screen with a follow-up psychological evaluation, if indicated, is required prior to proceeding with discectomy. If the surgery is being performed in an emergent or urgent situation, the screen is not necessary.

Recommendation 142. For a patient to be a candidate for open or minimally invasive lumbar discectomy, the provider must document all of the following:

- subjective report of severe radicular pain;
- positive straight leg raise or femoral stretch test and/or reflex, motor, or sensory changes on examination consistent with imaging findings; and
- objective findings of neural compression on imaging studies that specifically correlate with the subjective report and examination findings ([table 53](#)).

Recommendation 143. ESIs can be trialed prior to surgery if the patient has a preference to avoid surgery, is unable to participate in therapy after the first 2 weeks post-injury, and meets the criteria for ESI outlined in [recommendation 94](#).

Recommendation 144. The decision to proceed with an open or minimally invasive technique for lumbar discectomy is left to the treating surgeon's discretion, experience, and training in these techniques ([tables 54, 55](#)).

Acute Phase Requirements (Within 6 Weeks of Initial Care).

Recommendation 145. Early referral within 6 weeks for lumbar discectomy is permitted when the provider documents all criteria in recommendations [141](#) and [142](#) and at least 1 of the following:

- radicular pain-related functional impairment prevents participation in active therapy and ADLs;
- presence of progressive functional neurological deficits; or
- an opioid treatment is being considered.

Subacute and Chronic Phase Requirements (After 6 Weeks of Care).

Recommendation 146. Referral for lumbar discectomy is permitted for patients who meet the criteria in [recommendation 142](#) and both of the following:

- functional impairment despite 6 weeks of optimal medical management and participation in active therapy; and
- confounding psychosocial risk factors have been screened for and clinically addressed ([see the Behavioral and Psychological Interventions section](#)).

Recommendation 147. Laser discectomy and percutaneous discectomy are not recommended.

Evidence Tables.

Table 53.

Evidence Table: Effectiveness of Lumbar Discectomy

Summary:

In patients with persistently symptomatic subacute or chronic radiculopathy attributable to disc herniation, discectomy results in greater functional improvement and pain reduction compared to nonoperative management. However, it is reasonable to trial an ESI for patients who prefer to avoid surgery without concern for reducing later surgical success. An ESI may allow patients to avoid surgery.

Good evidence	Evidence statement	Design
	In patients with sciatica arising from a posterolateral disc herniation at L4-5 or at L5-S1 lasting 4 to 12 months, microdiscectomy leads to greater pain reduction and functional improvement than nonoperative treatment when these outcomes are assessed 6 to 12 months after the initiation of treatment.	RCT
Table 53 continued.		
<i>Evidence Table: Effectiveness of Lumbar Discectomy</i>		
Good evidence cont.	Among patients with herniated discs who have remained symptomatic after 6 weeks of noninvasive care and who are likely candidates for surgery, surgical discectomy successfully leads to symptom resolution in a large majority of patients. However, patients from the same population who receive ESI after 6 weeks of unsuccessful noninvasive care frequently experience symptom resolution sufficiently well to be able to avoid surgery. The magnitude of the surgery-sparing effect of ESI has not been established, but as many as half of patients may avoid surgery for up to 3 years. Patients who have an unsuccessful ESI after 6 weeks of noninvasive care may undergo discectomy at a later time, with no evidence that the delay in surgery reduces the success of the later operation.	Systematic review
	In the setting of lumbar microdiscectomy in which a large annular defect 4-6 mm in height and 6-10 mm in width is present, the risk of reherniation and reoperation may be significantly reduced by repair of the disc annulus with a mesh device, which is fixed to the adjacent vertebral body with a titanium anchor. The number needed to treat to prevent a reherniation may be approximately 8 and the number needed to treat to prevent a reoperation may be approximately 13.	RCT
	Open discectomy is likely to benefit patients with imaging-confirmed lumbar disc herniations who are failing to improve with several weeks of conservative treatment.	RCT
Some evidence	Evidence statement	Design
	The advantages of surgical over nonoperative treatment of herniated lumbar discs remain stable over an 8 year period following randomization to either surgery or nonoperative treatment, and the gains observed at 2 years do not deteriorate in the long term.	RCT

Table 54.

Evidence Table: Minimally Invasive Microdiscectomy versus Open Lumbar Discectomy

Summary:

Minimally invasive and open discectomy techniques appear to result in similar clinical outcomes, but there may be lower risk of surgical site infection with minimally invasive techniques.

Some evidence	Evidence statement	Design
	Minimally invasive discectomy is associated with a lower risk of surgical site infection than conventional microdiscectomy and open discectomy.	Systematic review
	Clinical outcomes such as leg pain, back pain, and functional disability are similar between minimally invasive discectomy and conventional microdiscectomy and open discectomy.	Systematic review
Table 54 continued. <i>Evidence Table: Minimally Invasive Microdiscectomy versus Open Lumbar Discectomy</i>		
Some evidence cont.	Minimally invasive discectomy has not been shown to improve function, extremity pain, or axial pain compared to open discectomy in the setting of cervical and lumbar primary discectomy.	Systematic review

Table 55.

Evidence Table: Intraoperative and Perioperative Factors in Discectomy Outcome

Some evidence	Evidence statement	Design
	In the setting of percutaneous lumbar discectomy, an epidural saline solution containing 40 mg of triamcinolone may reduce leg pain and disability during the first postoperative week, leading to earlier hospital discharge. However, there is insufficient information regarding the safety of epidural steroid administration with respect to wound healing and the risk of reherniation.	RCT
	Preoperative administration of dexamethasone reduced acute pain during movement 2 to 24 hours postoperatively from lumbar disc surgery as measured by the VAS.	RCT
	Intraoperative application of epidural corticosteroids such as DepoMedrol reduces postoperative pain for up to 1 month in patients who have undergone lumbar discectomy at either L3/L4 or L4/L5.	RCT
	Following a surgical intervention for first time lumbar disc herniation, patients are likely to achieve functional and symptomatic benefits from a 12 week, 20 session program of a physical therapy based rehabilitation intervention. This program emphasizes supervised exercise activities aimed at the core stabilizers, reduces fear of movement, and also coaches patients on the best ergonomic practices for return to work.	RCT

Table 56.

Evidence Table: Postoperative Factors Influencing Discectomy Outcome

Good evidence	Evidence statement	Design
	Among patients with a preoperative diagnosis of lumbar disk herniation, a program of early comprehensive physiotherapy commenced within 4 weeks after discectomy (or microdiscectomy) did not increase the rate of reherniation, reoperation, and revision surgery as compared to a control treatment.	Systematic review
Some evidence	Evidence statement	Design
	In the setting of elective open single-lumbar discectomy for herniated discs, where morphine is delivered by patient-controlled anesthesia, a low-dose separate infusion of 0.25 mcg/kg/hour of naloxone in the 24 hours after surgery may reduce morphine consumption and reduce postoperative pain, nausea, and pruritus.	RCT
	In the setting of lumbar discectomy for herniated discs, a postoperative rehabilitation program consisting of in-hospital instructions for transfers (bed to chair, etc) and ADLs can lead to successful recovery if supplemented by an instructional booklet with information about core strengthening and mobilization. The addition of an immediate exercise rehabilitation program in the first 6 to 8 weeks following surgery adds very little additional benefit and is not likely to be cost-effective.	RCT

Section 8.b.ii. Decompression for Central or Lateral Spinal Stenosis

Introduction. Multiple procedures described in this section offer surgical access to decompress neural elements by partial or total removal of various parts of the spinal elements. These procedures include:

- laminotomy,
- laminectomy,
- foraminotomy,
- facetectomy, and
- ligamentum flavum resection.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Decompressive Surgery.

- Unaddressed depression, unaddressed concomitant disorder influencing walking capacity (e.g., hip or knee arthrosis, neurologic disease, and cardiopulmonary disease), and unaddressed cardiovascular comorbidity are relative contraindications.
- Scoliosis is a relative contraindication for laminectomy.
- Laminectomy is contraindicated for patients with dynamic spinal instability.

Complications of Decompressive Surgery.

- Complications include graft dislodgement, nerve injury, post-surgical instability, cerebrospinal fluid leakage, hematoma, hemorrhage, infection, spinal cord injury causing paralysis, perineural fibrosis, ROM loss, incomplete decompression, and development of spinal instability requiring fusion surgery.

Recommendations.

Core Requirements.

Recommendation 148. A psychological screen with a follow-up psychological evaluation, if indicated, is required prior to proceeding with decompressive surgery. If the surgery is being performed in an emergent or urgent situation, the screen is not necessary ([table 57](#)).

Recommendation 149. Surgical indications include all of the following:

- radicular symptoms or symptoms of neurogenic claudication, often with clinical evidence of radiculopathy that correlates with the patient's pain and findings;
- evidence of nerve root or spinal compression on imaging studies; and
- failure of non-surgical care, including a trial of 6 weeks of active therapy ([table 58](#)).

Evidence Tables.

Table 57.
Evidence Table: Factors Influencing Surgical Outcomes

Summary:

Patients undergoing spinal surgery who have perioperative depression may have poorer functional outcomes than those without depression.

Some evidence	Evidence statement	Design
	Depression at the time of surgery and in the early recovery period is associated with poorer functional recovery at 2 years, even though it does not appear to be associated with worse pain at 2 years.	Cohort study
	Depression at the time of operation and in the early postoperative period predicts poorer functional recovery from lumbar spinal stenosis surgery.	Cohort study

Table 58.
Evidence Table: Decompressive Surgery Versus Non-Operative Care for Spinal Stenosis

Summary:

Some patients with lumbar spinal stenosis respond favorably to non-operative management, while others eventually progress to surgery. Surgical treatment is more likely to result in functional improvement and pain relief for the first several years compared with non-operative management, but the benefits become equivalent after 5-8 years.

Good evidence	Evidence statement	Design
	Surgical treatment of lumbar spinal stenosis leads to better symptomatic and functional outcomes, but patients with non-surgical treatment may also improve slightly. The non-operative improvement appears to be less likely for stenosis than for herniated discs. In a randomized spinal stenosis trial with crossover, 1/3 of those in the surgery group did not have surgery, and about 40% of those in the non-surgical group eventually had surgery.	RCT

	In the setting of lumbar spinal stenosis with neurogenic claudication, an initial treatment plan for a decompressive laminectomy and an initial treatment plan for nonoperative treatment with active physical therapy, educational counseling, and home exercise are likely to be equally successful, provided that these plans are allowed to change if circumstances change, that is, if surgery is done when nonoperative treatment is not meeting with success, and if plans for surgery are changed if a patient is improving with nonoperative treatment.	RCT
	Although surgery for spinal stenosis generally leads to greater benefits than nonoperative treatment for the first several years of treatment, the results of the treatment options tend to converge over a period of 5 to 8 years.	RCT
Table 58 continued. <i>Evidence Table: Decompressive Surgery Versus Non-Operative Care for Spinal Stenosis</i>		
Good evidence cont.	In patients with a confirmed diagnosis of stenosis who elect for spinal decompression surgery, a physiotherapist-led program consisting of active rehabilitation was found to improve function up to 12 months after the surgery. The programs ranged from 30 to 90 minutes per week or on a biweekly basis, lasting up to 12 weeks.	Systematic review
Some evidence	Evidence statement	Design
	Although decompressive laminectomy for spinal stenosis with neurogenic claudication is beneficial in terms of pain relief and improved function, the proportion of patients who consider themselves to be greatly improved from baseline tends to decline over a period of 8 years.	RCT

Table 59. <i>Evidence Table: Decompressive Surgery</i>		
	Evidence statement	Design
Good evidence	In patients with lumbar spinal stenosis, with or without spondylolisthesis, the 2-year disability outcomes as measured by the ODI do not differ in a clinically important manner between decompression surgery alone and decompression with added fusion surgery.	RCT
	In comparison with facet-sparing laminectomy, surgical procedures that preserve the spinous processes and other posterior elements of the spine are at least as effective as laminectomy in reducing disability and leg pain when used to treat symptomatic lumbar spinal stenosis.	Systematic review
	Compared to a fluoroscopically guided interlaminar ESI, a minimally invasive percutaneous partial resection of the ligamentum flavum leads to greater functional improvement and reduced pain at 6 and 12 months in patients with lumbar spinal stenosis and neurogenic claudication accompanied by confirmed hypertrophy of the ligamentum flavum. There is good evidence that these improvements remain stable 24 months following the procedure. However, there is no evidence that the surgical procedure reduces medication use compared to the ESI.	RCT

Some evidence	Evidence statement	Design
	In patients with spinal stenosis associated with degenerative spondylolisthesis not exceeding 3 mm of motion on flexion-extension films, an operation involving decompressive laminectomy and instrumented posterolateral fusion leads to slightly greater improvements in general health-related quality of life than decompressive laminectomy alone, with no evidence of clinically important differences in back-specific disability as measured by the ODI.	RCT
Table 59 continued. <i>Evidence Table: Decompressive Surgery</i>		
Some evidence cont.	In comparison with facet-sparing laminectomy, surgical procedures that preserve the spinous processes and other posterior elements of the spine may reduce postoperative instability of the operated segment when treating lumbar spinal stenosis.	Systematic review

Section 8.b.iii. Spinal Fusion

Introduction. Spinal fusion is a procedure that unites 2 or more vertebral bodies together to restrict motion and removes a degenerative disc to relieve symptoms of coexistent nerve root compression. The procedure often involves the use of bone grafts, and sometimes instrumentation, to produce a rigid connection between 2 or more adjacent vertebrae. Recombinant human bone morphogenetic protein-2 (rhBMP-2) may also be used in fusion. Discography is rarely used as an invasive diagnostic procedure for fusion candidates.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Lumbar Fusion and Discography.

- Morbid obesity is a relative contraindication to fusion ([table 60](#)).
- Absolute contraindications to the use of rhBMP-2 include known sensitivities to rhBMP-2, bovine type 1 collagen, or other components of the formulation.
- Contraindications to discography include infection, bleeding diathesis, significant spinal stenosis, clinical myelopathy, known allergic reaction to injectate, effacement of the cord, thecal sac, or circumferential absence of epidural fat.

Complications of Lumbar Fusion and Discography.

- Potential complications include instrumentation failure, pseudoarthrosis, superficial or deep wound infection, iliac crest bone graft donor site pain graft extrusion, acceleration of adjacent segment disease, and nerve damage.
- Potential complications associated with the use of rhBMP-2 include:
 - development of sensitization to rhBMP-2 or to absorbable collagen sponge, and
 - neurovascular encroachment due to swelling and ectopic bone formation associated with the use of rhBMP-2 outside of the anterior cage.
- Potential complications of lumbar discography include discitis, nerve damage, chemical meningitis, pain exacerbation, disc damage, and anaphylaxis ([table 62](#)).

Recommendations.

Core Requirements.

Recommendation 150. Lumbar fusion is not recommended in the first 6 months of symptoms, except for fracture, dislocation, or for some patients with functional loss due to stenosis and instability.

Recommendation 151. Psychological evaluation is required to assess suitability for a lumbar fusion. Documentation should include the following items with associated treatment recommendations:

- psychological factors that might influence elective surgical treatment outcomes, or
- psychological factors that might complicate surgical recovery.

Confounding depression or anxiety must be addressed prior to proceeding with surgery. Presurgical psychological evaluation should not be done by a psychologist employed by the physician performing the procedure. See the Behavioral and Psychological Interventions section (table 61).

Recommendation 152. The following are required prior to proceeding with spinal fusion:

- all pain generators are adequately defined and treated;
- all physical medicine and manual therapy interventions are completed;
- imaging studies demonstrate spinal stenosis with instability or disc pathology, requiring decompression;
- spine pathology is limited to 2 levels; and
- psychological evaluation, as discussed in recommendation 151 (tables 64, 65, 66, 67).

Recommendation 153. Diagnostic indications for fusion include the following:

- neural arch defect with associated stenosis or instability;
- spondylolytic spondylolisthesis;
- degenerative spondylolisthesis 4 mm or greater;
- surgically induced segmental instability;
- symptomatic spinal stenosis in the presence of spondylolisthesis (>2 mm); or
- primary mechanical low back pain / functional spinal unit failure with objective evidence of 2 or more of the following:
 - internal disc disruption (poor success rate if more than 1 disc involved),
 - painful motion segment,
 - disc resorption,
 - facet syndrome, and/or
 - ligamentous tear.

Recommendation 154. Discography is not generally recommended, but it is permitted when confirmatory information is necessary prior to proceeding with lumbar fusion (table 62).

Recommendation 155. Tobacco cessation for at least 6 weeks prior to fusion is required, unless the surgeon documents reasoning that the benefits of proceeding with surgery outweigh the risks.

Recommendation 156. The choice of bone graft donor site, grafting substance, operative technique, and surgical approach are at the discretion of the treating surgeon (tables 68, 69).

Recommendation 157. Strategic perioperative medication use is recommended for pain management to minimize opioid use and optimize surgical outcome (table 70).

Recommendation 158. Postoperative participation in an active therapy program that includes core stabilization, strengthening, and endurance is recommended. See the Active Therapies section (table 71).

Evidence Tables.

Table 60. <i>Evidence Table: Complications Related to Lumbar Fusion</i>		
Some evidence	Evidence statement	Design
	Morbid obesity increases hospital length of stay, mortality, and postoperative complications after spinal fusion surgery, with concomitant increases in hospital costs.	Cross sectional study

Table 61. <i>Evidence Table: Factors Influencing Surgical Outcomes</i>		
Summary: Depression at the time of surgery predicts a worse functional outcome.		
Some evidence	Evidence statement	Design
	Depression at the time of surgery and in the early recovery period is associated with poorer functional recovery at 2 years, even though it does not appear to be associated with worse pain at 2 years.	Cohort study
Table 61 continued. <i>Evidence Table: Factors Influencing Surgical Outcomes</i>		
Some evidence cont.	Depression at the time of operation and in the early postoperative period predicts poorer functional recovery from lumbar spinal stenosis surgery.	Cohort study

Table 62. <i>Evidence Table: Discography</i>		
Summary: Radiographically identified spondylolisthesis more accurately identifies patients who are likely to benefit from fusion than a positive discogram. Discography may result in an increased risk of later disc herniation.		
Good evidence	Evidence statement	Design
	A positive discogram does not identify patients who benefit from fusion, in contrast to radiographically identified spondylolisthesis, which does identify patients likely to benefit. Only 27% of “discogenic” pain patients had successful outcomes of fusion, even when predictors of a poor outcome were excluded from surgery; 72% of spondylolisthesis patients had success with the same surgery.	Cohort study
Some evidence	Evidence statement	Design
	Provocative discography, facet joint blocks, and temporary external transpedicular fixation do not adequately screen patients with nonspecific low	Systematic review

	back pain for fusion success. The tests tend to be sensitive but not specific.	
	Discography with a small-bore needle increases the risk of later disc herniation at the level of the injected disc, and this risk should be taken into account when deciding on referral for discography.	Cohort study

Table 63.
Evidence Table: Factors Predicting Response to Fusion

Good evidence	Evidence statement	Design
	Patients undergoing rehabilitation comprised of exercise and CBT have lower levels of self-reported disability and reduced fear avoidance behavior than patients receiving usual care for up to 12 months following lumbar fusion surgery.	Systematic review

Table 64.
Evidence Table: Nonoperative Treatment Versus Lumbar Fusion

Summary:

For patients with severe symptoms due to lumbar stenosis and spondylolisthesis, decompression and fusion provide greater benefit than nonoperative treatment. For chronic non-radicular back pain without stenosis or instability, intensive exercise combined with cognitive interventions may be as effective as posterolateral fusion.

Good evidence	Evidence statement	Design
	Decompression and fusion, with or without instrumentation, of lumbar stenosis with degenerative spondylolisthesis leads to better 2 year outcomes for patients whose symptoms are severe. However, patients who choose non-operative treatment can also expect their symptoms to improve with nonsurgical treatment, and nonoperative treatment is acceptable if this is the patient preference.	RCT
Some evidence	Evidence statement	Design
	Intensive exercise for approximately 25 hours per week for 4 weeks, combined with cognitive interventions emphasizing the benefits of maintaining usual activity, produces functional results similar to those of posterolateral fusion in patients with chronic non-radicular back pain and no stenosis or instability after 1 year.	RCT
	Intensive exercise coupled with CBT is as effective as posterolateral fusion for chronic un-operated low back pain.	RCT
	Fusion is likely to have a higher beneficial effect compared to multidisciplinary rehabilitation for patients with isthmic spondylolisthesis, as differentiated from those without the condition who suffered from chronic low back pain.	Systematic review
	In patients scheduled for lumbar spinal fusion surgery, a 4-week preoperative program of multidisciplinary CBT, emphasizing the interactions between cognition and pain perception, coping strategies for dealing with pain, pacing principles, return to work, and details about the fusion operation, does not	RCT

	appear to reduce pain scores in the immediate postoperative period. However, such a CBT program as preparation for surgery does appear to be advantageous in the first 3 days after surgery by leading to earlier mobilization and independent ambulation, even though the intensity of pain scores is not significantly reduced.	
	Lumbar fusion produces better symptomatic and functional results in patients with chronic nonradicular pain when several months of conservative treatment have not produced a satisfactory outcome.	Systematic review

Table 65.

Evidence Table: Lumbar Fusion With or Without Decompression

Summary:

In patients with degenerative spondylolisthesis and minimal instability, the addition of fusion to decompression surgery:

- may improve general health-related quality of life;
- provides no added benefit for function, pain, rates of complication or rates of reoperation; and
- increases operative blood loss, operative time and length of stay.

In patients with lumbar spinal stenosis, the addition of fusion to decompression surgery provides no added benefit for function or pain.

Strong evidence	Evidence statement	Design
	In patients with degenerative spondylolisthesis and minimal instability (motion of ≤ 3 mm on flexion-extension films), decompression alone is as beneficial as decompression plus fusion for outcomes such as disability, leg and back pain, and rates of complication and reoperation.	Systematic review
Good evidence	Evidence statement	Design
	In patients with degenerative spondylolisthesis and minimal instability (motion of ≤ 3 mm on flexion-extension films), decompression alone is associated with less operative blood loss, shorter operative time, and shorter length of stay than decompression plus fusion.	Systematic review
	In patients with lumbar spinal stenosis, with or without spondylolisthesis, the 2-year disability outcomes as measured by the ODI do not differ in a clinically important manner between decompression surgery alone and decompression with added fusion surgery.	RCT
	In the setting of symptomatic lumbar spinal stenosis, the addition of a fusion procedure to a decompression procedure adds little additional benefit with respect to pain and disability, both in the short term and in the long term.	Systematic review
Some evidence	Evidence statement	Design
	In patients with spinal stenosis associated with degenerative spondylolisthesis not exceeding 3 mm of motion on flexion-extension films, an operation involving decompressive laminectomy and instrumented posterolateral fusion leads to slightly greater improvements in general health-related quality of life than decompressive laminectomy alone, with no evidence of clinically important differences in back-specific disability as measured by the ODI.	RCT

Table 66.
Evidence Table: Lumbar Fusion Versus Other Surgical Options

Summary:

Fusion surgery, the use of interspinous spacers, and total disc replacement (TDR) result in similar improvements in function and pain. For patients with painful degenerative disc disease, but without facet arthritis, spondylolysis, spondylolisthesis, osteoporosis, spinal stenosis, or deformity, a TDR leads to greater functional improvement and pain relief than fusion. In lumbar fusion, the use of a single cage or 2 cages results in equal rates of radiographic fusion. However, a single cage results in less intraoperative blood loss, shorter operating time, and fewer complications.

Strong evidence	Evidence statement	Design
	Disc replacement is not inferior to fusion at 24 months for relief of back pain, reduction of disability, and provision of patient satisfaction.	Meta analysis
	In the setting of lumbar interbody fusion, the use of a single cage achieves equal radiographic fusion as the use of 2 cages, with less intraoperative blood loss, shorter operating time, and fewer complications.	Systematic review
Good evidence	Evidence statement	Design
	Reoperation rates are higher with the use of interspinous spacers than with laminectomy or laminotomy in the setting of symptomatic lumbar spinal stenosis, but reoperation rates are not different between spacers and fusion operations.	Systematic review
	In the setting of symptomatic degenerative lumbar disease from a variety of causes, motion-preserving devices, such as interspinous spacers or TDRs, can be expected to lead to pain relief and disability improvement comparable to the improvements seen with fusion surgery.	Systematic review
	TDR is more likely than fusion to lead to long term complete pain relief in patients who have painful degenerative disc disease, no facet arthritis, spondylolysis or spondylolisthesis, osteoporosis, spinal stenosis, or deformity. TDR also leads on average to greater improvement in the ODI than fusion over 5 years. Both fusion and TDR lead to lasting symptomatic and functional benefits over 5 years in carefully selected patients.	RCT
	An artificial disc is non-inferior to allograft fusion with a cage for single level disease.	Systematic review
Some evidence	Evidence statement	Design
	Pain and disability outcomes at 2 years are similar for 2-level lumbar degenerative disc disease between hybrid operations involving stand-alone anterior lumbar interbody fusion (ALIF) at L4-5 plus disc replacement at L5-S1, compared to non-hybrid operations involving interbody fusion at both levels or TDR at both levels.	Meta analysis

Table 66 continued.

Evidence Table: Lumbar Fusion Versus Other Surgical Options

Some evidence cont.	In the setting of stable spinal stenosis with Grade I spondylolisthesis, decompression combined with a Coflex interlaminar stabilization device achieves similar clinical benefits as decompression combined with instrumented posterolateral fusion.	RCT
	In patients with symptomatic spinal stenosis at L4-5, posterior stabilization with Dynesys leads to greater motion preservation at the operated segment, and a reduced amount of excess movement at adjacent levels, than does posterior lumbar interbody fusion (PLIF) at 3 years postoperatively.	RCT
	Dynesys and PLIF lead to comparable improvements in leg pain, back pain, and ODI scores at 3 years.	RCT
	Dynesys requires shorter operative time, shorter hospital stays, and less blood loss than PLIF.	RCT
	A 2-level lumbar disc replacement is not inferior to circumferential fusion in patients with 2 level degenerative disc disease and less than grade I spondylolisthesis 24 months after surgery.	RCT
	A trial comparing TDR with circumferential fusion found superior function with lumbar disc replacement compared to fusion. However, the use of narcotics remained high even 2 years after a successful operation.	RCT

Table 67.

Evidence Table: Lumbar Fusion and Adjacent Segment Degeneration

Summary:

Lumbar surgery with motion-preserving devices reduces the incidence of adjacent level disease and reoperation within 2-6 years.

Good evidence	Evidence statement	Design
	Over a period of 2 to 6 years, surgery with motion-preserving devices in the lumbar spine is likely to lead to a lower incidence of adjacent segment disease and to require fewer reoperations than spinal fusion.	Systematic review
Some evidence	Evidence statement	Design
	In the setting of lumbar fusion, the risk of reoperation for adjacent segment disease is likely to be increased when there is a diagnosis of degenerative scoliosis and when the fusion extends from L4 to S1.	Review
	TDR of the lumbar spine reduces the 5-year risk of adjacent level degeneration compared to circumferential fusion.	RCT

Table 68. Evidence Table: Lumbar Fusion Graft Site and Other Issues		
Strong evidence	Evidence statement	Design
	In the setting of lumbar spinal fusion, the use of bone morphogenetic protein (BMP) is associated with lower rates of reoperation over the following 2 years as compared to iliac crest bone graft.	Systematic review
Good evidence	Evidence statement	Design
	rhBMP-2 has a greater influence on promoting radiographic fusion in smokers than in nonsmokers when compared to iliac crest bone graft.	Meta analysis
	rhBMP-2 promotes fusion better than iliac crest bone graft in normal weight and overweight patients, but rhBMP-2 may not influence fusion rates in obese and severely obese patients.	Meta analysis
Some evidence	Evidence statement	Design
	When instrumented single-level fusion is done for lumbar spondylolisthesis, local bone graft requires less operative time, yields equal rates of fusion and clinical improvement, and avoids persistent local symptoms at the iliac crest graft donor site.	RCT

Table 69. Evidence Table: Lumbar Fusion - Other Technical Issues		
Good evidence	Evidence statement	Design
	In the setting of spine fusion surgery, the use of antifibrinolytics such as tranexamic acid reduces intraoperative blood loss and total blood loss, leading to lower transfusion requirements, while not being accompanied with significant adverse effects compared to no antifibrinolytic use.	Systematic review
	In the setting of transforaminal lumbar interbody fusion (TLIF), fusion rates are slightly lower with unilateral than with bilateral pedicle screw fixation.	Systematic review
	In the setting of TLIF, cage migration occurs more commonly with unilateral than with bilateral pedicle screw fixation, and device-related complication rates also are likely to be higher.	Systematic review
	In the setting of TLIF, there is less blood loss with unilateral than with bilateral pedicle screw fixation, and operative times are also shorter.	Systematic review
	In the setting of TLIF, pain and functional outcomes are similar between unilateral and bilateral pedicle screw fixation; it is possible that the lower fusion rates and higher rates of cage migration with unilateral fixation are offset by less soft tissue dissection, shorter operative times, and less blood loss with unilateral fixation.	Systematic review

Table 69 continued.

Evidence Table: Lumbar Fusion - Other Technical Issues

Some evidence	Evidence statement	Design
	In the setting of posterior spinal fusion using posterior iliac crest bone graft, the application of a sponge soaked in 0.25% bupivacaine at the time of bone harvest may reduce donor site pain for the first several months after surgery, compared to a sponge soaked in saline.	RCT
	In patients undergoing posterior lumbar surgery for stenosis or spondylolisthesis, a loading dose of 30mg/kg tranexamic acid over 15 minutes and a maintenance infusion of 2 mg/kg/hour, was found to reduce perioperative blood loss as compared to a placebo.	RCT
	In the setting of instrumented posterolateral fusion of the lumbar spine for degenerative spondylolisthesis, the addition of autologous bone marrow concentrate to allograft can be expected to significantly increase the rate of radiographic fusion 2 years after the operation is done.	RCT

Table 70.

Evidence Table: Perioperative Medication Use in Fusion

Summary:

For patients undergoing spinal fusion surgery, preoperative analgesia may reduce postoperative pain and opioid use. Postoperative non-steroidal anti-inflammatory drugs (NSAIDs) may increase the risk of nonunion at higher doses.

Good evidence	Evidence statement	Design
	In the setting of lumbar spinal fusion, NSAIDs at an equivalence of no more than 120 mg of ketorolac for 14 days postoperatively do not appear to increase the risk of nonunion of the operated spinal segment.	Systematic review
Some evidence	Evidence statement	Design
	In the setting of PLIF surgery, a single saline infusion of 800 mg of ibuprofen one half hour prior to surgery may lead to less pain and morphine use in the immediate postoperative period and may thereby be an option for preoperative care of the surgical patient.	RCT
	In the setting of 1-level or 2-level spinal fusion surgery, the administration of pre-emptive analgesia 8 hours preoperatively with acetaminophen, ketorolac, and pregabalin may confer benefits in the 48 hours following the procedure in the form of less pain, better ambulation, and less morphine use.	RCT

Table 70 continued.

Evidence Table: Perioperative Medication Use in Fusion

Some evidence cont.	Among patients undergoing posterior lumbar spinal fusion, the addition of preemptive analgesia using a single intravenous dose of ketorolac (30 mg) or parecoxib (40 mg) resulted in better immediate postoperative pain control as compared to a control (saline) group. However, the difference among the groups did not persist in measures from one-hour post-operatively up to 24 hours. There was no apparent opioid-sparing effect with preemptive analgesia.	RCT
	The preoperative administration of gabapentin in doses as low as 300 mg may reduce postoperative pain and opioid consumption in the setting of spinal laminectomy and fusion. The incidence of many postoperative complications such as nausea, headache, and dizziness is probably similar for gabapentin and placebo. Gabapentin may increase somnolence but may decrease the incidence of urinary retention pruritus.	Systematic review
	In the setting of lumbar fusion without rhBMP-2, postoperative NSAIDs with an equivalent dose of more than 300 mg of diclofenac are associated with an increased risk of nonunion of the operated spinal segment. There appears to be a dose-dependent effect of postoperative NSAIDs on the risk of nonunion.	Systematic review

Table 71.

Evidence Table: Post-operative Rehabilitation

Some evidence	Evidence statement	Design
	It is appropriate to defer active rehabilitation for 12 weeks after instrumented fusion, as the group beginning 12 weeks postoperatively had greater improvements in disability compared to the 6 week group.	RCT

Section 8.b.iv. Lumbar Total Disc Replacement (TDR)

Introduction. Lumbar total disc replacement (TDR) is a surgical procedure where a degenerated disc is replaced with a prosthetic device. The endplates are positioned under intraoperative fluoroscopic guidance for optimal placement in the sagittal and frontal planes. The prosthetic device physiologically distributes the mechanical load of the vertebrae and maintains ROM.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Lumbar TDR.

- Absolute and relative contraindications include:
 - significant spinal deformity and scoliosis;
 - symptomatic facet joint arthrosis;
 - spinal instability at the pathologic or adjacent level requiring fusion;
 - deficient posterior elements;
 - infection;
 - any contraindication to an anterior abdominal approach, including multiple prior abdominal procedures;
 - previous compression or burst fracture;
 - spinal canal stenosis;

- spondylolysis;
- spondylolisthesis > 3 mm;
- osteopenia, osteoporosis, or any metabolic bone disease;
- chronic steroid or other medication use that interferes with bone or soft tissue healing;
- allergy to device materials;
- pregnancy or desire to become pregnant;
- extreme obesity (e.g., BMI > 40 kg/m² or over 100 pounds overweight);
- active malignancy; and
- generalized chronic pain.

Complications of Lumbar TDR.

- Complications include nerve and vascular injury, dural tears, retrograde ejaculation, malpositioning/suboptimal positioning of prosthesis, implant failure, complex regional pain syndrome, complications from abdominal surgery (e.g., hernia or adhesions), and need for reoperation.

Recommendations.

Core Requirements.

Recommendation 159. Psychological evaluation is required to assess suitability for lumbar disc replacement. Documentation should include the following items with associated treatment recommendations:

- psychological factors that might influence elective surgical treatment outcomes, or
- psychological factors that might complicate surgical recovery.

Confounding depression or anxiety must be addressed prior to proceeding with surgery. Presurgical psychological evaluation should not be done by a psychologist employed by the physician performing the procedure. See the Behavioral and Psychological Interventions section.

Recommendation 160. Lumbar TDR is reserved for patients who meet all of the following criteria:

- symptomatic 1- or 2-level disc disease established by objective testing (e.g., CT scan or MRI followed by provocation discography),
- symptoms unrelieved after 6 months of active nonsurgical treatment,
- all pain generators are adequately defined and treated,
- all physical medicine and manual therapy interventions are completed,
- imaging studies demonstrate disc pathology requiring decompression, and
- psychosocial evaluation as outlined in recommendation 159, with confounding issues addressed (tables 72, 73, 74, 75).

Recommendation 161. If there are signs suggestive of facet-mediated pain, the pain must be evaluated with medial branch blocks before proceeding with disc replacement.

Recommendation 162. A trial of multi-disciplinary therapy must occur prior to proceeding with surgery (table 73).

Recommendation 163. Surgeons performing total disc arthroplasty must be trained in anterior spinal techniques and lumbar disc arthroplasty, and they must have attended lumbar arthroplasty training courses or undergone lumbar arthroplasty training in fellowship. It is highly recommended that the surgeon is mentored and proctored, and it is required that they have an assistant surgeon with anterior access experience.

Evidence Tables.

Table 72. <i>Evidence Table: Degenerative Imaging Findings and Lumbar Disc Replacement</i>		
Some evidence	Evidence statement	Design
	In patients with chronic non-radicular low back pain electing for lumbar disc prosthesis surgery, degenerative MRI findings may not be related to associated pain and disability.	Cross sectional

Table 73. <i>Evidence Table: Total Disc Arthroplasty Versus Conservative Care</i>		
Summary: Both TDR and multidisciplinary therapy may improve function.		
Some evidence	Evidence statement	Design
	Disc replacement has a slight advantage over intensive multidisciplinary treatment - 60 hours over 5 weeks.	RCT
	In patients with symptomatic degenerative disc disease at L4-5 or at L5-S1 lasting more than 1 year, persisting after 6 months of physical therapy or chiropractic treatment, an initial treatment regimen consisting of rehabilitation with a multidisciplinary team of specialists may effectively improve function, defined as a 15 point improvement in the ODI, in up to half of patients, making it a reasonable initial treatment option. However, a TDR is overall more effective than multidisciplinary rehabilitation for pain and function, and may improve function by the same amount in about 70% of patients.	RCT

Table 74. <i>Evidence Table: Total Disc Arthroplasty Versus Spinal Fusion</i>		
Summary: TDR, fusion surgery, and the use of interspinous spacers result in similar improvements in function and pain. TDR results in a greater segmental ROM after 1-2 years compared with fusion. TDR leads to greater functional improvement and pain relief than fusion for patients with painful degenerative disc disease and without facet arthritis, spondylolysis, spondylolisthesis, osteoporosis, spinal stenosis, or deformity.		
Strong evidence	Evidence statement	Design
	Disc replacement is not inferior to fusion at 24 months for relief of back pain, reduction of disability, and provision of patient satisfaction.	Meta analysis
Good evidence	Evidence statement	Design
	An artificial disc is non-inferior to allograft fusion with a cage for single level disease.	Systematic review
	There is greater segmental ROM after 1-2 years with arthroplasty as compared with fusion.	Meta analysis

Table 74 continued.

Evidence Table: Total Disc Arthroplasty Versus Spinal Fusion

Good evidence cont.	TDR is more likely than fusion to lead to long term complete pain relief in patients who have painful degenerative disc disease, no facet arthritis, spondylosis or spondylolisthesis, osteoporosis, spinal stenosis, or deformity. TDR also leads on average to greater improvement in the ODI than fusion over 5 years. Both fusion and TDR lead to lasting symptomatic and functional benefits over 5 years in carefully selected patients.	RCT
	In the setting of symptomatic degenerative lumbar disease from a variety of causes, motion-preserving devices such as interspinous spacers or TDRs can be expected to lead to pain relief and disability improvement comparable to the improvements seen with fusion surgery.	Systematic review
Some evidence	Evidence statement	Design
	A 2-level lumbar disc replacement is not inferior to circumferential fusion in patients with 2 level degenerative disc disease and less than grade I spondylolisthesis 24 months after surgery.	RCT
	A trial comparing TDR with circumferential fusion found superior function with lumbar disc replacement compared to fusion. However, the use of narcotics remained high even 2 years after a successful operation.	RCT
	An artificial disc is non-inferior to circumferential fusion with iliac crest autograft for single level disease.	Systematic review
	Pain and disability outcomes at 2 years are similar for 2-level lumbar degenerative disc disease between hybrid operations involving stand-alone ALIF at L4-5 plus disc replacement at L5-S1, compared to non-hybrid operations involving interbody fusion at both levels or TDR at both levels.	Meta analysis

Table 75.

Evidence Table: Adjacent Segment Degeneration and Lumbar TDR

Summary:

Lumbar surgery with motion-preserving devices reduces the incidence of adjacent level disease and reoperation within 2-6 years.

Good evidence	Evidence statement	Design
	Over a period of 2 to 6 years, surgery with motion-preservation devices in the lumbar spine is likely to lead to a lower incidence of adjacent segment disease and to require fewer reoperations than spinal fusion.	Systematic review

Table 75 continued.

Evidence Table: Adjacent Segment Degeneration and Lumbar TDR

Some evidence	Evidence statement	Design
	TDR of the lumbar spine reduces the 5-year risk of adjacent level degeneration compared to circumferential fusion.	RCT

Section 8.b.v. Interspinous Spacers

Introduction. Interspinous process decompression theoretically relieves narrowing of the spinal canal and neural foramen in extension, thereby reducing the symptoms of neurogenic intermittent claudication, secondary to lumbar spinal stenosis.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Lumbar Interspinous Spacers.

- Absolute and relative contraindications include:
 - anatomy that prevents implantation due to significant lumbar instability, ankylosis, acute fracture of the spinous process or pars interarticularis;
 - allergy to titanium or titanium alloy;
 - significant scoliosis;
 - fixed motor deficit;
 - cauda equina syndrome;
 - neural compression causing neurogenic bowel or bladder dysfunction;
 - previous lumbar surgery at the proposed spacer level;
 - significant peripheral neuropathy;
 - anterolisthesis greater than 1.0 (on a scale from 1-4) at the affected level;
 - sustained pathological fractures;
 - severe osteoporosis of the vertebrae or hips;
 - severe foraminal stenosis;
 - extreme obesity;
 - active infection or systemic disease;
 - Paget's disease or metastasis to the vertebrae;
 - steroid use for more than 1 month within 12 months preceding surgery; and
 - adjacent level disease is a relative contraindication.

Complications of Lumbar Interspinous Spacers.

- Complications include symptomatic spinous process fractures, new radicular defects, recurrent back pain, device extrusion or malposition, device failure with need for further surgery, infection, hematoma, and bilateral foot drop.

Recommendations.

Core Requirements.

Recommendation 164. Psychological evaluation is required to assess suitability for lumbar interspinous spacers. Documentation should include the following items with associated treatment recommendations:

- psychological factors that might influence elective surgical treatment outcomes, or
- psychological factors that might complicate surgical recovery.

Confounding depression or anxiety must be addressed prior to proceeding with surgery. Presurgical psychological evaluation should not be done by a psychologist employed by the physician performing the procedure. See the Behavioral and Psychological Interventions section (table 76).

Recommendation 165. Interspinous spacers are indicated for patients 50 years or older with intermittent neurogenic claudication caused by lumbar spinal stenosis who meet all of the following criteria:

- all pain generators are adequately defined and treated;
- all physical medicine and manual therapy interventions are completed over 6 months;
- impaired physical function is correlated with physical findings;

- CT or MRI demonstrates stenosis (e.g., evidence of thickened ligamentum flavum, narrowed lateral recess and/or central canal narrowing);
- spine pathology is limited to 1 or 2 levels;
- psychological evaluation as discussed in [recommendation 164](#);
- ability to sit for 50 minutes without pain;
- ability to walk up to 50 feet or more; and
- relief in flexion from symptoms of leg/buttock/groin pain, with or without back pain ([table 77](#)).

Evidence Tables.

Table 76. <i>Evidence Table: Factors Influencing Surgical Outcomes</i>		
Summary: Patients undergoing spinal surgery who have perioperative depression may have poorer functional outcomes than those without depression.		
Some evidence	Evidence statement	Design
	Depression at the time of surgery and in the early recovery period is associated with poorer functional recovery at 2 years, even though it does not appear to be associated with worse pain at 2 years.	Cohort study
	Depression at the time of operation and in the early postoperative period predicts poorer functional recovery from lumbar spinal stenosis surgery.	Cohort study

Table 77. <i>Evidence Table: Lumbar Interspinous Spacers</i>		
Some evidence	Evidence statement	Design
	An interspinous spacer device is superior to continuing nonoperative treatment after 6 months of conservative care has not resolved neurogenic claudication.	RCT

Section 8.b.vi. Minimally Invasive Sacroiliac (SI) Joint Fusion

Introduction. Minimally invasive sacroiliac (SI) joint fusion is performed from a lateral approach under fluoroscopic guidance. The gluteus muscle is bluntly dissected and a pin is inserted across the SI joint so that a cannulated drill and broach can create a triangular wedge-shaped cavity in the ilium and sacrum through which titanium implants (typically 2-4) are inserted. The therapeutic goal of minimally invasive SI joint fusion is to provide stabilization and minimize micromotion/rotation of the instrumented SI joint. Open SI joint fusion for acute traumatic pelvic disruption is not discussed in these guidelines.

Contraindications / Complications / Side Effects and Adverse Events.

Relative and Absolute Contraindications to Minimally Invasive SI Joint Fusion.

- A contradiction to minimally invasive SI joint fusion is acute, traumatic instability of the SI joint due to trauma.
- Generalized pain behavior (e.g., somatoform disorder and presence of Waddell's signs) is an absolute contraindication.
- Systemic inflammatory arthropathies are a contraindication.

- Generalized pain syndromes (e.g., fibromyalgia) are relative contraindications.

Complications of Minimally Invasive SI Joint Fusion Requiring Surgical Revision.

- Complications of minimally invasive SI joint fusion that require surgical revision include symptomatic malposition, symptom recurrence, postoperative hematoma, device loosening, sacral nerve root impingement, suboptimal device positioning, iliac fracture, iliac vessel injury, and postoperative radicular pain.
- Long-term complication rates are unknown.

Complications and Adverse Events Related to Minimally Invasive SI Joint Fusion.

- Complications and adverse events include infection, wound healing issues, nerve damage, recurrent pain, urinary tract infection, bursitis, and postoperative issues.
- Long-term complication rates are unknown.

Recommendations.

Core Requirements.

Recommendation 166. Minimally invasive SI joint fusion is not recommended in the acute or subacute period.

Recommendation 167. Prior authorization is required before surgery. The requesting provider should document how the patient satisfies the clinical indications outlined in this section.

Recommendation 168. Psychological evaluation is required to assess suitability for minimally invasive SI joint fusion. Documentation should include the following items with associated treatment recommendations:

- psychological factors that might influence elective surgical treatment outcomes, or
- psychological factors that might complicate surgical recovery.

Confounding depression or anxiety must be addressed prior to proceeding with surgery. Presurgical psychological evaluation should not be done by a psychologist employed by the physician performing the procedure. See the Behavioral and Psychological Interventions section.

Recommendation 169. Minimally invasive SI joint fusion should be reserved for patients who meet all of the following indications:

- at least 6 months of persistent functional impairment and pain, unresponsive to intensive conservative therapies, including medication, manual therapy, active therapeutic exercise (e.g., optimizing lumbosacral and pelvic ROM and identification/treatment of muscular imbalance);
- patient localization of typically unilateral symptoms as indicated by Fortin finger test with reported pain at the posterior superior iliac spine;
- physical examination findings consistent with SI joint origin pain, including 3 positive physical exam maneuvers (e.g., compression test, thigh thrust test, sacral thrust test, Patrick's, Gaenslen test, distraction test, Gillett test);
- imaging studies (e.g., CT or MRI) of the lumbar spine, and where clinically indicated, of the hip, and/or pelvis that fail to reveal other sources that might better explain the symptoms (e.g., inflammatory arthropathy, evidence of neural compression, facet disease, disc disease, degenerative labral tears, tumor, infection, fracture); and
- psychological evaluation as outlined in recommendation 168 (see the Behavioral and Psychological Interventions section); and

- positive response to initial and confirmatory fluoroscopically-guided SI joint local anesthetic blocks with arthrogram confirming needle placement in the SI joint, consisting of:
 - documented improvement in previously impaired SI joint function and provocative physical examination maneuvers within expected time frame of local anesthetic (e.g., spinal ROM; tolerance and time limits for sitting, standing, walking, and lifting; Patrick's sign; Gaenslen; distraction; gapping and compression tests); and
 - 80% improvement in accepted pain scales (e.g., VAS or NRS), consistent with:
 - the expected duration of the injected local anesthetic phase, and
 - a post-injection pain diary with at least 8 hourly response recordings, but preferably daily for 1 week (table 78).

Recommendation 170. The surgeon must have specific training and expertise in performing this procedure.

Recommendation 171. Systemic inflammatory arthropathies, including ankylosing spondylitis and rheumatoid arthritis, are not indications for minimally invasive SI joint fusion.

Recommendation 172. Tobacco cessation for at least 6 weeks prior to fusion is required, unless the surgeon documents reasoning that the benefits of proceeding with surgery outweigh the risks.

Recommendation 173. Imaging guidance is required to perform implant placement.

Recommendation 174. An individualized postoperative rehabilitation program is appropriate for patients at the discretion of the surgeon.

Evidence Tables.

Table 78. <i>Evidence Table: Minimally Invasive SI Joint Fusion for Chronic SI Joint Pain</i>		
Summary: For patients with chronic low back pain and a high likelihood of true SI joint pathology, surgical fusion with triangular titanium implants provides greater improvement in function and pain at 6 months than continued conservative treatment. These benefits may continue for at least 4 years after the operation.		
Good evidence	Evidence statement	Design
	In patients with chronic low back pain originating from the SI joint, a fusion procedure using triangular titanium implants leads to greater improvements in pain and function at 6 months than continued conservative treatment with optimum medical management and individualized physical therapy.	RCT
Some evidence	Evidence statement	Design
	In the setting of chronic SI joint pain, a surgical fusion procedure involving triangular titanium implants is likely to confer greater benefits in pain reduction and functional improvement than continued nonsurgical pain management with medications, steroid injection, and RF neural ablation, with benefits appearing to be maintained 2, 3, and 4 years after the operation.	RCT

Section 8.b.vii. Kyphoplasty

Introduction. Kyphoplasty is a surgical procedure for the treatment of symptomatic thoracic or lumbar vertebral compression fractures and occasionally post-traumatic compression fractures and minor burst fractures that do not significantly compromise the posterior cortex of the vertebral body.

Kyphoplasty involves the percutaneous insertion of a trocar and inflatable balloon or expanding polymer into the vertebral body, which re-expands the body, elevating the endplates and reducing the compression deformity. Polymethylmethacrylate (PMMA) bone cement is injected under low pressure into the cavity created by balloon inflation.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Kyphoplasty.

- Absolute and relative contraindications include:
 - asymptomatic vertebral body compression fracture;
 - patient improvement with medical treatment;
 - presence of neurologic compromise related to fracture;
 - high-velocity fractures with a significant burst component;
 - significant posterior vertebral body wall fracture;
 - severe vertebral collapse (vertebra plana);
 - infection; and
 - uncorrectable coagulopathy.

Complications of Kyphoplasty.

- Complications include new vertebral compression fracture, cement leakage, infection and extravertebral extravasation of the bone cement, with rare serious complications such as cemental pulmonary embolism, radiculopathy, and paraplegia.
- Procedure-related deaths have been reported.

Recommendations.

Core Requirements.

Recommendation 175. Kyphoplasty may occur within the first 12 weeks of care after vertebral compression fracture when all of the following criteria are met:

- compression fracture with vertebral height loss between 15% and 85%,
- failure to respond to at least 4 weeks of conservative management, and
- functional impairment prevents performance of ADLs ([table 79](#)).

Evidence Table.

Table 79.		
<i>Evidence Table: Kyphoplasty Versus Conservative Management for Compression Fractures</i>		
Summary: Kyphoplasty results in rapid functional improvement in the first month after compression fracture as compared to non-operative treatment, but the difference between groups diminishes over time.		
Good evidence	Evidence statement	Design
	Kyphoplasty provides rapid improvement in function in the initial months after the fracture, as compared to non-operative treatment or analgesics alone.	RCT

Some evidence	Evidence statement	Design
	In the setting of osteoporitic vertebral compression fractures, the Kiva (trademarked vertebral augmentation system) coiled implant is noninferior to conventional balloon kyphoplasty in terms of pain relief, functional improvement, and safety at 1 year after the procedure is done.	RCT

Section 8.b.viii. Vertebroplasty

Introduction. Vertebroplasty is a minimally invasive surgical procedure for the treatment of painful thoracolumbar vertebral compression fractures secondary to osteoporosis or other metabolic bone disease. In this procedure, bone cement is injected with high pressure into the vertebral body under constant fluoroscopic guidance. This guidance is used throughout the procedure to minimize cement leakage. Traditionally, a low-viscosity acrylic bone cement, PMMA, is used, but other types of bone cement such as high-viscosity PMMA, glass polymers, hydroxyapatite, and calcium phosphate are also commercially available. The procedure is usually performed under intravenous sedation or light general anesthesia. The goal of the procedure is to stabilize the spine and to relieve pain.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Vertebroplasty.

- Absolute and relative contraindications include:
 - asymptomatic vertebral body compression fracture,
 - patient improvement with conservative measures alone,
 - presence of fracture-related neurologic compromise,
 - high-velocity fractures with a significant burst component,
 - posterior vertebral body wall fracture,
 - severe vertebral collapse (vertebra plana),
 - spinal canal stenosis,
 - allergy to bone cement or opacification agents,
 - infection, and
 - uncorrectable coagulopathy.

Complications of Vertebroplasty.

- Procedure-related deaths have been reported.
- Complications include extravertebral extravasation of the bone cement, with rare serious complications such as cemental pulmonary embolism, radiculopathy and paraplegia.
- New vertebral compression fractures may occur following vertebroplasty.

Side Effects and Adverse Events Related to Vertebroplasty.

- Side effects and adverse events include localized pain at the site of the procedure.

Recommendations.

Core Requirements

Recommendation 176. Vertebroplasty may be considered for a select subgroup of patients with vertebral compression fractures within 4 and 12 weeks of pain onset when the following criteria have been met:

- compression fracture has been radiographically confirmed to cause pain and limit activity;

- symptoms are localized clinically to the level of the vertebral fracture;
- inability to perform ADLs;
- failure to respond to at least 4 weeks of conservative management;
- patient is healthy enough to undergo surgery, if necessary, for decompression;
- vertebral height loss between 15% and 85%; and
- an intact posterior wall (table 80).

Evidence Tables.

Table 80.
Evidence Table: Vertebroplasty for Compression Fracture

Summary:

For patients with acute osteoporotic vertebral fractures and mild to moderate pain, vertebroplasty and sham vertebroplasty (infiltration of the periosteum with local anesthesia) have clinically equivalent effects on function and pain. Vertebroplasty results in greater improvement in function and pain than continued medical management and physical therapy. Sham vertebroplasty has not been compared with continued medical management.

Strong evidence	Evidence statement	Design
	In the setting of acute osteoporotic vertebral compression fractures with mild to moderate pain, there are no clinically important differences between true and sham vertebroplasty for pain, function, or quality of life. The addition of PMMA cement does not enhance the therapeutic effects of a local anesthetic infiltration of the periosteum of the affected vertebra. Percutaneous vertebroplasty should not be a standard treatment for these fractures.	Systematic review
Good evidence	Evidence statement	Design
	Osteoporotic vertebral fractures improve equally with both vertebroplasty and with well-simulated sham vertebroplasty that includes infiltration of the periosteum with local anesthesia. This study has been criticized for selection bias due to the failure of most patients to be willing to enroll in a randomized trial.	RCT
	Vertebroplasty does not differ from sham vertebroplasty in patients with acute (less than 6 months duration) osteoporotic vertebral fractures demonstrated by MRI criteria of edema and or/a fracture line, but that may apply only to patients whose pain is not severe enough to lead to hospitalization. This study has been criticized for selection bias due to the failure of most patients to be willing to enroll in a randomized trial.	RCT
	In the setting of acute osteoporotic vertebral compression fractures, percutaneous vertebroplasty is associated with greater pain relief and disability improvement than continued medical management combined with physical therapy. It is uncertain whether this difference represents a placebo (expectation) effect for the procedure or whether it represents a therapeutic effect of the local anesthetic. The effect of an anesthetic infiltration of the periosteum of the affected vertebra has not been tested as a stand-alone treatment in any randomized trial published to date.	Systematic review
	In patients with acute osteoporotic vertebral fractures accompanied by edema on MRI, a vertebroplasty procedure and a sham vertebroplasty procedure	RCT

	yield rapid decreases in pain and disability that persist for up to 12 months. After a periosteal infiltration with a local anesthetic is done, the addition of a PMMA cement injection does not appear to add significant additional benefits.	
	Vertebroplasty improves pain scores more rapidly than individualized pharmacological therapy for patients with acute osteoporotic vertebral fractures with effects detectable in the first day and persisting up to 1 year.	RCT
Some evidence	Evidence statement	Design
	In patients with severe pain (7 or more on a 0-10 scale) arising from acute osteoporotic vertebral compression fractures, treatment with vertebroplasty is more effective than sham vertebroplasty in reducing pain at 14 days post-procedure, and in patients hospitalized for osteoporotic vertebral compression fractures, vertebroplasty is likely to reduce the length of hospital stay.	RCT
	Vertebroplasty is associated with earlier pain relief and functional mobility than pharmacologic treatment alone, but there may be a risk of new clinically significant vertebral fractures when more than 1 vertebra is treated.	RCT

Section 8.b.ix. Spinal Cord Stimulators

Introduction. Spinal cord stimulation is the delivery of low-voltage electrical stimulation to the spinal cord or peripheral nerves to inhibit or block the sensation of pain. The system uses implanted electrical leads and a battery powered implanted pulse generator. See the Chronic Pain Disorder MTGs.

Recommendations.

Core Requirements.

Recommendation 177. Neurostimulation is reserved for patients who meet the criteria outlined in the Chronic Pain Disorder MTGs.

Evidence Tables. See the Chronic Pain Disorder MTGs for evidence.

Section 8.b.x. Epiduroscopy with Epidural Lysis of Adhesions

Introduction. Epiduroscopy and epidural lysis of adhesions involves the introduction of a fiberoptic endoscope into the epidural space via the sacral hiatus. A saline irrigation is performed with or without epiduroscopy to distend the epidural space and result in lysis of adhesions.

Contraindications / Complications / Side Effects and Adverse Events.

Complications of Epiduroscopy.

- Saline irrigation on epiduroscopy and lysis of adhesions is associated with risks of elevated pressures that may impede blood flow and venous return, possibly causing ischemia of the cauda equina and retinal hemorrhage.
- Other epiduroscopy complications associated with instrumented lysis include catheter shearing, need for catheter surgical removal, infection (including meningitis), hematoma, and possible severe hemodynamic instability during application.

Recommendations.

Core Requirements.

Recommendation 178. Epiduroscopy and mechanical lysis of adhesions and epiduroscopy-directed steroid injections are not recommended.

Evidence Tables. None. The above recommendation was based on consensus.

Section 9. Medications

Introduction. Medications are used in the treatment of low back injuries to control acute pain, chronic pain, and inflammation. Use of medications will vary widely due to the spectrum of injuries. If medications are being considered for long-term chronic pain management, refer to the Chronic Pain Disorder Medical Treatment Guidelines (MTGs) medication section.

Contraindications / Complications / Side Effects and Adverse Events.

- See the specific medication section. The medication lists within these sections do not provide complete information on side effects, potential complications, drug interactions, or drug monitoring. For more complete information, refer to the Chronic Pain Disorder MTGs medication section or a medication reference text.

Recommendations.

Core Requirements.

Recommendation 179. Medication reconciliation is required at the initial visit and periodically during treatment to avoid medication errors and to discuss side effects, drug interactions, and expected functional goals. Reconciliation includes the following elements:

- current medication name, dosage, frequency, and route;
- patient understanding of indication;
- potential interaction of prescription and over the counter medications;
- drug allergies;
- comorbid medical issues;
- history of substance abuse; and
- checking the Colorado Prescription Drug Monitoring Program (PDMP).

The medications documented as a part of the reconciliation will reflect those that the provider deems directly relevant to the claim-related condition.

Recommendation 180. A therapeutic trial of medications is recommended to evaluate the effect on functional status. The length of a medication trial will depend on the individual medication, and the patient should be informed on the time to expected benefit. If no functional benefit is observed at that time, the medication should be discontinued.

Recommendation 181. Medications should be initiated at the lowest dose expected to result in functional improvement and then titrated based on clinical response.

Evidence Tables. None. The above recommendations were based on consensus.

Section 9.a. Non-Steroidal Anti-Inflammatory Drugs (NSAIDs) and Acetaminophen

Introduction. Non-steroidal anti-inflammatory drugs (NSAIDs) are medications for low back pain and inflammation. Acetaminophen is an analgesic commonly used to treat mild pain.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Acetaminophen and NSAIDs.

- Acetaminophen
 - Concomitant use of other drug products containing acetaminophen exceeding the maximum recommended daily dose of acetaminophen.
 - Allergy to acetaminophen or any of the inactive ingredients.
 - Use with caution in patients with alcoholic liver disease, patients with hepatic impairment or active liver disease, and patients with known glucose-6-phosphate-dehydrogenase (G6PD) deficiency.
- NSAIDs (table 81)
 - Comorbid conditions that result in an increased risk for gastrointestinal (GI), renal, or cardiovascular adverse reactions.
 - Patients at risk for GI bleeding, including patients who use alcohol, smoke, are older than 65 years old, take corticosteroids or anti-coagulants, and patients on longer durations of therapy.
 - Celecoxib is contraindicated in sulfonamide allergic patients.
 - Contraindicated in patients with aspirin-exacerbated respiratory disease due to cross-reactivity risk and potential for a severe asthma attack.
- Topical salicylate and non-salicylate medications may result in alterations in bleeding time; they should be used with caution in patients on warfarin therapy.

Side Effects and Adverse Events Related to Specific Classes of Acetaminophen and NSAIDs.

- Acetaminophen may result in hepatotoxicity, and chronic use may result in chronic kidney disease, hypersensitivity or anaphylactic reactions, serious and potentially fatal skin reactions, hypertension, chronic daily headaches, and peptic ulcer disease.
- NSAIDs may result in abnormal or worsening renal function, including renal failure; abnormal liver function; GI bleeding, particularly in patients at higher risk for a bleed; increased risk of cardiovascular events; gastric or duodenal perforation and ulceration; anaphylactoid reaction; platelet function abnormalities; and fluid retention and edema. Postoperative NSAIDs may increase the risk of nonunion at higher doses (table 85).
- Topical agents may result in localized skin reactions.

Recommendations.

Core Requirements.

Recommendation 182. Acetaminophen or NSAIDs are recommended for initial analgesic treatment of uncomplicated low back pain (tables 82, 83).

Recommendation 183. Co-prescription of a proton pump inhibitor, histamine H2-receptor antagonists (H2-blockers), or prostaglandin analog with NSAIDs is recommended to reduce risk of duodenal or gastric ulceration in patients with concurrent antiplatelet or corticosteroid therapy.

Recommendation 184. Cyclooxygenase-2 (COX-2) inhibitors are not recommended as a first-line agent for short-term use in low-risk patients, but they can be used for patients who do not tolerate traditional NSAIDs.

Recommendation 185. Topical NSAIDs are recommended when oral NSAID use is contraindicated due to systemic side effects. It must be started with the lowest dose expected to result in functional improvement and then titrated until functional improvement is noted ([table 84](#)).

Recommendation 186. Perioperative use of acetaminophen and/or NSAIDs, either alone or in combination with other medications, to optimize analgesia is recommended ([table 85](#)).

Recommendation 187. See the [Chronic Pain Disorder MTGs medication section](#) if acetaminophen or NSAIDs are being considered for long-term use.

Time Frames.

Time Frames for Medications		
	Optimum duration	Maximum duration
Acetaminophen	up to 10 days	Extended use on a case-by-case basis*
NSAIDs	7 days	1 year*
*Use of this substance long-term (for 3 days per week or greater) may be associated with rebound pain upon cessation.		

Evidence Tables.

Table 81. Evidence Table: Adverse Events and NSAIDs		
Good evidence	Evidence statement	Design
	Naproxen has a more favorable cardiovascular profile than other NSAIDs when used over a long period for chronic pain.	Meta analysis
	For general low back pain, specific adverse effects of dyspepsia and GI bleeding were more frequent with NSAIDs than placebos.	Systematic review
	Celecoxib in a dose of 200 mg per day, administered over a long period, does not have a worse cardiovascular risk profile than naproxen at a dose of up to 1000 mg per day or ibuprofen at a dose of up to 2400 mg per day.	RCT
	Celecoxib has a more favorable safety profile than ibuprofen or naproxen with respect to serious GI adverse events, and has a more favorable safety profile than ibuprofen with respect to renal adverse events.	RCT
Some evidence	Evidence statement	Design
	Fewer adverse events occurred in NSAIDs treatment arms as compared to placebo in patients experiencing chronic low back pain.	Systematic review

Table 82.

Evidence Table: Acetaminophen

Summary:

For acute nonspecific low back pain, acetaminophen and placebo provide similar pain relief and functional improvement. While acetaminophen as a monotherapy is not superior to placebo, its use as 1 component of a multimodal analgesic regimen is generally accepted.

Strong evidence	Evidence statement	Design
	In the setting of acute nonspecific low back pain, acetaminophen at a dose of up to 4 grams per day is no more effective than placebo for pain relief and for improvement of disability.	Meta analysis

Table 83.

Evidence Table: NSAIDs for Low Back Pain

Summary:

NSAIDs may reduce acute low back pain and short term disability when compared to placebo, but the clinical benefit is small.

Good evidence	Evidence statement	Design
	In the setting of acute low back pain lasting less than 12 weeks, NSAIDs lead to a small reduction in short term pain intensity, approximately 10 points on a 100 point scale, compared to placebo.	Meta analysis
	In the setting of acute low back pain lasting less than 12 weeks, NSAIDs lead to a small reduction in short term disability, approximately 2 points on a 24 point scale, compared to placebo.	Meta analysis
	For recent low-back pain, oral NSAIDs are equally effective as placebo or muscle-relaxants.	Systematic review
	For persistent low back pain, oral NSAIDs are more effective than placebo or acetaminophen.	Systematic review
Some evidence	Evidence statement	Design
	NSAIDs are not more beneficial than placebo when evaluating pain-intensity and disability outcomes; the change in these scores does not reach a minimal clinical important difference in patients experiencing chronic low back pain.	Systematic review

Table 84.

Evidence Table: Topical NSAIDs for Analgesia

Summary:

Topical NSAIDs are more effective than placebo for acute musculoskeletal injuries and have fewer systemic adverse events than oral NSAIDs.

Strong evidence	Evidence statement	Design
	Topical NSAIDs are more effective than placebo vehicles, such as gels or creams, in the setting of acute musculoskeletal injuries.	Meta analysis
Some evidence	Evidence statement	Design
	Topical NSAIDs are associated with fewer systemic adverse events than oral NSAIDs.	Meta analysis

Table 85.

Evidence Table: Perioperative Medication Use for Optimal Analgesia

Summary:

For patients undergoing spinal fusion surgery, preoperative analgesia may reduce postoperative pain and opioid use. Postoperative NSAIDs may increase the risk of nonunion at higher doses.

Good evidence	Evidence statement	Design
	In the setting of lumbar spinal fusion, NSAIDs at an equivalence of no more than 120 mg of ketorolac for 14 days postoperatively do not appear to increase the risk of nonunion of the operated spinal segment.	Systematic review

Table 85 continued.

Evidence Table: Perioperative Medication Use for Optimal Analgesia

Some evidence	Evidence statement	Design
	In the setting of posterior lumbar interbody fusion (PLIF) surgery, a single saline infusion of 800 mg of ibuprofen one half hour prior to surgery may lead to less pain and morphine use in the immediate postoperative period and may thereby be an option for preoperative care of the surgical patient.	RCT
	In the setting of 1-level or 2-level spinal fusion surgery, the administration of preemptive analgesia 8 hours preoperatively with acetaminophen, ketorolac, and pregabalin may confer benefits in the 48 hours following the procedure in the form of less pain, better ambulation, and less morphine use.	RCT
	Among patients undergoing posterior lumbar spinal fusion, the addition of preemptive analgesia using a single intravenous dose of ketorolac (30 mg) or parecoxib (40 mg) resulted in better immediate postoperative pain control as compared to a control (saline) group. However, the difference among the groups did not persist in measures from one-hour post-operatively up to 24 hours. There was no apparent opioid-sparing effect with preemptive analgesia.	RCT

	In the setting of lumbar fusion without bone morphogenetic protein (BMP), postoperative NSAIDs with an equivalent dose of more than 300 mg of diclofenac are associated with an increased risk of nonunion of the operated spinal segment. There appears to be a dose-dependent effect of postoperative NSAIDs on the risk of nonunion.	Systematic review
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Section 9.b. Muscle Relaxants

Introduction. Muscle relaxants are a heterogeneous class of medications with varying mechanisms of action used to treat muscle spasm associated with low back injury.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Muscle Relaxants.

- Cyclobenzaprine should not be used when a patient has a history of cardiac dysrhythmia or when there is concurrent use of monoamine oxidase inhibitors.
- Metaxalone should not be used when there is a history of significantly impaired renal or hepatic disease, pregnancy, and predisposition to drug induced hemolytic anemia.
- Methocarbamol should not be used when the patient has a hypersensitivity to the medication or there is possible renal compromise.
- Tizanidine should not be used when there is concurrent use of ciprofloxacin or fluvoxamine or hepatic disease.

Side Effects and Adverse Events Related to Muscle Relaxants.

- Cyclobenzaprine may cause sedation, anticholinergic reactions, and blurred vision. Patients should also be monitored for suicidal ideation and drug abuse.
- Metaxalone may cause sedation and hematologic abnormalities.
- Methocarbamol may cause decreased cognition, lightheadedness, and GI side effects.
- Tizanidine may cause hypotension, sedation, hepatotoxicity, hallucinations, psychosis, and dry mouth.

Recommendations.

Core Requirements.

Recommendation 188. Muscle relaxants may be used as an adjunct to rest and physical therapy for relief of muscle spasm associated with acute, painful, musculoskeletal conditions ([table 86](#)).

Recommendation 189. Tizanidine (alpha-2 adrenergic agonist) is only approved by the Food and Drug Administration (FDA) for the treatment of true centrally mediated spasticity associated with musculoskeletal disorders. Use for musculoskeletal conditions, such as neck or back pain, would be “off label.”

Recommendation 190. Benzodiazepines are not recommended for use in combination with opioids due to the elevated risk of death due to respiratory depression.

Recommendation 191. See the [Chronic Pain Disorder MTGs medication section](#) if muscle relaxants are being considered for long-term use.

Time Frames.

Time Frames for Muscle Relaxants	
Optimum duration	Maximum duration
1 week	2 weeks [†]
[†] or longer, if only used at night.	

Evidence Tables.

Table 86.		
<i>Evidence Table: Muscle Relaxants</i>		
Summary: For patients with acute low back pain, muscle relaxants provide greater short-term pain relief than placebo. However, they do not provide additional benefit over NSAIDs alone, and they increase adverse events.		
Strong evidence	Evidence statement	Design
	Non-benzodiazepine muscle relaxants are more effective than placebo for providing short-term pain relief in acute low back pain. They should be used with caution because of central nervous system side effects.	Meta analysis
Good evidence	Evidence statement	Design
	In patients with acute nontraumatic musculoskeletal low back pain who have optimized their use of NSAIDs such as naproxen, the addition of cyclobenzaprine 5mg to be taken as needed adds no significant functional benefit in the short term, but it does increase the frequency of adverse events such as dizziness, drowsiness, and GI side effects.	RCT
Table 86 continued.		
<i>Evidence Table: Muscle Relaxants</i>		
Good evidence cont.	In the setting of acute nonradicular, nontraumatic musculoskeletal low back pain of 2 weeks duration or less, the addition of diazepam to naproxen adds no additional pain relief or reduction of functional disability beyond that of placebo.	RCT

Section 9.c. Oral Steroids

Introduction. Oral steroids are used for pain and inflammation.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Oral Steroids.

- Absolute and relative contraindications include poorly controlled diabetes mellitus or hypertension; heart failure with peripheral edema; cataract or glaucoma; peptic ulcer disease; presence of infection; and low bone mineral density or osteoporosis.

Complications of Oral Steroids.

- Complications include gastritis, ulcer formation, and GI bleeding.

Side Effects and Adverse Events Related to Oral Steroids.

- Side effects and adverse events fluid retention, mood disturbance, and hyperglycemia.

Recommendations.

Core Requirements.

Recommendation 192. Oral steroids are not recommended for the treatment of acute low back pain with or without radiculopathy ([table 87](#)).

Evidence Tables.

Table 87.

Evidence Table: Oral Steroids for Acute Low Back and/or Radicular Pain

Summary:

For patients with acute low back pain with or without radicular pain, oral steroids do not provide clinically important improvements in function or pain, nor reductions in the rate of surgery. They also increase the risk for serious adverse effects.

Good evidence	Evidence statement	Design
	In patients with an acutely herniated lumbar disc with radicular symptoms, a 15-day course of tapering oral prednisone may produce a small improvement in spine function compared to placebo, but this improvement is of very uncertain clinical importance. Also, there is no evidence that oral prednisone reduces the rate of back surgery in the following year. The benefits of a small functional improvement should be weighed against the frequent occurrence of steroid-related adverse effects such as insomnia, nervousness, and increased appetite. There is no evidence that oral prednisone reduces sciatica pain compared to placebo.	RCT

Table 87 continued.		
Evidence Table: Oral Steroids for Acute Low Back and/or Radicular Pain		
Good evidence cont.	In patients who have not taken oral steroids in the past year, short-term use of an oral steroid can increase the risk of fracture, sepsis, and venous thromboembolism in the subsequent 5 to 90 days, with the greatest increased risk occurring in the first 5-30 days after the prescription is filled.	Cohort study
Some evidence	Evidence statement	Design
	Among patients presenting to the emergency department with acute onset of musculoskeletal or radicular low back pain or acute exacerbation of chronic low back pain, a 5-day course of 50 mg oral prednisone results in similar pain reduction at 5 days post-discharge when compared to a placebo given over 5 days.	RCT
	A short course of oral corticosteroids, lasting 14 days or fewer, is associated with an increased incidence of GI bleeding, sepsis, and heart failure in the period following their administration.	Cohort study

Table 88.		
Evidence Table: Other Medications		
Strong evidence	Evidence statement	Design
	Bisphosphonates taken alone or in combination with calcium and/or vitamin D are more beneficial than placebo taken alone or with calcium and/or vitamin D for the prevention and treatment of spinal glucocorticoid induced osteoporosis, with data extending to 24 months of use.	Systematic review

Section 9.d. Gabapentinoids

Introduction. Gabapentinoids are anticonvulsant and nerve pain medications that may be used to treat neuropathic symptoms. They include gabapentin and pregabalin.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Gabapentinoids.

- Renal insufficiency is a relative contraindication to gabapentin use, but dosage can be adjusted to accommodate use in the setting of renal dysfunction.

Side Effects and Adverse Events Related to Gabapentinoids.

- Gabapentinoids may cause dizziness, sedation, and respiratory depression in older patients who receive gabapentin along with other analgesics or sedatives.
- Coadministration of opioids and pregabalin or gabapentin may increase the risk of opioid related mortality.
- Gabapentinoids may be associated with an increased risk of mental health disturbance (e.g., depression, suicide), unintentional overdose, and motor vehicle accidents.

Recommendations.

Core Requirements.

Recommendation 193. A trial of oral gabapentin, with a goal of using the lowest dose expected to result in functional improvement, can be a first line intervention before an epidural steroid injection (ESI) for a patient who meets the acute ESI indications but wishes to avoid ESI ([table 89](#)). See the Epidural Steroid Injection section.

Recommendation 194. Perioperative use of gabapentin, either alone or in combination with other medications, to optimize analgesia is recommended ([table 90](#)).

Recommendation 195. If a gabapentinoid is being considered for long-term, chronic pain management, refer to the [Chronic Pain Disorder MTGs medication section](#).

Evidence Tables.

Table 89. Evidence Table: Gabapentinoids		
Summary: For patients with lumbosacral radiculopathy due to herniated disc or spinal stenosis, oral gabapentin is equally effective at reducing leg pain as ESI. This finding is limited because the interventions were not compared to placebo and were not paired with physical therapy.		
Strong evidence	Evidence statement	Design
	Gabapentin is more effective than placebo in the relief of painful diabetic neuropathy and postherpetic neuralgia.	Meta analysis
Good evidence	Evidence statement	Design
	In the setting of lumbosacral radiculopathy from herniated disc or spinal stenosis, oral gabapentin without physical therapy and an ESI without physical therapy are approximately equally effective in the first 3 months, but no conclusions can be supported that either intervention is superior to placebo.	RCT

Table 90. Evidence Table: Perioperative Gabapentin Use		
Summary: For patients undergoing spinal surgery, preoperative gabapentin may reduce postoperative pain and opioid use, without increasing complications.		
Some evidence	Evidence statement	Design
	The preoperative administration of gabapentin in doses as low as 300 mg may reduce postoperative pain and opioid consumption in the setting of spinal laminectomy and fusion.	RCT
	The incidence of many postoperative complications such as nausea, headache, and dizziness is probably similar for gabapentin and placebo. Gabapentin may increase somnolence but may decrease the incidence of urinary retention and pruritus.	RCT

Table 90 continued.

Evidence Table: Perioperative Gabapentin Use

Some evidence cont.	In patients with chronic lumbosacral radiculopathy undergoing lumbar spinal operation, combined administration of pregabalin and dexamethasone conferred a significant reduction in rescue analgesic requirement up to 2 days postoperatively compared to placebo.	RCT
	In the setting of 1-level or 2-level spinal fusion surgery, the administration of pre-emptive analgesia 8 hours preoperatively with acetaminophen, ketorolac, and pregabalin may confer benefits in the 48 hours following the procedure in the form of less pain, better ambulation, and less morphine use.	RCT

Section 9.e. Antidepressants

Introduction. Antidepressants are used for the treatment of pain, dysesthesias, sleep disorders, and depression. If medications are being considered for chronic pain management, refer to the [Chronic Pain Disorder MTGs medication section](#).

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Antidepressants.

- Tricyclic antidepressants should be avoided in patients with the following:
 - at risk of unintentional overdose;
 - heart disease;
 - underlying cardiac conduction system disease;
 - dysrhythmia;
 - prostatic hypertrophy;
 - at risk for suicide;
 - uncontrolled hypertension or orthostatic hypotension;
 - those 65 years or older, particularly if a fall risk; and
 - seizure disorder.
- Serotonin norepinephrine reuptake inhibitors (SNRIs) should not be used in patients with seizures or eating disorders.

Complications of Antidepressants.

- Tricyclic antidepressants may cause acute hepatitis, neuroleptic malignant syndrome, and tardive dyskinesia.
- SNRIs may result in serotonin syndrome.
- Venlafaxine may cause hypertension, glaucoma, sexual dysfunction, and cardiac issues.

Side Effects and Adverse Events Related to Antidepressants.

- Tricyclic antidepressants (e.g., amitriptyline, nortriptyline) may cause anticholinergic effects (e.g., sedation, dry mouth and associated periodontal conditions, orthostatic hypotension, constipation, urinary retention); decreased seizure threshold; sexual dysfunction; diaphoresis; tremor; antihistamine effects; alpha-1-adrenergic receptor blockade; and cardiac effects, including increasing intraventricular conduction, prolonged QT interval, prolonged conduction through the atrioventricular node.
- SNRIs (e.g., duloxetine, venlafaxine) may cause nausea, dizziness, sweating, loss of appetite, dry mouth, insomnia, drowsiness, constipation, abnormal bleeding, fatigue, sexual dysfunction, and suicidal ideation and attempts in adolescents and young adults.

Recommendations.

Core Requirements.

Recommendation 196. Tricyclic antidepressants are the recommended first-line agent for neuropathic pain, particularly in the setting of insomnia, but they are not recommended as a first-line agent for depression.

Recommendation 197. SNRIs are recommended as a second-line agent for neuropathic pain if a tricyclic offers inadequate relief. However, duloxetine may be considered a first-line agent for a patient who is a candidate for pharmacologic treatment of both chronic pain and depression.

Recommendation 198. Selective serotonin reuptake inhibitors (SSRIs) are recommended for treating depression, but they are not recommended for neuropathic pain.

Recommendation 199. Evaluation and ongoing monitoring for suicidal ideation and mood swings are required for all patients being considered for antidepressant medications.

Recommendation 200. A screening electrocardiogram may be indicated for select patients prior to initiating treatment with a tricyclic or SNRI antidepressant to assess cardiovascular risk.

Time Frames.

Time Frames for Antidepressant Medications	
Optimum duration	Maximum duration
up to 6 months	up to 12 months, with monitoring

Evidence Tables. None. The above recommendations were based on consensus.

Section 9.f. Opioids

Introduction. Opioids are powerful analgesics reserved for severe, acute pain associated with major trauma, burn, sickle cell disease, and end-of-life cancer pain. If opioid medications are being considered for chronic pain management, refer to the [Chronic Pain Disorder MTGs medication section](#).

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Opioids.

- Opioids should not be prescribed with benzodiazepines, antihistamines, or other central nervous system depressants or when the history is consistent with regular alcohol or other substance use.
- Tramadol should be used with caution in patients who have a history of seizures, are taking medications that lower the seizure threshold, or are taking medications that impact serotonin reuptake and could increase the risk for serotonin syndrome, such as monoamine oxidase inhibitors, SSRIs, selective serotonin agonists (triptans), tricyclic antidepressants, and alcohol. Also use caution in patients who are taking medications that result in QT prolongation. Tramadol has been associated with death in those with an emotional disturbance or concurrent use of alcohol and other opioids. Significant renal and hepatic dysfunction requires dosage adjustment.

Side Effects and Adverse Events Related to Opioids.

- Opioid medications may commonly cause nausea, vomiting, drowsiness, unsteadiness, constipation, and confusion. Occasional side effects include dry mouth, sweating, pruritus, hallucinations, and myoclonus. Adverse events include opioid-induced hyperalgesia, respiratory depression, dependence, opioid use disorder, overdose, or death. Prolonged opioid use may result in hypogonadism (tables 91, 92). Abrupt discontinuation may precipitate withdrawal.
- Certain comorbid medical conditions can increase the risk for opioid overdose (e.g., obesity, pulmonary disease, obstructive sleep apnea, congestive heart failure, history of alcohol or substance use disorder, advanced age, renal or hepatic dysfunction).

Recommendations.

Core Requirements.

Recommendation 201. Opioid medications are not generally recommended for the treatment of low back pain. Rare exceptions include either of the following:

- acute, severe functionally limiting pain in a patient for whom other non-opioid medications are contraindicated; or
- acute, severe functionally limiting pain that is refractory to non-opioid medications and nonpharmacologic treatment and an absence of risk factors for potential misuse or abuse ([table 93](#)).

Recommendation 202. Long-acting opioids are not recommended for the treatment of acute, subacute, or postoperative pain.

Recommendation 203. Opioid medications, including tramadol, are not generally recommended for use in patients with a history of opioid dependence. However, if an opioid medication is deemed clinically appropriate, a referral to a pain specialist is permitted.

Short-term Prescribing Requirements.

Recommendation 204. Prior to dispensing an opioid medication, the following steps are required:

- documented results of a rapid risk assessment for developing opioid use disorder (e.g., Opioid Risk Tool [ORT]),
- risk assessment of developing opioid related adverse events,
- review data on the Colorado PDMP,
- education on the short- and long-term risks and side effects of opioid therapy,
- realistic goals of opioid therapy and the anticipated course of recovery,
- establish the lowest effective dose and shortest duration of therapy,
- education on the safe storage and disposal of opioid medications, and
- develop a discontinuation plan for opioids prior to prescribing.

Prescribing Requirements Beyond 7 Days.

Recommendation 205. Whenever opioids are prescribed for more than 7 days, providers must follow all recommendations for screening and follow-up of chronic pain use. [See the Chronic Pain Disorder MTGs.](#)

Time Frames.

Time Frames for Opioids	
Optimum duration	Maximum duration
≤ 3 days	7 days [‡]
[‡] Whenever there is use of opioids for > 7 days, providers should follow all recommendations for screening and follow-ups of chronic pain use.	

Evidence Tables.

Table 91.
Evidence Table: Risk of Initiating Opioid Medications

Summary:

Short-term opioid prescriptions increase the risk of disability and the risk of developing an opioid use disorder. The use of 50 morphine milligram equivalents (MME) or greater average daily dose or the use of long-acting opioids increases the risk of opioid-related death.

Good evidence	Evidence statement	Design
	In the setting of common low back injuries, when baseline pain and injury severity are taken into account, a prescription for more than 7 days of opioids in the first 6 weeks is associated with an approximate doubling of disability 1 year after the injury.	Cohort study
	In the setting of new onset chronic noncancer pain, there is a clinically important relationship between opioid prescription and subsequent opioid use disorder. Compared to no opioid use, short-term opioid use approximately triples the risk of opioid use disorder in the next 18 months. Use of opioids for over 90 days is associated with very pronounced increased risks of the subsequent development of an opioid use disorder, which may be as much as 100-fold when doses greater than 120 MME are taken for more than 90 days. The absolute risk of these disorders is very uncertain but is likely to be greater than 6.1% for long duration treatment with a high opioid dose.	Cohort study
	In generally healthy patients with chronic musculoskeletal pain, treatment with long-acting opioids, compared to treatments with anticonvulsants or antidepressants, is associated with an increased risk of death of approximately 69%, most of which arises from non-overdose causes, principally cardiovascular in nature. The excess cardiovascular mortality principally occurs in the first 180 days from starting opioid treatment.	Cohort study
	Prescription opioids in excess of 200 MME average daily doses are associated with a near tripling of the risk of opioid-related death, compared to average daily doses of 20 MME. Average daily doses of 100-200 mg and doses of 50-99 mg per day may be associated with a doubling of mortality risk, but these risk estimates need to be replicated with larger studies.	Cohort study

Some evidence	Evidence statement	Design
	Compared to an opioid dose under 20 MME per day, a dose of 20-50 mg nearly doubles the risk of death, a dose of 50 to 100 mg may increase the risk more than fourfold, and a dose greater than 100 mg per day may increase the risk as much as sevenfold. However, the absolute risk of fatal overdose in chronic pain patients is fairly low, and may be as low as 0.04%.	Cohort study

Table 92.
Evidence Table: Adverse Effects of Opioid Medications

Strong evidence	Evidence statement	Design
	Adverse events such as constipation, dizziness, and drowsiness are more frequent with opioids than with placebo.	Systematic review
Good evidence	Evidence statement	Design
	Opioids produce significantly more adverse effects than placebo such as constipation, drowsiness, dizziness, nausea, and vomiting.	Systematic review

Table 93.
Evidence Table: Opioid Medications Effectiveness

Summary:

The majority of patients with musculoskeletal pain, including low back pain, do not experience clinically meaningful pain relief or functional improvement with opioid treatment as compared to placebo. For patients with neuropathic pain, there is no evidence that opioid treatment improves function or quality of life. For neuropathic pain patients, opioid treatment is more likely to result in pain relief as compared to placebo, but there is no evidence that opioids are superior to gabapentin or nortriptyline for this indication. Opioids produce significantly more adverse effects than placebo.

Strong evidence	Evidence statement	Design
	In the setting of chronic nonspecific low back pain, the short and intermediate term reduction in pain intensity of opioids, compared with placebo, falls short of a clinically important level of effectiveness for a majority of patients, although some patients may experience a clinically meaningful analgesic benefit.	Systematic review
Good evidence	Evidence statement	Design
	In patients with chronic low back pain, or chronic pain from osteoarthritis of the hip and knee, opioid therapy does not lead to better pain outcomes in terms of daily functioning or in pain intensity.	RCT
	In patients with acute nontraumatic musculoskeletal low back pain who have optimized their use of NSAIDs such as naproxen, the addition of oxycodone 5 mg/acetaminophen 325 mg to be taken as needed adds no significant functional benefit in the short term, but it does increase the frequency of adverse events such as dizziness, drowsiness, and GI side effects.	RCT
	Opioids are more efficient than placebo in reducing neuropathic pain by clinically significant amounts during the first 8 weeks of treatment.	Systematic review

Lack of evidence	Lack of evidence statement	Design
	There is a lack of evidence that opioids improve function and quality of life more effectively than placebo.	Systematic review
Table 93 continued. <i>Evidence Table: Opioid Medications Effectiveness</i>		
Lack of evidence cont.	There is a lack of evidence that opioids are superior to gabapentin or nortriptyline for pain reduction.	Systematic review

Table 94. <i>Evidence Table: Pre- and Post-Operative Medication Use</i>		
Some evidence	Evidence statement	Design
	In the setting of PLIF surgery, a single saline infusion of 800 mg of ibuprofen one half hour prior to surgery may lead to less pain and morphine use in the immediate postoperative period and may thereby be an option for preoperative care of the surgical patient.	RCT
	In the setting of elective open single-lumbar discectomy for herniated discs, where morphine is delivered by patient-controlled anesthesia, a low-dose separate infusion of 0.25 mcg/kg/hour of naloxone in the 24 hours after surgery may reduce morphine consumption and reduce postoperative pain, nausea, and pruritus.	RCT
	In the setting of 1-level or 2-level spinal fusion surgery, the administration of pre-emptive analgesia 8 hours preoperatively with acetaminophen, ketorolac, and pregabalin may confer benefits in the 48 hours following the procedure in the form of less pain, better ambulation, and less morphine use.	RCT
	The preoperative administration of gabapentin in doses as low as 300 mg may reduce postoperative pain and opioid consumption in the setting of spinal laminectomy and fusion. The incidence of many postoperative complications such as nausea, headache, and dizziness is probably similar for gabapentin and placebo. Gabapentin may increase somnolence but may decrease the incidence of urinary retention pruritus.	Systematic review

Section 9.g. Tobacco Cessation and Supplements

Introduction. Nicotine replacement therapy is used to relieve nicotine withdrawal symptoms by providing nicotine without the use of tobacco. The mechanism of bupropion in smoking cessation is not entirely understood, but it is believed to act by enhancing central nervous system noradrenergic and dopaminergic release. Varenicline reduces the symptoms of nicotine withdrawal by binding to the receptor that mediates the reinforcing effects of nicotine dependence.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Tobacco Cessation Therapies.

- Use nicotine replacement therapy with caution in patients with unstable cardiovascular disease.

- Bupropion is contraindicated in those with seizure disorders.

Side Effects and Adverse Events Related to Tobacco Cessation Therapies.

- Nicotine replacement therapy can cause GI symptoms, headache, and local irritation with topical products.
- Bupropion can cause insomnia, agitation, dry mouth, and headache.
- Varenicline can cause nausea, insomnia, neuropsychiatric disorders, and abnormal dreams.
- Review package insert for supplement contraindications and side effects.

Recommendations.

Core Requirements.

Recommendation 206. Tobacco cessation, including medication and behavioral support, are recommended when tobacco use is expected to negatively impact claim-related medical outcomes. Medications may include nicotine patches, gum, inhaler, lozenges or nasal spray, bupropion, or varenicline (table 95).

Evidence Tables.

Table 95.		
<i>Evidence Table: Smoking and Non-Operative Spine Care</i>		
Some evidence	Evidence statement	Design
	Patients who smoke respond less favorably to non-operative spine care than nonsmokers, and quitting smoking during treatment results in greater improvement than patients who continue smoking during treatment.	Cohort study

Table 96.		
<i>Evidence Table: Supplements</i>		
Good evidence	Evidence statement	Design
	Glucosamine does not improve pain related disability in those with chronic low back pain and degenerative changes on radiologic studies. Therefore, it is not recommended for chronic lower spinal or non-joint pain.	RCT
Some evidence	Evidence statement	Design
	In patients with chronic low back pain not associated with nerve root compression or radicular symptoms, treatment with pregnenolone may have a moderate short term benefit in reducing pain of moderate intensity as compared with placebo. No information is available regarding functional benefits.	RCT

Section 10. Interdisciplinary Rehabilitation Programs

Introduction. These guidelines discuss an interdisciplinary approach to low back pain treatment in the acute and subacute timeframes. Interdisciplinary rehabilitation programs are the gold standard of treatment for individuals with chronic low back pain who have not responded to less intensive modes of treatment or individuals who require concurrent treatment for chemical dependency. See the Chronic Pain Disorder Medical Treatment Guidelines (MTGs) for additional information, including indications, recommendations, and time frames.

Appendix

Functional Tests

Name of Test	Body Part	Description
Oswestry Low Back Pain Disability Questionnaire (ODI)	Lumbar Spine	Functional questions with sleeping, lifting, walking, sitting, and standing
Quebec Back Pain Disability Scale	Spine	Functional and emotional questions
Neck Disability Index	Cervical Spine	10 questions addressing pain, personal care, headaches, and functional deficits
36-Item Short Form Survey (SF-36) and 12-Item Short Form Survey (SF-12)	Physical Health	Assesses activities of daily living (ADLs)
Dallas Pain Questionnaire	Spine	Chronic pain
Brief Pain Inventory	General	Sleep, walking, ADLs

Psychological Screens

Name of Test	Description
Brief Battery for Health Improvement, 2 nd Edition (BBHI 2)	Measures pain, functioning, somatization, depression, anxiety, and defensiveness; brief measure of risk factors for delayed recovery
Distress and Risk Assessment Method (DRAM)	Measures depression and somatic symptoms of anxiety, risk factors commonly associated with chronic pain
Center for Epidemiological Studies Depression Scale (CES-D)	Measures depression, 20 items
Beck Depression Inventory-II (BDI-II)	Measures depression, 21 items
Primary Care Evaluation for Mental Disorders (PRIME-MD) <i>Must be filled out by a provider</i>	2 components: paper and pencil screen for patient and follow-up interview by physician. Assesses mood, anxiety, somatoform tendencies, and alcohol and eating disorders

Zung Depression Inventory	Measures depression, brief measure
Patient Health Questionnaire (PHQ) and PHQ-9	Self-administered version of the PRIME-MD. Assesses mood, anxiety, somatoform tendencies, and alcohol and eating disorders
Generalized Anxiety Disorder Scale (GAD-7)	Assesses generalized anxiety, 7 questions
Behavioral Health Index-Multimedia Version (BHI-MV)	Screens for addiction

Editor's Notes

7 CCR 1101-3 has been divided into smaller sections for ease of use. Versions prior to 01/01/2011, and rule history, are located in the first section, 7 CCR 1101-3. Prior versions can be accessed from the History link that appears above the text in 7 CCR 1101-3. To view versions effective after 01/01/2011, select the desired part of the rule, for example 7 CCR 1101-3 Rules 1-17, or 7 CCR 1101-3 Rule 18: Exhibit 1.

History

[For history of this section, see Editor's Notes in the first section, 7 CCR 1101-3]